

SAFETY PRECAUTIONS

☒	Stopped engine
☒	Block the starting mechanism
☒	Shut off starting air supply
☒	Engage turning gear
☐	Shut off cooling water
☒	Shut off fuel oil
☐	Shut off lubricating oil
☐	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit
D09-01	One shim alters p-max	2.5	bar
D09-05	Nominal shim thickness	0.5	mm
D09-10	Max. number of shims	16	pcs.

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P90951 refers to chapter 909.

Plate	Item No.	Description
P90951	200	Hexagon head screw for topcover

Before adjustment of the fuel pump is carried out, see *Volume I, Chapter 706*, also the fuel cam lead should be checked, see *Procedure 909-3.1*.

$p_{\max}$ adjustment:

1. Before Maximum Combustion Pressure ($p_{\max}$) is adjusted, loosen the drain oil pipe from the top cover and dismount the top cover fixing nuts.

Remove the two screws in the threaded holes of the top cover.

2. Screw the two dismantling screws (from the tool panel) into the threaded holes of the top cover, thus lifting the top cover to provide space for inserting or removing shims.
3. To increase $p_{\max}$, remove shims (advance injection).

To decrease $p_{\max}$, insert shims (delay injection).

One shim will alter $p_{\max}$ approximately as given in Data D09-01.

For nominal shim thickness, see Data D09-05.

For maximum inserted number of shims, see Data D09-10.

Note down the new total shim thickness (number of shims).

4. Remove the dismantling screws, and press down the top cover by hand. Mount the two original screws to protect the threaded holes. Mount and tighten the nuts for the top cover, see *Procedure 909-5*.

Note!

The individual $p_{\max}$ value must not deviate more than 3 bar from the average value for all cylinders.

1.

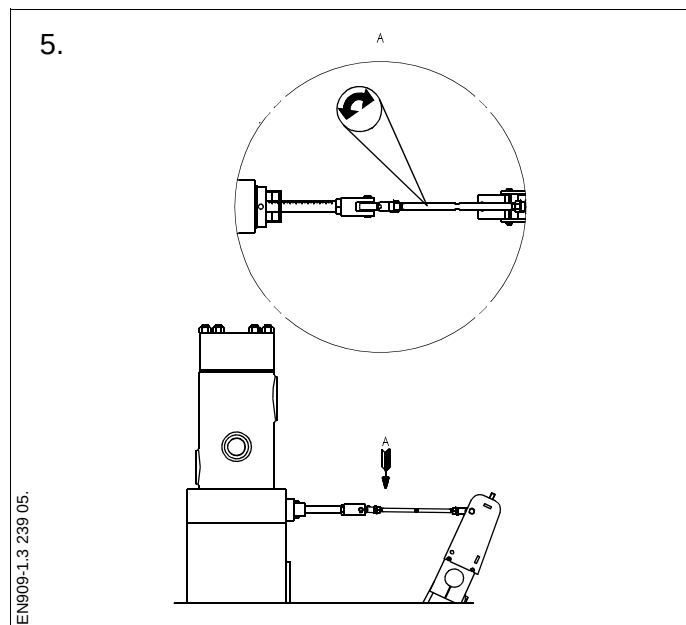
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3.

GN909-1.3 205 02

4.

GN909-1.3 205 03



If the required $p_{\max}$ adjustment cannot be attained by inserting or removing shims, it is possible to carry out the adjustment on the fuel cam disc. Contact MAN B&W Diesel for advice.

MIP adjustment:

5. The Mean Indicated Pressure (p_i) for a certain cylinder is adjusted on the fuel pump index arm.

Increase index for higher p_i .

Decrease index for lower p_i .

Note!

The mean indicated pressure is one of the most important parameters, in obtaining good engine performance.

The mean indicated pressure should not deviate more than 0.5 bar from the average value for all cylinders. See Volume I, Chapter 706.

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Ref.	Description	Value	Unit

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Plate	Item No.	Description
P90651	44	Impact wrench for turning cam
P90951	45	Measuring tool for fuel pump

Fuel cam lead:

The lead **C** of the fuel cam is defined as the number of mm the plunger is lifted from its bottom position when the main piston is in TDC.

1. To measure the fuel cam lead (top lift):
Dismantle the suction valve and the puncture valve. See *Procedures 909-7.2 and 909-8.2*.

Make sure that the reversing mechanism is in AHEAD direction.

2. Push down the measuring pin through the top cover, until it rests against the pump plunger.

Turn the engine until the lowest point of the cam is found, use a calliper to find the point.

3. Measure the distance **x** from the top of the measuring pin to the top cover.

Note down the result as **x2**.

4. Turn ahead until the main piston of the cylinder concerned is in TDC.

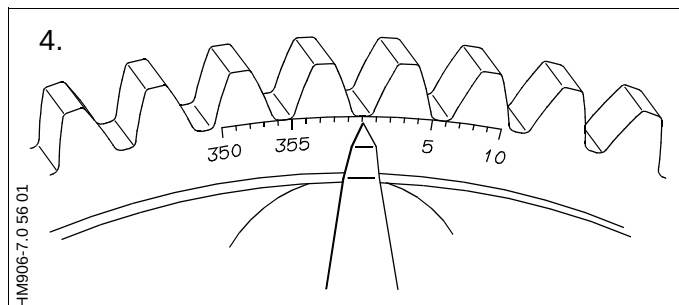
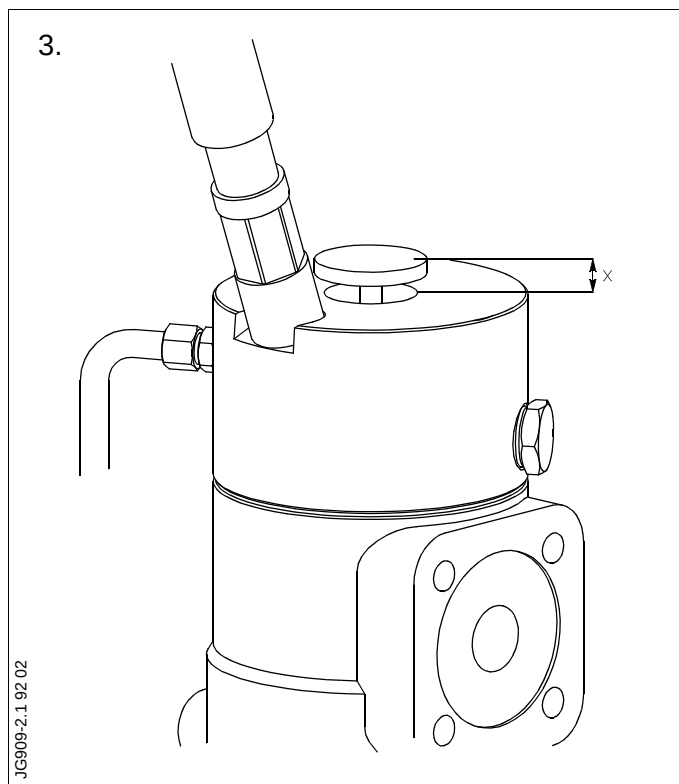
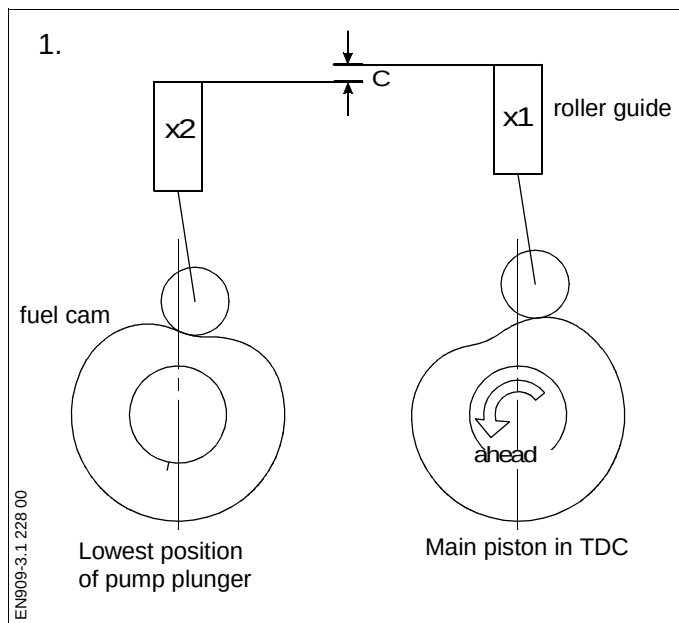
Again measure the distance **x** from the top of the measuring tool to the top cover. Note down the result as **x1**.

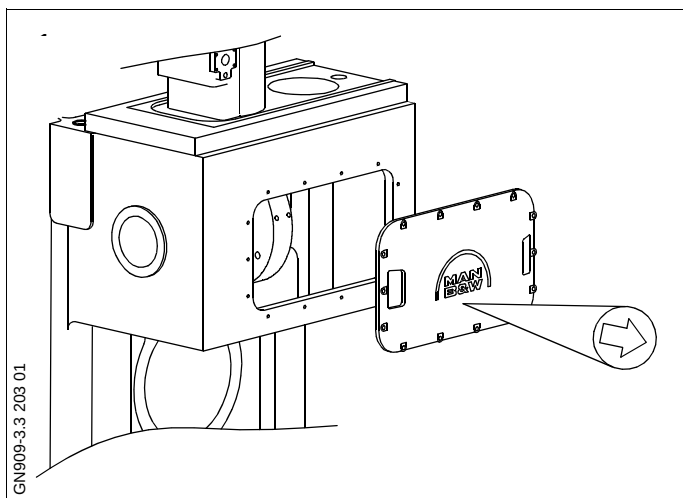
Calculate the fuel cam lead **C** (top lift) as:

$$C = x1 - x2 \text{ (mm)}$$

5. Note down the result for comparison with the testbed results and for future reference.
For adjustment of the fuel cam lead, see Procedure 909-3.3.

Remount the suction valve and the puncture valve.



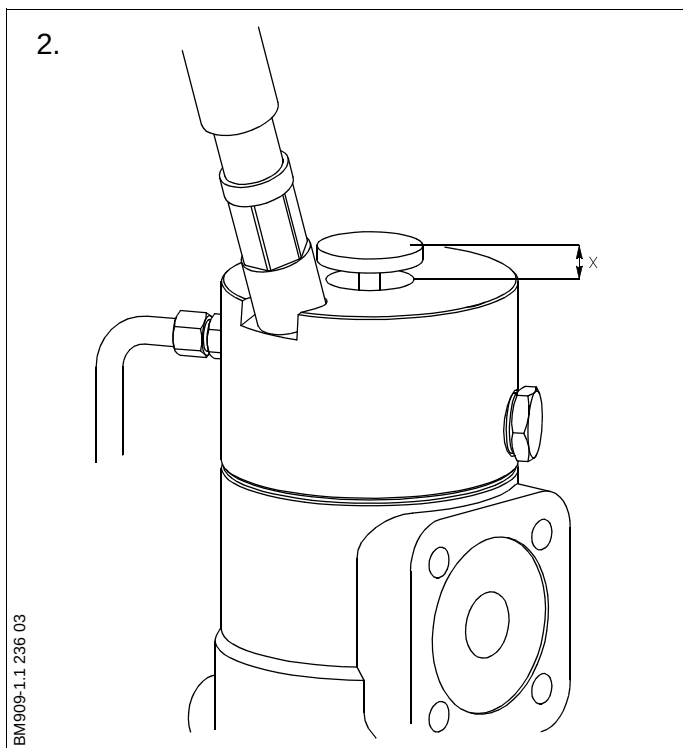


Normally, adjustment of the fuel cam should not be necessary. *Before adjustment of fuel equipment, see Volume I, Chapter 706.*

1. Remove the inspection cover from the camshaft housing.

Note!

Before dismantling any part of the fuel pump, make sure that the pump has been relieved of pressure and that all oil has been drained off.



2. Dismount the puncture valve and the suction valve from the fuel pump top cover, and mount the measuring tool.
See Procedure 909-3.1.

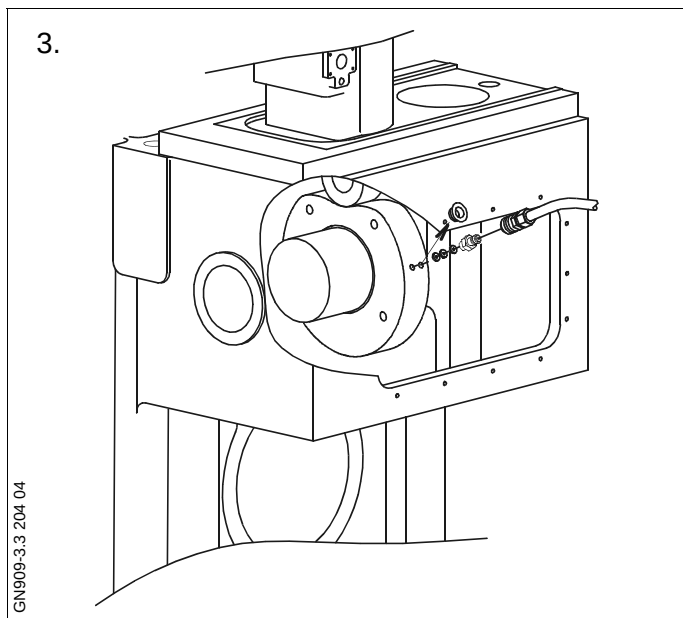
3. Turn the camshaft until there is access to the oil ducts in the fuel cam through the inspection hole.

If fitted, remove the plugs from the oil ducts (using, for instance, a screwdriver).

Insert three copper gaskets in each oil duct.

Mount snap-on couplings in the oil ducts, but do not tighten them.

Fit hoses between the snap-on couplings and the distributor block and between the distributor block and the hydraulic high-pressure pump.



4. Mount the special spanner on the fuel cam disc, ensuring that the two pins enter the holes in the cam.

Apply a light pressure to the hydraulic system and, after venting the system, tighten the snap-on couplings.

Raise the hydraulic pressure until oil seeps out along the camshaft under the cam disc.

Keep the cam pressurised for at least five minutes before turning the cam.

Keep the cam pressurised during turning.

Turn the cam disc, using the fitted spanner, until the desired change of lead is obtained.

To increase lead and p_{max} :
– turn the cam disc AHEAD.

To reduce lead and p_{max} :
– turn the cam disc ASTERN.

See *Procedure 909-1*.

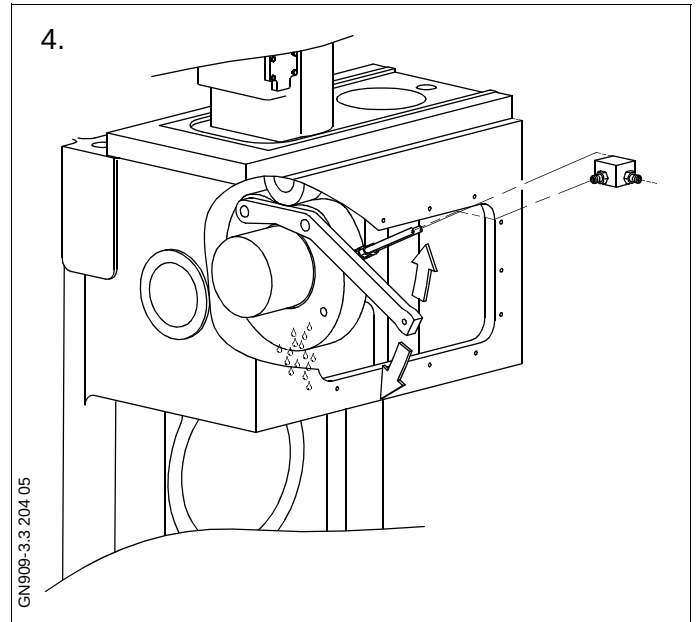
5. After completing the desired turning of the cam disc, relieve the hydraulic system of pressure and dismount the spanner and the hydraulic equipment.

Wait at least 15 minutes – the cam must be allowed time to “settle” – before mounting the plugs again in the oil ducts of the cam disc.

After carrying out adjustment, measure the fuel cam lead again.

Note down and file the new results for purposes of comparison with future measurements and adjustment.

6. Mount the inspection cover on the camshaft housing.
7. Mount a new O-ring on the puncture valve. Mount the puncture valve and the fuel oil high pressure pipes.
See *Procedure 909-8.4* and *909-14.4*.



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Data

Ref.	Description	Value	Unit
D09-18	Pump housing nuts, tightening torque	300	Nm
D09-19	Pump housing nuts, tightening angle	62	°
D09-20	Pump housing	75	kg

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P90951 refers to chapter 909.

Plate	Item No.	Description
P90951	70	Lifting tool for fuel pump housing

Note!

Before dismantling any part of the fuel pump, make sure that the pump has been relieved of pressure and that all oil has been drained off.

1. Dismantle the fuel pump top cover/plunger barrel assembly.
See Procedure 909-5.2.

2. Remove the fuel oil inlet pipe between the fuel pump and the fuel oil inlet valve.

Discard the gaskets.

3. Remove the nuts at the base of the fuel pump.

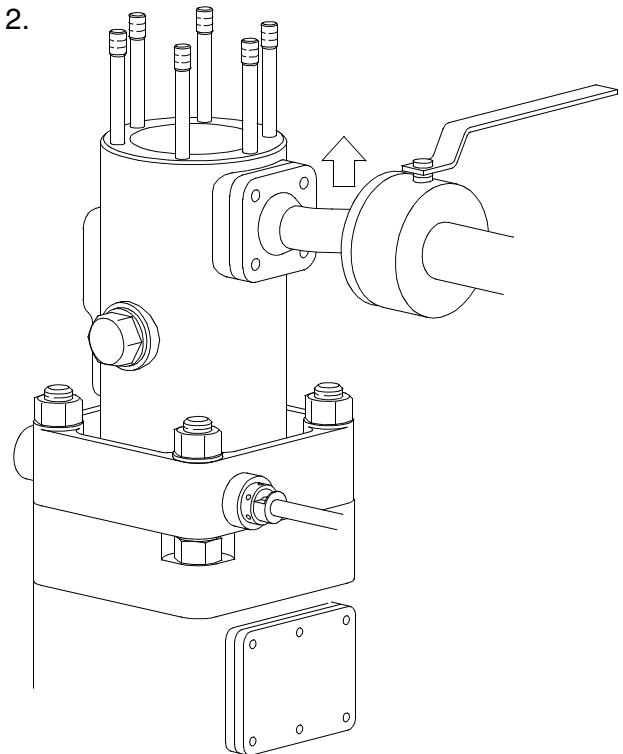
Mount the lifting tool and lift the fuel pump housing away. *See Data.*

Note!

Take care not to damage the threads of the studs.

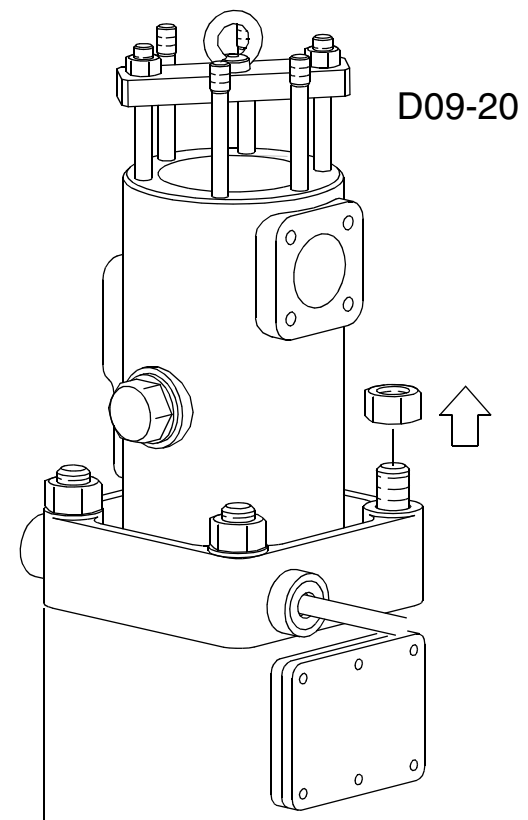
Land the fuel pump on wooden planks.

2.



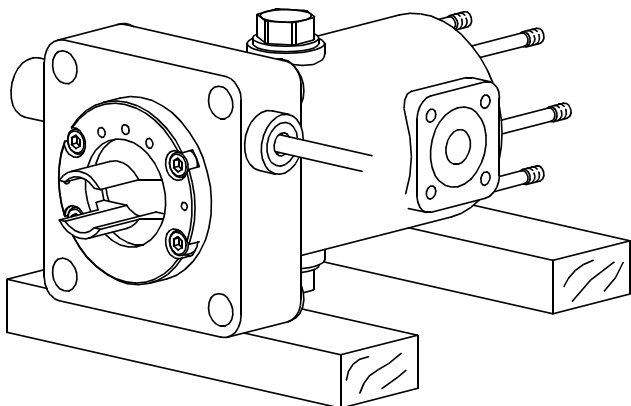
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3.



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1.



1. Tilt the fuel pump housing into a horizontal position on the wooden planks.

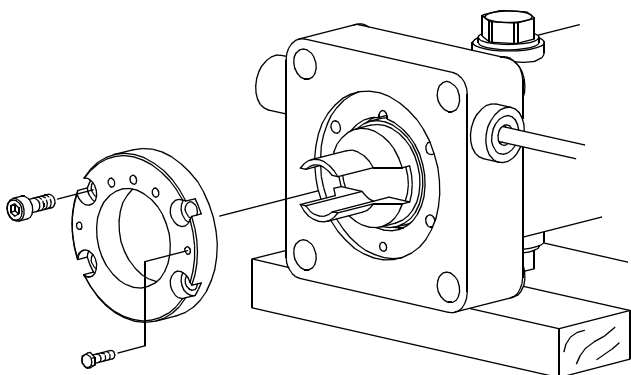
2. Remove the bottom plate from the fuel pump housing.

If the bottom plate sticks, loosen it by screwing two M10x70 mm screws into the threaded holes of the bottom plate.

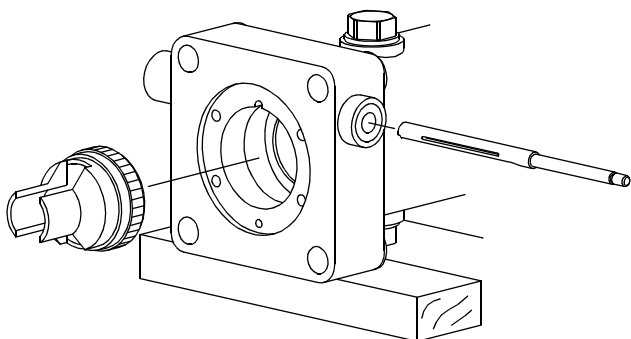
3. Dismantle the:

- Fuel oil index arm.
- Regulating guide.

2.



3.



4. Dismantle and inspect the erosion plugs on both sides of the fuel pump housing.

If heavily eroded, replace the erosion plugs.

5. Clean all the parts in diesel oil or kerosene. Blow dry with compressed air.

Warning!

Always use eye protection when working with compressed air.

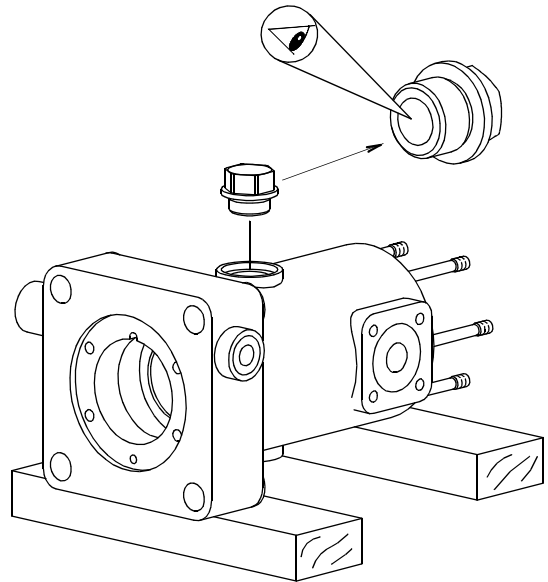
6. Mount a new sealing ring in the pointer for the fuel oil index arm.

Note!

Soak the sealing ring in lubricating oil before mounting.

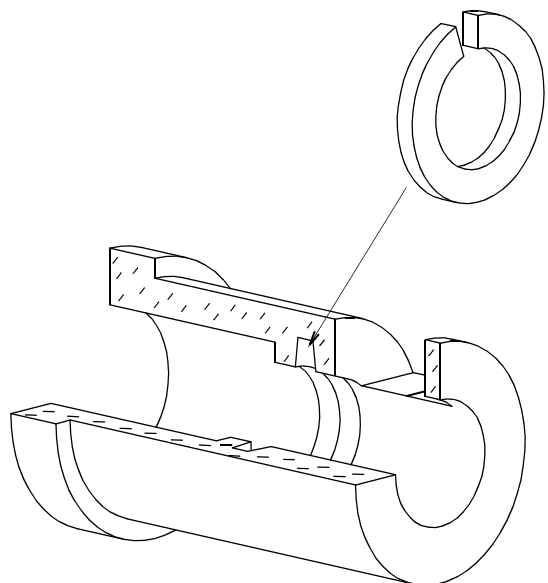
7. Clean all inside surfaces of the fuel pump housing.

4.



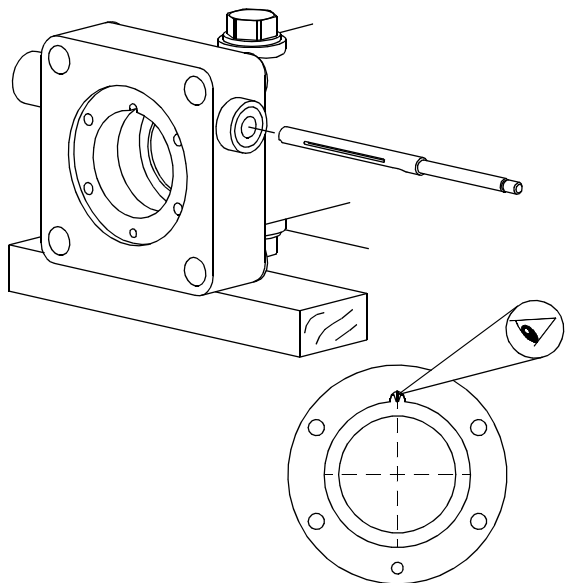
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6.



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9.



8. Lubricate the following with molybdenum disulphide grease (MoS_2):

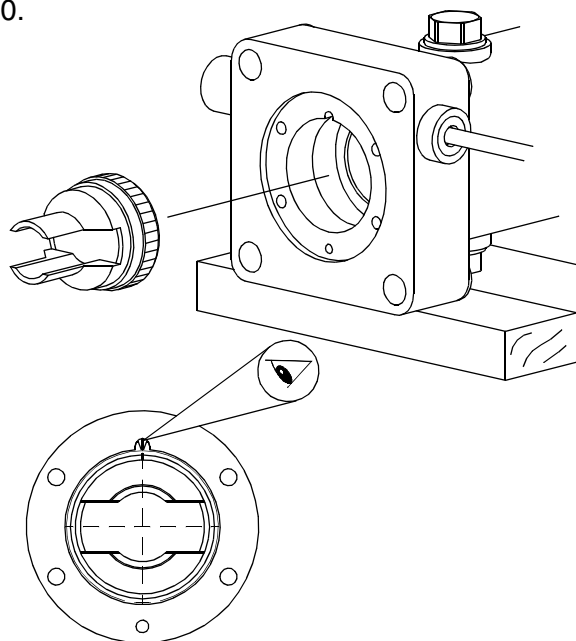
- All internal surfaces of the fuel pump housing.
- The sliding surfaces of all the parts of the regulating system.
- The teeth of the:
 - Regulating guide.
 - Fuel index arm.

9. Mount the fuel oil index arm.

Adjust the position of the fuel oil index arm so that the scratch mark on the fuel oil index arm is aligned with the centre of the fuel pump housing.

Mount the fuel oil index pointer.

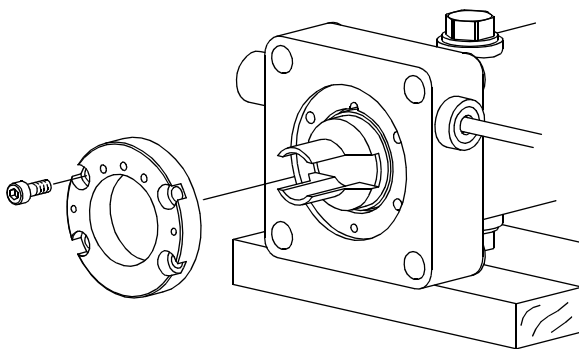
10.



10. Mount the regulating guide in the fuel pump housing, ensuring that the scratch mark on the regulating guide is aligned with the scratch mark on the fuel oil index arm.

11. Mount the bottom plate on the fuel pump housing.

11.



Note!

Before mounting the fuel pump housing, make sure that all sliding faces and threads are cleaned and lubricated with molybdenum disulphide grease (MoS₂).

See Procedure 913-11.

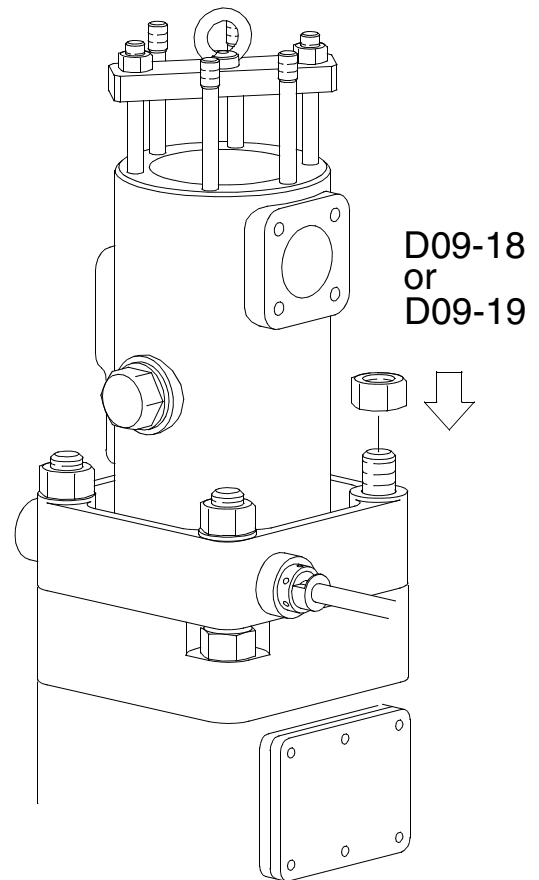
1. Lower the fuel pump housing on to the pump base, taking care not to damage the threads of the studs.

Remove the lifting tool.

Mount and tighten the nuts at the base of the fuel pump housing. See Data.

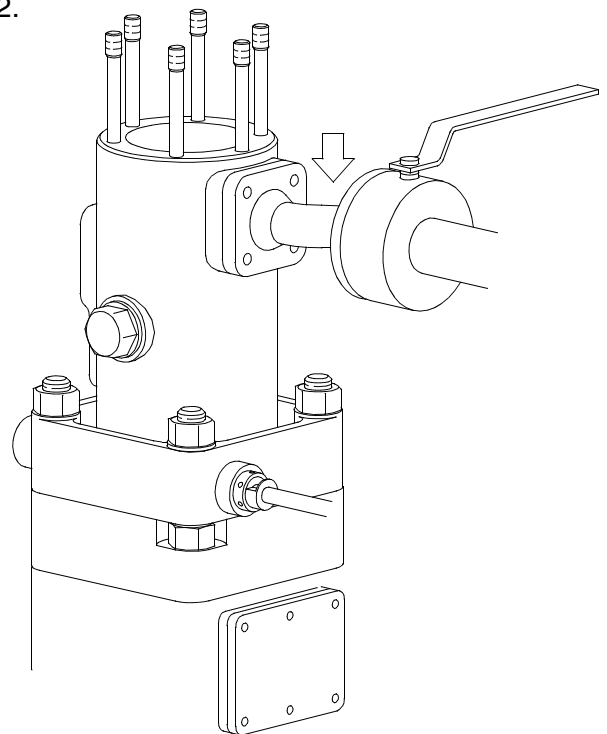
2. Mount the fuel oil inlet pipe between the fuel pump and the fuel inlet valve.
3. Mount the fuel pump top cover/plunger-barrel assembly.
See Procedure 909-5.4.

1.



EN909-4.4 205 01

2.



EN909-4.4 205 02

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Data

Ref.	Description	Value	Unit
D09-23	Top cover studs, screwing-in torque	100	Nm
D09-24	Top cover nuts, tightening torque	210	Nm
D09-25	Pump barrel/top cover, tightening torque	30	Nm
D09-26	Outlet seat, max. grinding diameter	18	mm
D09-27	Top cover	30	kg
D09-28	Plunger/barrel assembly + top cover	60	kg

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P90951	69	Lifting tool for fuel pump
P90951	94	Milling tool for fuel oil pipe seats
P90951	200	Hexagon head screw for topcover
P90951	297	Extraction tool for suction valve

1. Close the oil inlet valve to the fuel oil pump and drain the pump.

Dismount the drain oil pipe from the top cover and the air pipe connection to the puncture valve.

Dismount the high pressure pipes.
See *Procedure 909-14*.

Dismount the puncture valve from the top cover. See *Procedure 909-8*.

Dismount the fuel index pointer.

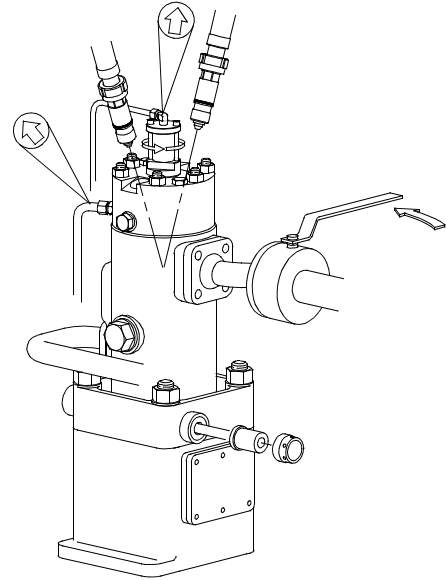
2. Dismount the suction valve by means of the extractor tool.

3. Remove the top cover fixing nuts, as well as the two screws in the top cover.

Loosen the top cover by means of the two dismantling screws from the tool panel.

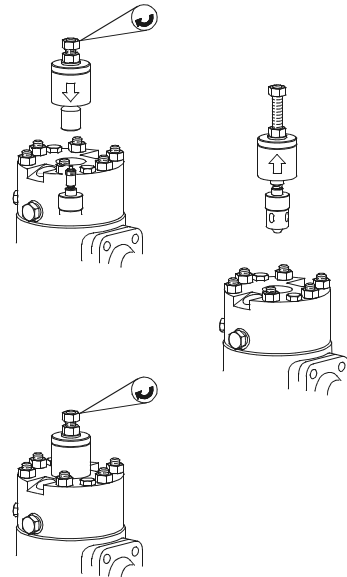
Release the link for the regulating drive and pull out the toothed rack as far as possible.

1.



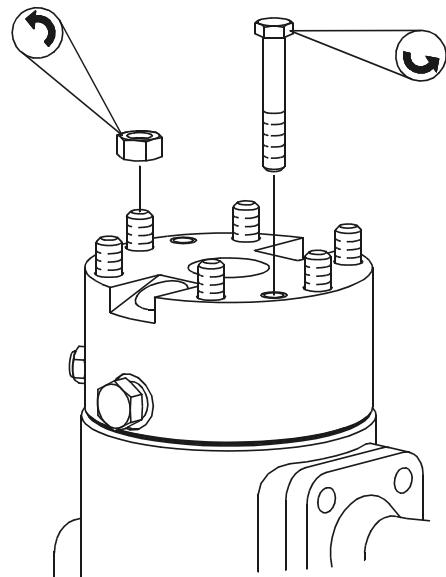
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2.

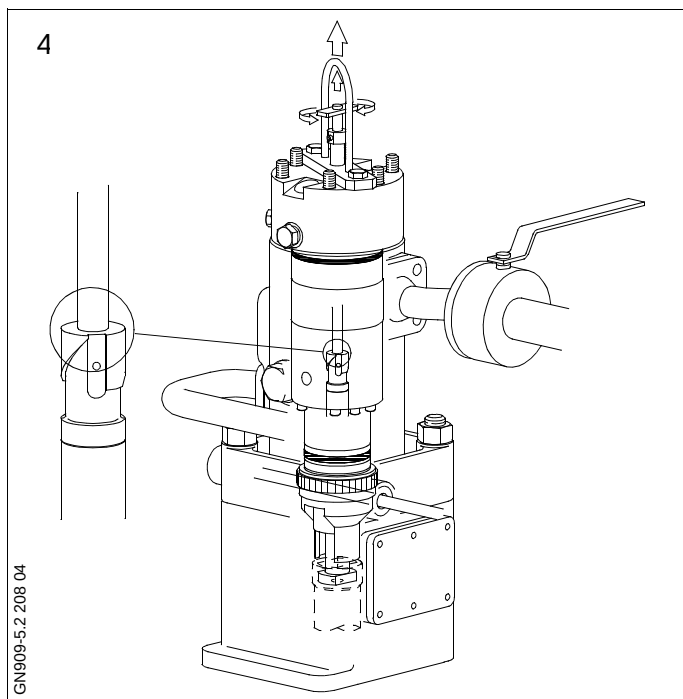


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3.



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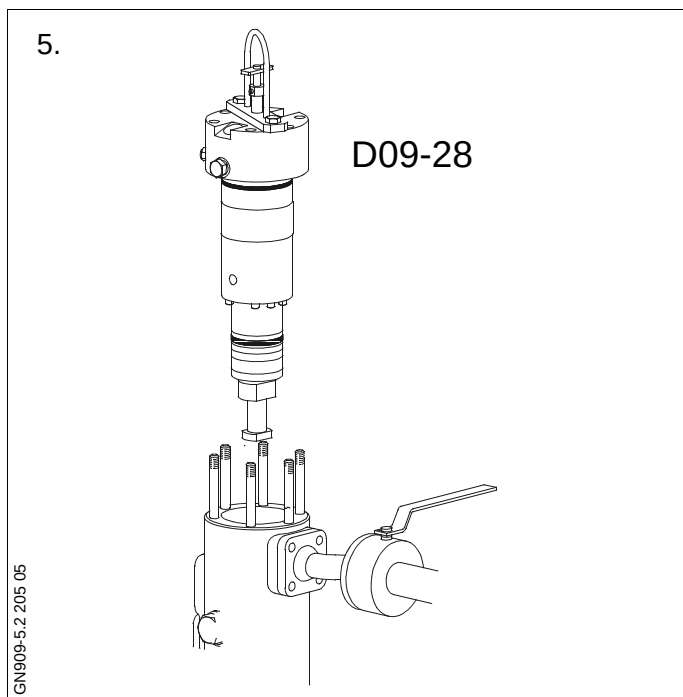


4. Mount the lifting tool on the fuel pump top cover. Screw the centre screw of the tool into the pump plunger. Ensure that the stop ring on the centre screw is loose.

Pull the centre screw of the tool to lift the plunger clear of the bayonet joint in the roller guide and the milled recess of the regulating guide.

To do this it may be necessary to move the toothed rack and the centre screw of the tool in relation to each other.

When the plunger has been lifted clear of the roller guide and regulating guide, lock the centre screw in this position with the stop ring.



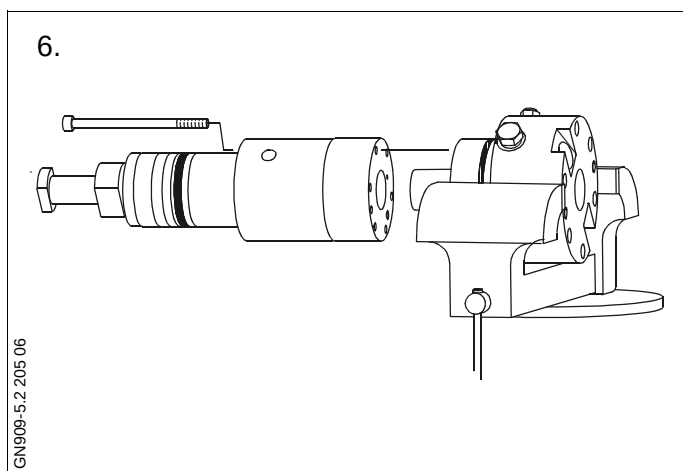
5. Lift the top cover and barrel with plunger out of the fuel pump housing as one unit. Remove the shims and tag them for reinstallation.

6. Set up the pump unit in a bench vice provided with 'soft' jaws. Dismount the lifting tool. Take care that the plunger does not slide out of the barrel.

Press the plunger to the top of the barrel and remove the barrel/plunger assembly from the top cover.

Send the barrel/plunger assembly to an MAN B&W authorised workshop for overhaul, or overhaul it on board.

See *Procedure 909-6.3*.



1. Remove and discard the sealing rings.

Carefully clean and examine the top cover.

Reconditioning of the seatings for the high-pressure pipes is carried out by milling according to the following procedure:

Fill the oil ducts in the top cover with Vaseline or heavy grease.

During the work the miller is guided by the guide screwed into the thread for the high-pressure pipe.

Turn the miller by means of, e.g. a tap wrench, while tightening the guide screw lightly to provide a suitable pressure between miller and seat.

During the milling, add drilling oil emulsion liberally.

For the max. milling/grinding diameter of the seat, see Data D09-26.

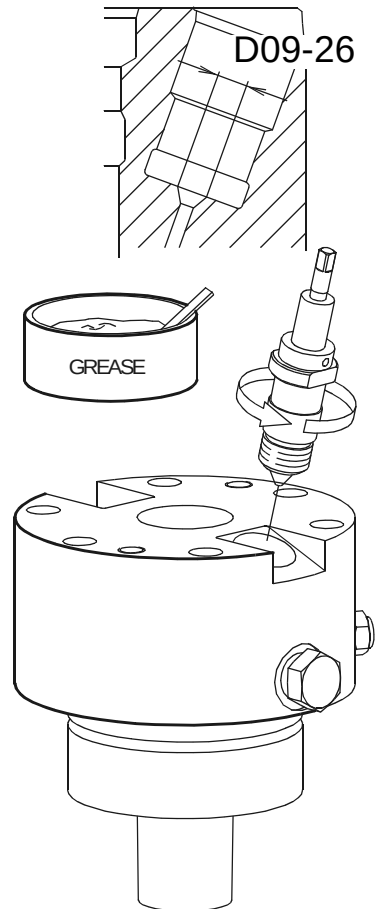
2. After completing the milling, blow out the Vaseline/grease from the oil ducts by means of compressed air.

Warning!

Always use eye protection when working with compressed air.

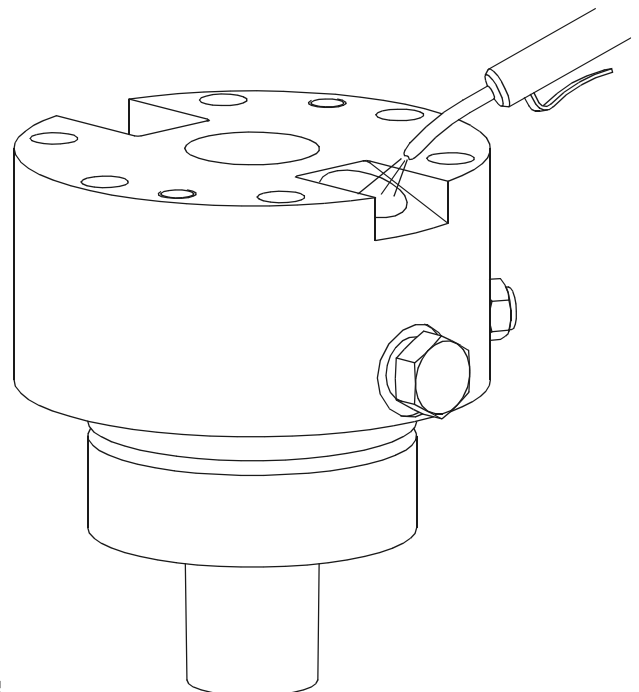
Clean the top cover with diesel oil, and blow dry with compressed air.

1.

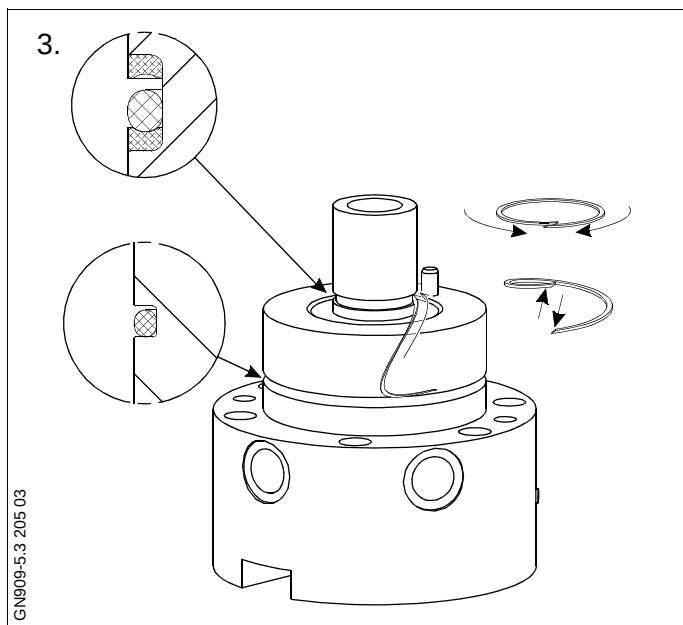


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2.



GN909-5.3 205 02



3. Mount the sealing rings in the ring grooves.

Sealing between the top cover and the fuel pump housing is provided by an O-ring.

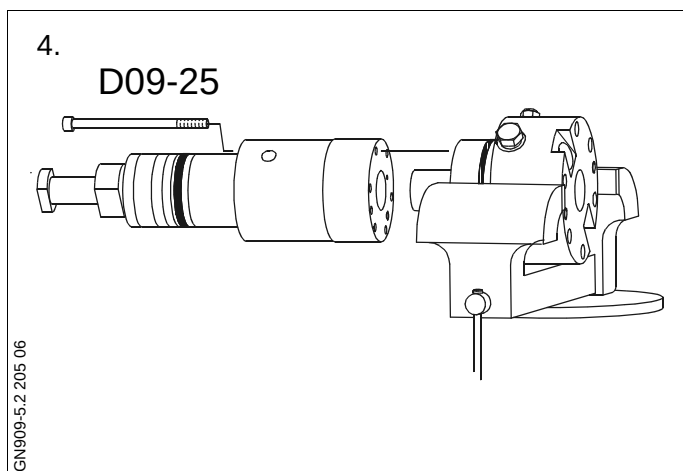
Sealing between the top cover and pump barrel is provided by two back-up rings and an O-ring in between.

Prepare the back-up rings by twisting the rings in the hand, in such a way that the diameter of the rings becomes smaller than the diameter of the ring groove in which the rings are to be mounted.

'Screw' the back-up rings onto the top cover, into the groove.

Finally, insert the O-ring **between** the back-up rings in the ring groove.

Mount the O-ring which seals between top cover/fuel pump housing.



4. Mount the overhauled top cover on top of the pump barrel, guided by the guide pin. Tighten the screws, see Data D09-25.

1. Mount the supplied measuring pin on the centre screw of the lifting tool as shown in the sketch. Loosen the stop ring on the centre screw.

Place the tool on the pump housing and press down the centre screw until the measuring pin reaches the thrust piece of the roller guide. Lock the centre screw in this position with the stop ring.

Keep the stop ring tightened in this position until the plunger has been correctly mounted.

Note!

The engine **must not** be turned until the mounting of the barrel/plunger assembly has been completed.

2. Remove the tool from the pump housing and remove the measuring pin from the centre screw.

Mount the tool on the barrel/plunger and top-cover assembly.

Tighten the tool to the top cover by means of the two screws, and attach the plunger to the centre screw.

The tool is now firmly attached to the assembly, enabling the correct position to be obtained during mounting.

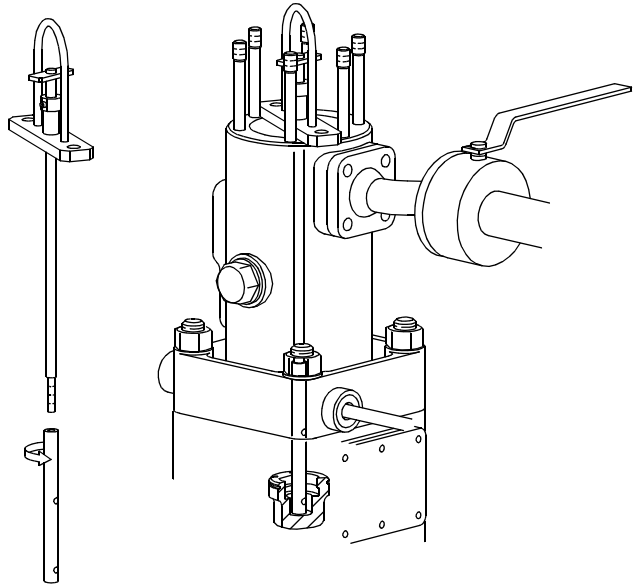
Pull the regulating toothed rack out as far as possible.

Pull the plunger as high up in the barrel as possible.

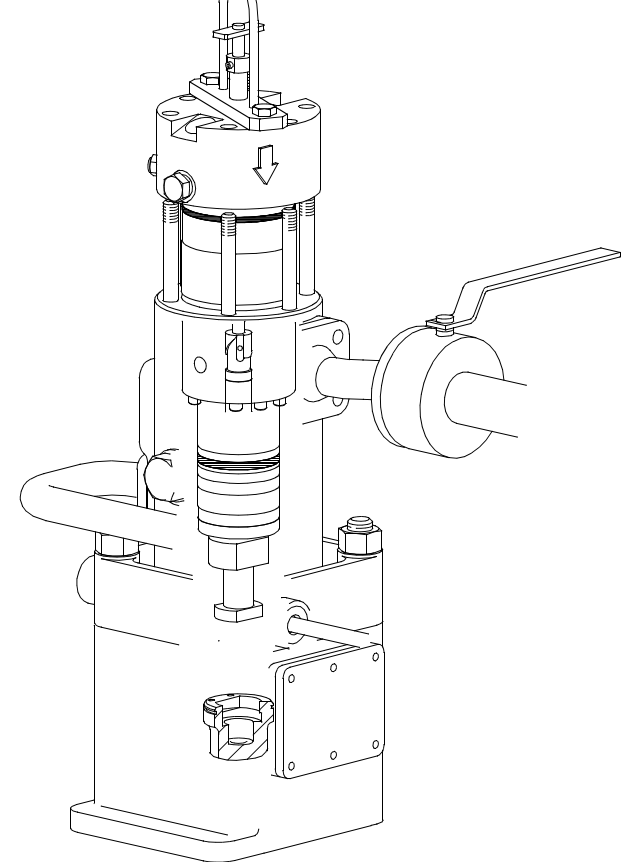
Carefully lower the assembly into the pump housing (taking special care not to damage the O-ring and back-up rings on the barrel).

Adjust the top cover so that the guide pin enters the hole in the pump housing. Lower the top cover on to the pump housing.

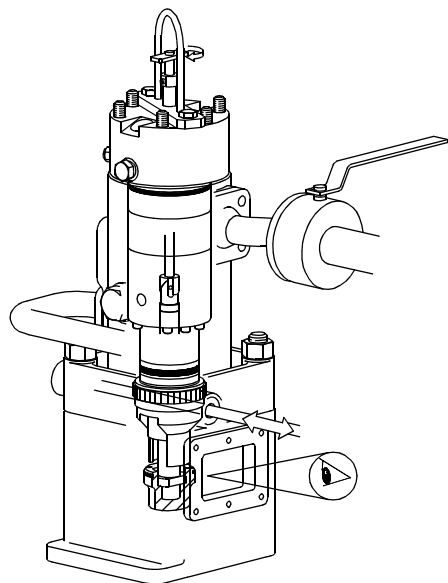
1.



2.



3.



3. Press the plunger down into contact with the thrust piece of the roller guide.

If necessary, turn the regulating guide a little (using the toothed rack) to make the foot and regulating block of the plunger fit properly in the cutout of the regulating guide.

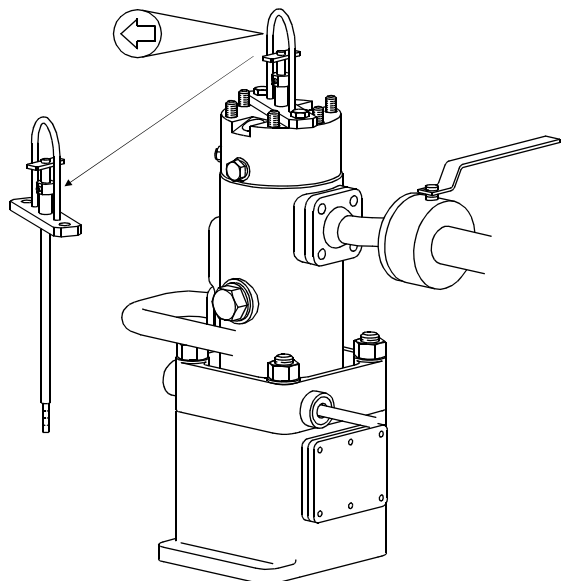
Check that the stop ring of the centre screw is in full contact with the flange of the tool, as when measuring.

When the plunger is in place, press the regulating toothed bar in, causing the plunger foot to interlock with the bayonet joint of the roller guide.

Check through the inspection cover on the fuel pump base and by pulling the centre screw of the tool, that the plunger is correctly engaged.

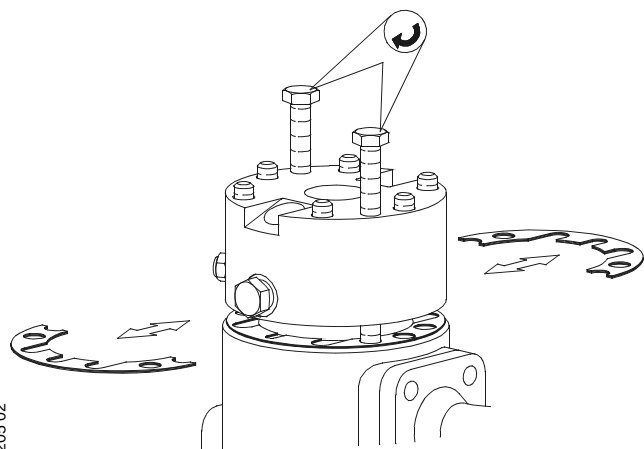
When pulling on the centre screw it must not be possible to lift the plunger more than a few mm.

4.



4. Unscrew the centre screw from the plunger and remove the tool from the top cover.
5. Screw the two dismantling screws from the tool panel into the threaded holes of the top cover, thus lifting the top cover to provide space for inserting shims.

5.



6. Insert shims between the pump housing and the top cover. (If the barrel assembly is the same as that dismantled, insert the same number of shims as were removed). If a new barrel assembly is mounted, insert the number of shims specified in the test bed report.

7. Replace the dismantling screws with the two screws for mounting the lifting tool, and press down the top cover.

Turn the engine one complete revolution, and check that the top cover does not lift. If the top cover lifts, the plunger is **not** correctly mounted and must be remounted.

Mount the top cover fastening nuts, and tighten them diagonally.
See Data D09-24.

If any of the fuel pump top cover studs have come loose, tighten them.
See Data D09-23.

8. Mount the suction valve and the puncture valve in the top cover.
See Procedure 909-8.

Connect the drain oil pipe to the top cover and the pipe connection to the puncture valve.

Mount the two high-pressure pipes on the top cover/fuel valves.
See Procedure 909-14.

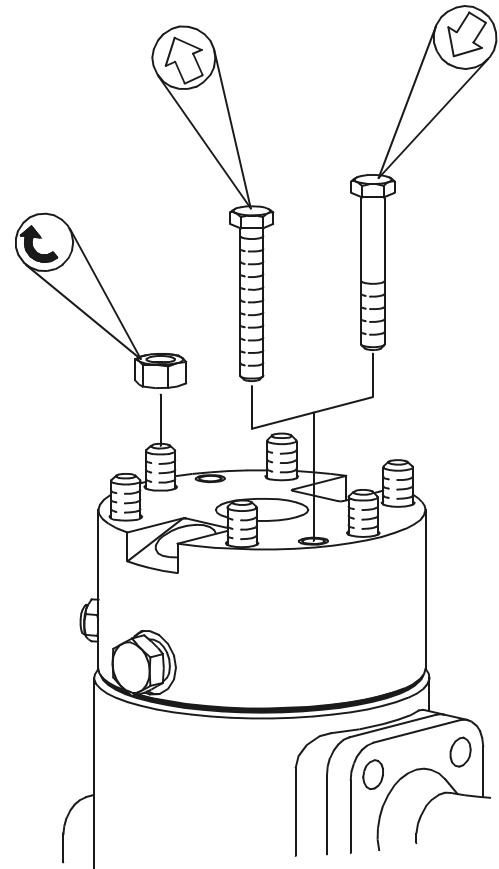
Mount the fuel index pointer

Mount the link for the regulating drive.

Open the fuel oil inlet.

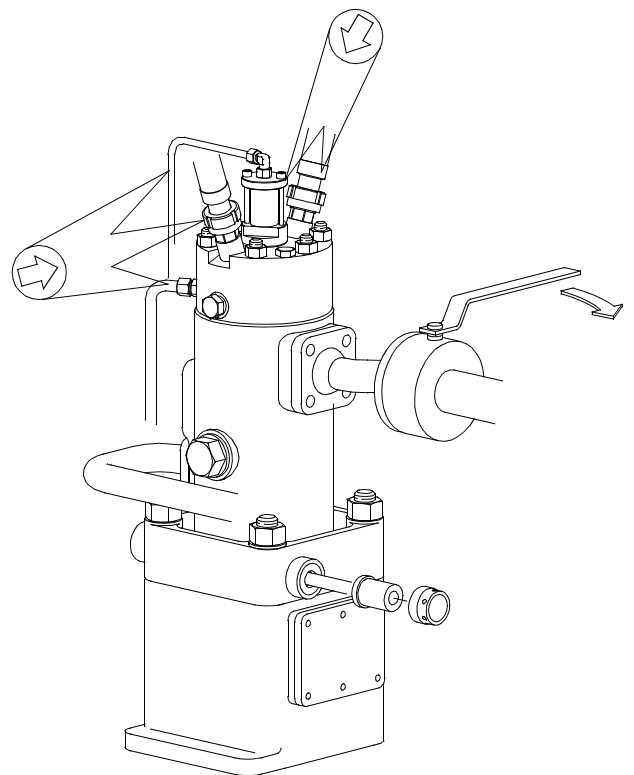
For fuel pump settings, see Procedure 909-1 and Volume I, OPERATION, Chapter 706.

7.



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9.



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SAFETY PRECAUTIONS

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☐	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit
D09-29	Barrel assembly	31	kg



The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P90951 refers to chapter 909.

Plate	Item No.	Description

1. Remove the top cover with the barrel/plunger assembly from the fuel pump housing, and dismantle the barrel/plunger assembly from the top cover. See *Procedure 909-5.2*.
2. Remove and discard the O-rings.

Clean the outside of the barrel.

Note!

It is recommended not to remove the plunger from the barrel, as this may cause scratches to the plunger.

For evaluation of pump barrel/plunger, see Volume I, OPERATION, Chapter 706.

3. The sealing ring on the pump barrel consists of an O-ring and a back-up ring..

Prepare the back-up ring by twisting the ring in the hand, in such a way that the diameter of the ring becomes smaller than the diameter of the barrel.

'Screw' the back-up ring on to the barrel, into the groove.

Finally, position the O-ring in the ring groove.

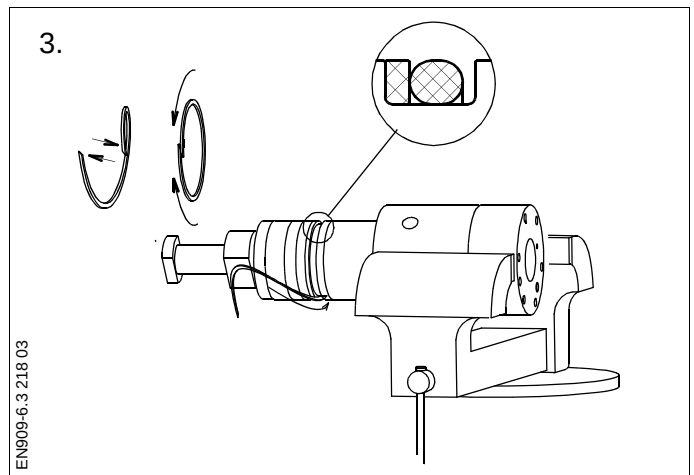
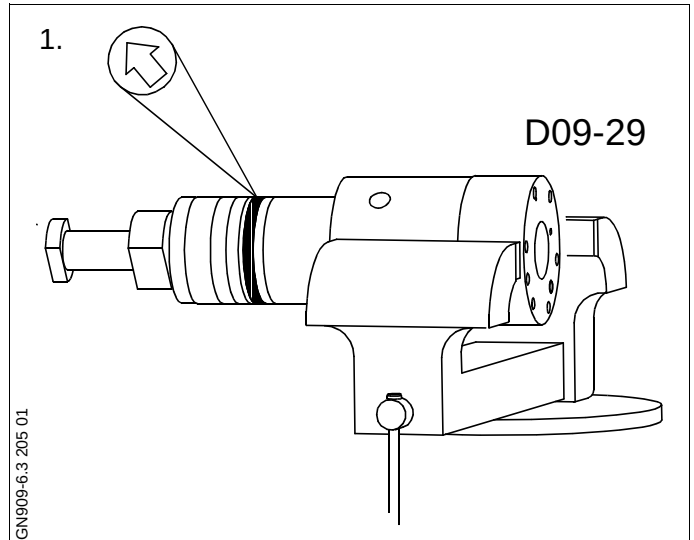
The O-ring must be above the back-up ring.

Note!

Take care that the plunger does not slide out of the barrel.

4. If the barrel assembly is not to be mounted immediately, cover all openings with plastic to prevent dirt from entering the barrel assembly during storage.

For mounting the fuel pump barrel assembly, see Procedure 909-5.4 (fuel pump top cover).



SAFETY PRECAUTIONS

☒	Stopped engine
☒	Block the starting mechanism
☒	Shut off starting air supply
☐	Engage turning gear
☐	Shut off cooling water
☒	Shut off fuel oil
☐	Shut off lubricating oil
☐	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit



The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P90951 refers to chapter 909.

Plate	Item No.	Description

1. Set up the suction valve in a bench vice with "soft" jaws and press down the spring by means of the cone to release the valve spindle from the two-part conical ring.

Clean the parts thoroughly in clean diesel oil and blow dry with compressed air.

2. Inspect the seat on the valve spindle and the seat on the valve housing for damage.

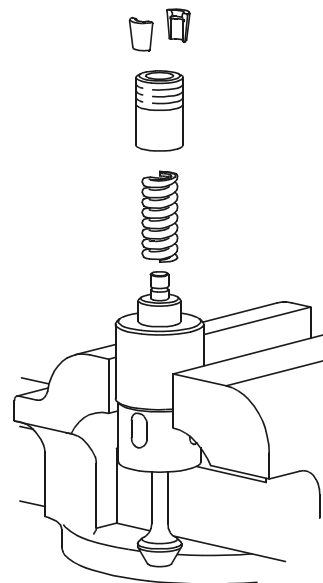
If the seats are damaged, a new or reconditioned suction valve must be fitted. (Send the damaged valve to an authorized MAN B&W repair shop for reconditioning).

3. After cleaning and inspecting all the parts, lubricate these with molybdenum disulphide (MoS_2) and assemble the suction valve.

Check the seats for tightness by filling the inlet hole with diesel oil and waiting 5 minutes. No oil may pass through the seats.

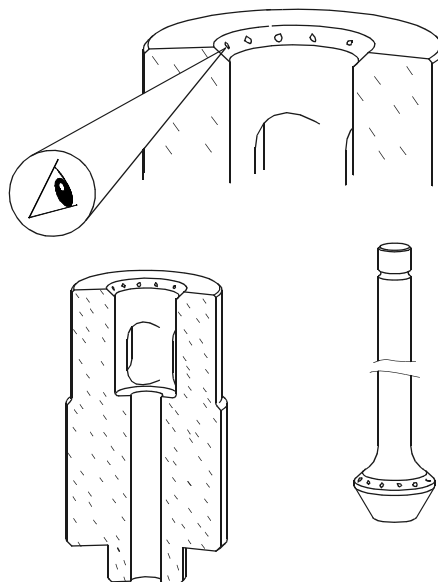
4. If the suction valve is not to be mounted on the engine immediately after the overhaul, cover all openings of the valve with plastic to prevent dirt from entering the valve during storage.

1.



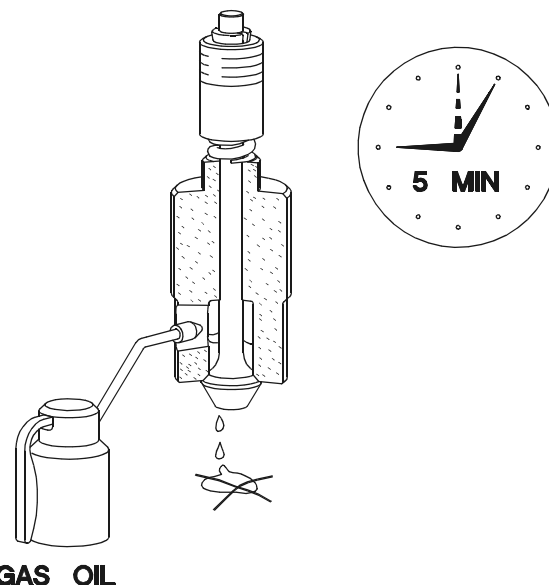
GN909-7.3 205 05

2.



GN909-7.3 205 06

3.



GN909-7.3 205 07

SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

☒	Stopped engine
☒	Shut off starting air supply – <i>At starting air receiver</i>
☒	Block the main starting valve
☒	Shut off starting air distributor/distributing system supply
☒	Shut off safety air supply – <i>Not ME engines</i>
☒	Shut off control air supply
☐	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
☐	Engage turning gear
☐	Shut off cooling water
☒	Shut off fuel oil
☐	Stop lubricating oil supply
☐	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit
D09-35	Puncture valve, tightening torque	650	Nm

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

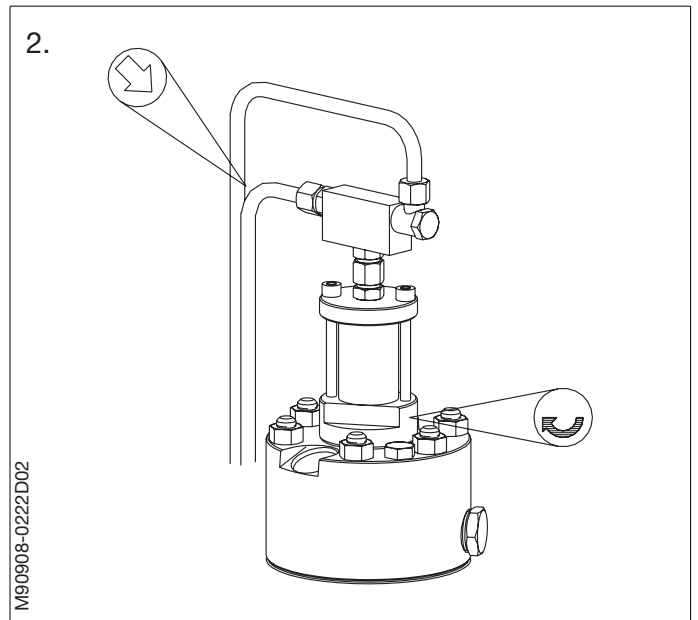
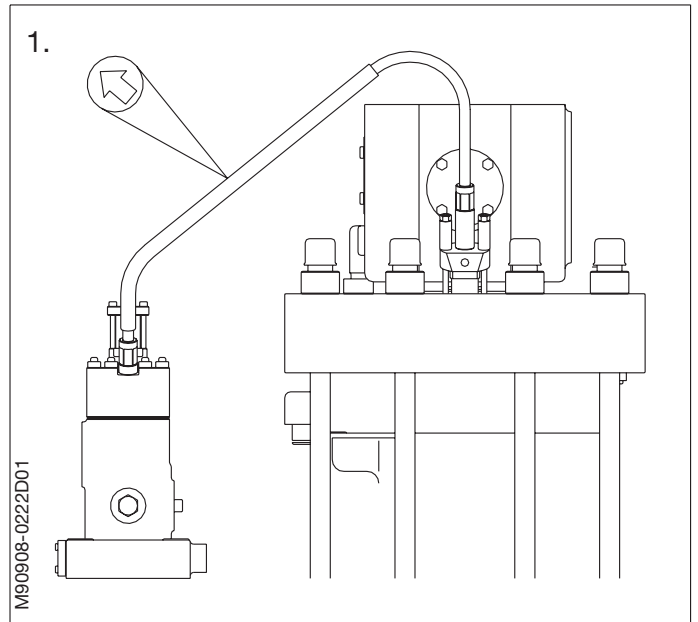
Plate	Item No.	Description
P90951 P90980	128	Crowfoot wrench head Mounting Tools - Air Piston Seals

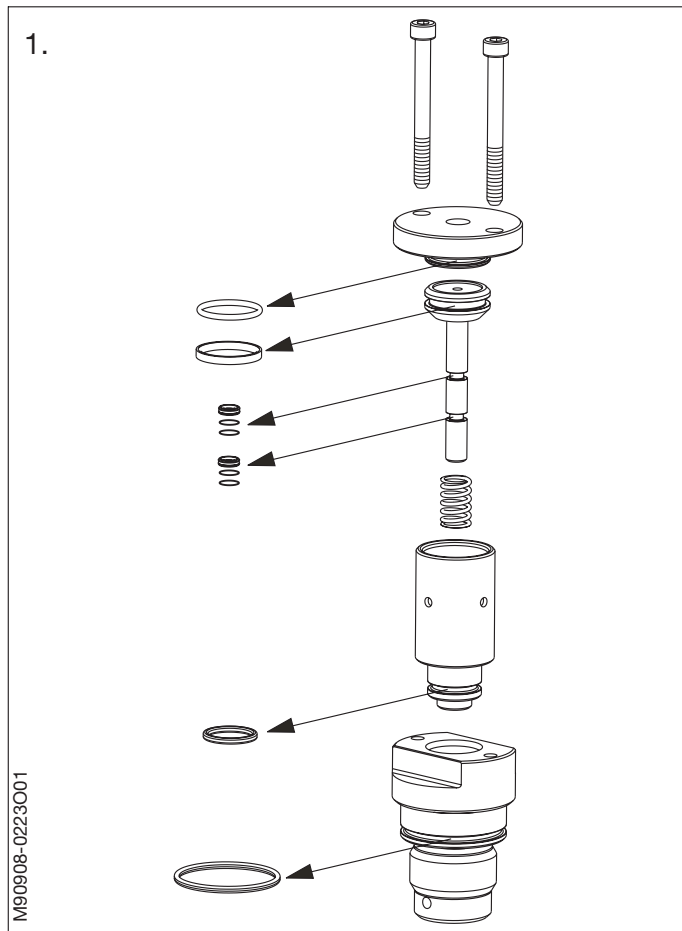
1.

Note!
Before dismantling any part of the fuel pump, make sure that the pump has been relieved of pressure and that all oil has been drained off.

Remove the fuel oil high pressure pipes.
See *Procedure 909-14.2*.
2. Disconnect the air pipes from the puncture valve.

Unscrew the puncture valve and remove it from the fuel pump top cover.





1. Set up the puncture valve in a bench vice with "soft" jaws.

Loosen and remove the two screws and dismount the cover.

Pull the upper part of the housing clear of the lower part of the housing.

Remove the spindle and the spring from the upper part of the housing.

Remove and discard all O-rings and sealing rings.

Clean all the parts in clean diesel oil.

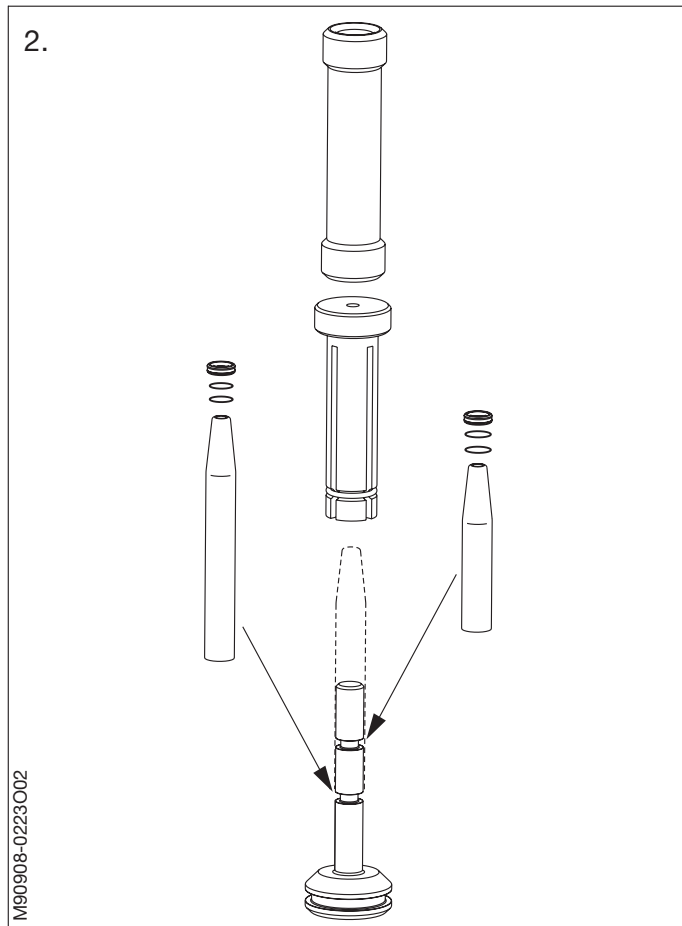
2. Mount the two O-rings in each ring groove on the rod of the air piston, and position the long cone on the rod.

Heat the sealing ring in 100°C hot water for minimum 5 minutes.

Place the sealing ring on the cone and, by means of the pusher, push the sealing ring on to the O-rings in the lowermost groove. Remove the pusher and the long cone.

Finally, compress the sealing ring by means of the compression tool.

3. Repeat step 2 on the uppermost ring groove on the rod of the air piston with the short cone.



4. Mount the O-ring in the ring groove on top of the piston. Position the cone on the piston head.

Heat the sealing ring in 100°C hot water for minimum 5 minutes.

Place the sealing ring on the cone and, using the pusher, push the sealing ring onto the O-ring in the groove.

Remove the pusher and the cone.

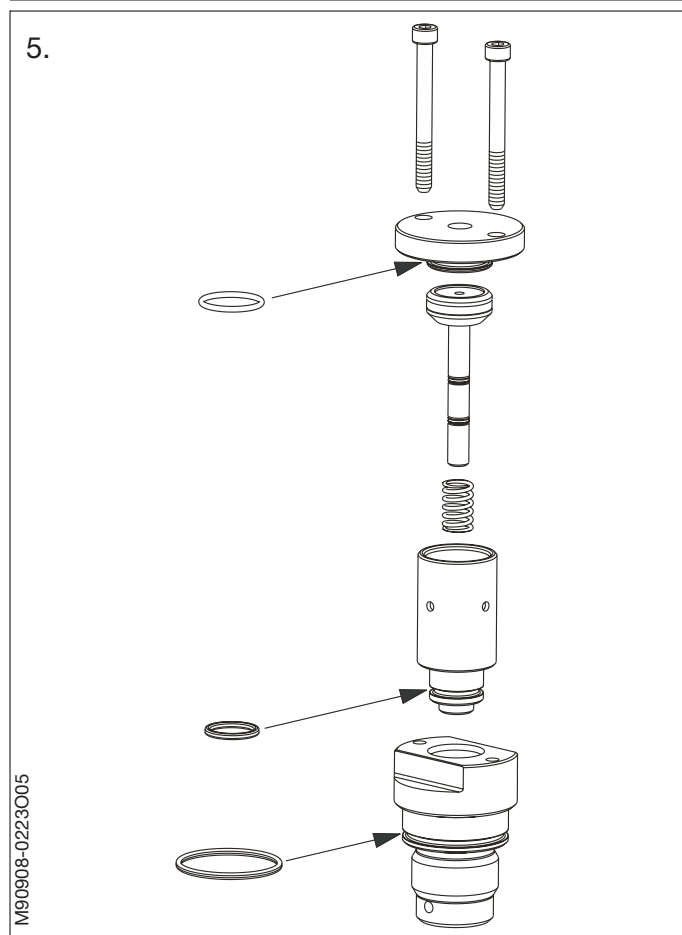
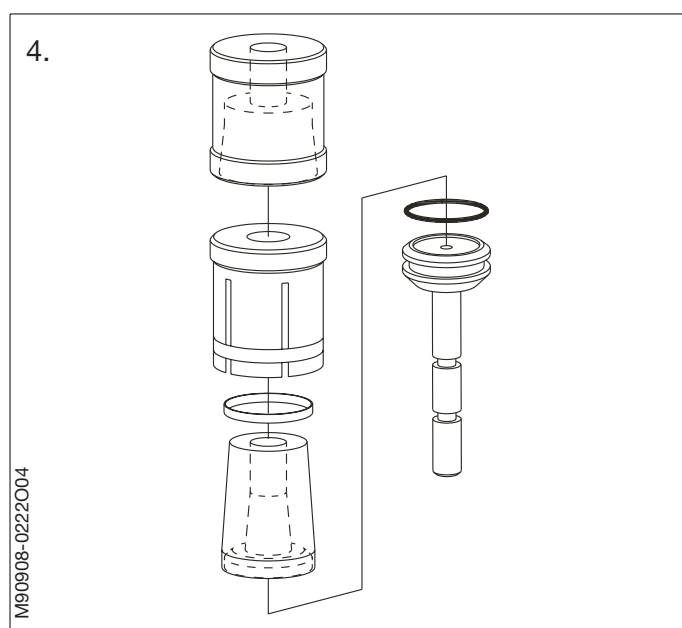
Finally, compress the sealing ring by means of the compression tool.

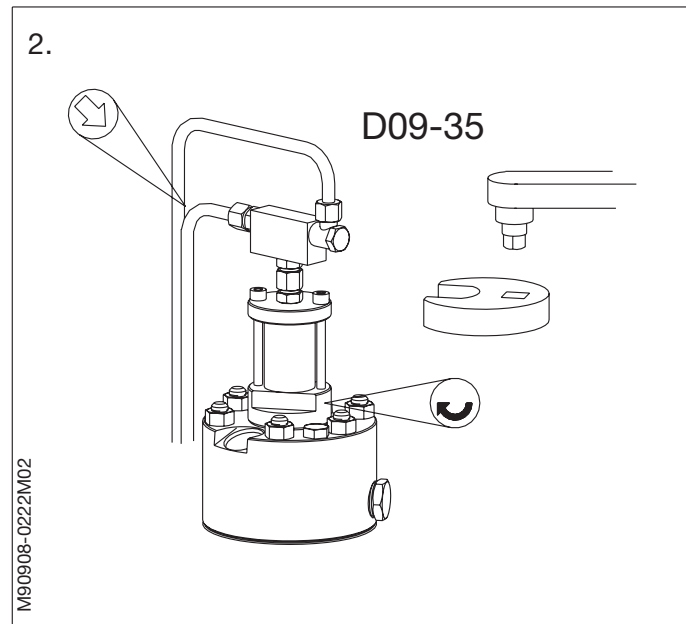
5. Before assembling the puncture valve, lubricate all parts with molybdenum disulphide, (MoS2).

Mount new O-rings on the upper and the lower part of the housing and the cover.

Mount the spring and air piston in the upper part of the housing and assemble the two housing parts and the cover by means of the two screws.

6. If the puncture valve is not to be mounted on the engine immediately after the overhaul, cover all openings of the valve with plastic to prevent dirt from entering the valve during storage.





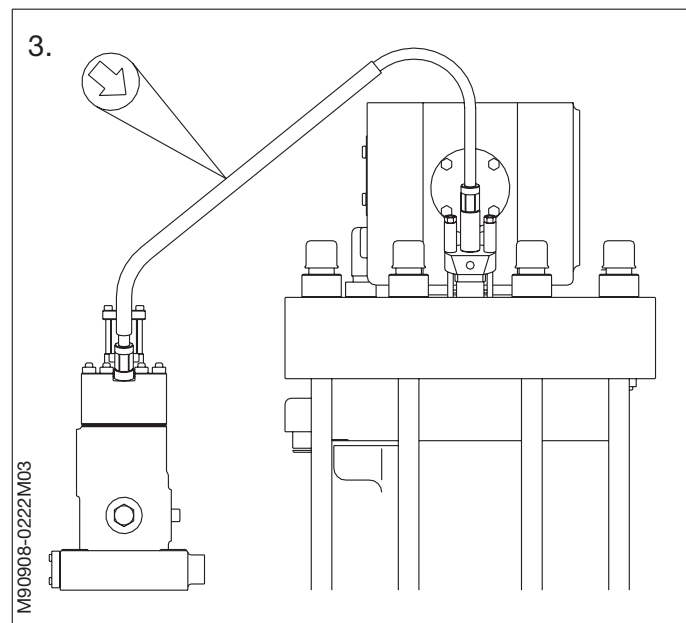
1. Before mounting the puncture valve, check that the bore in the top cover is clean and check the O-ring on the puncture valve.

Lubricate the thread on the puncture valve with molybdenum disulphide grease (MoS₂).

2. Screw in the puncture valve and tighten with a crowfoot wrench and a torque spanner to the specified torque, see *Data D09-35*.

Mount the air pipes for the puncture valve.

3. Mount the fuel oil high pressure pipes, see *Procedure 909-14.4*



SAFETY PRECAUTIONS

☒	Stopped engine
☒	Block the starting mechanism
☒	Shut off starting air supply
☒	Engage turning gear
☐	Shut off cooling water
☒	Shut off fuel oil
☐	Shut off lubricating oil
☐	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit
D09-39	Shock absorber	20	kg



The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P90951 refers to chapter 909.

Plate	Item No.	Description

1. To ensure satisfactory operation, the functioning of the fuel oil system shock absorber should be checked at regular intervals. Normally, inspection is necessary only at the intervals stated in the maintenance programme. The checks should be carried out while the engine is operating.
2. Remove the plug from the end cover of the shock absorber. Check that air is 'pulsating' through the threaded hole.

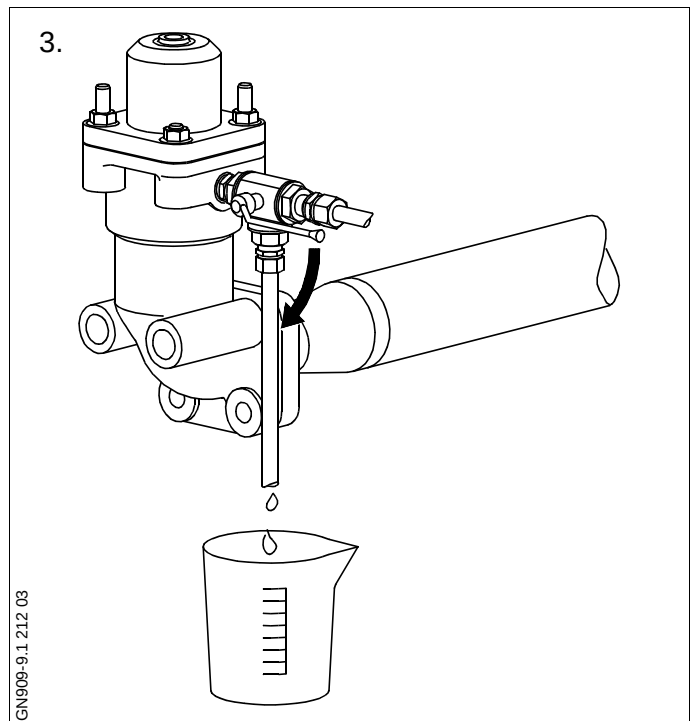
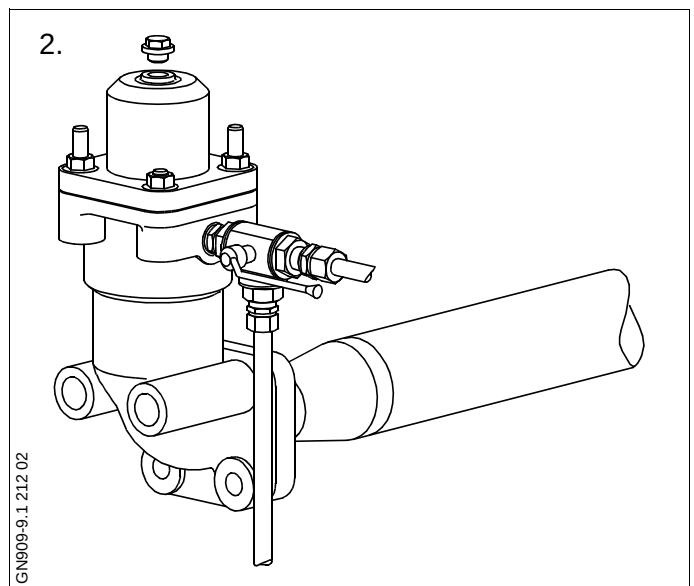
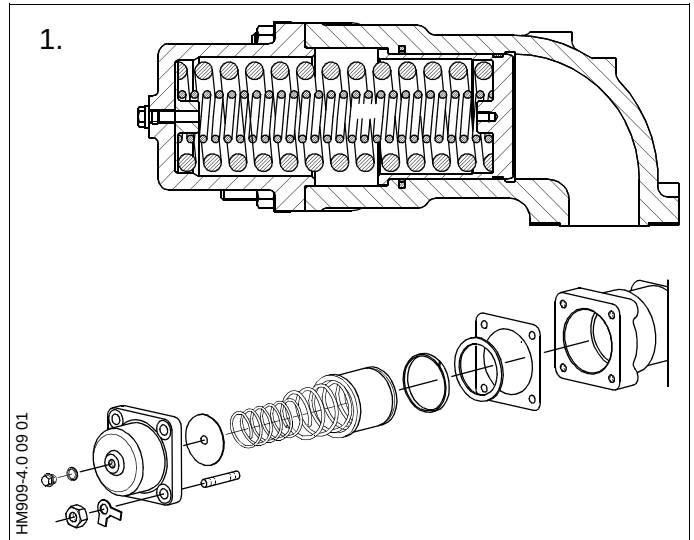
Short bursts of air should be felt against the hand, corresponding to the strokes of the fuel pump.

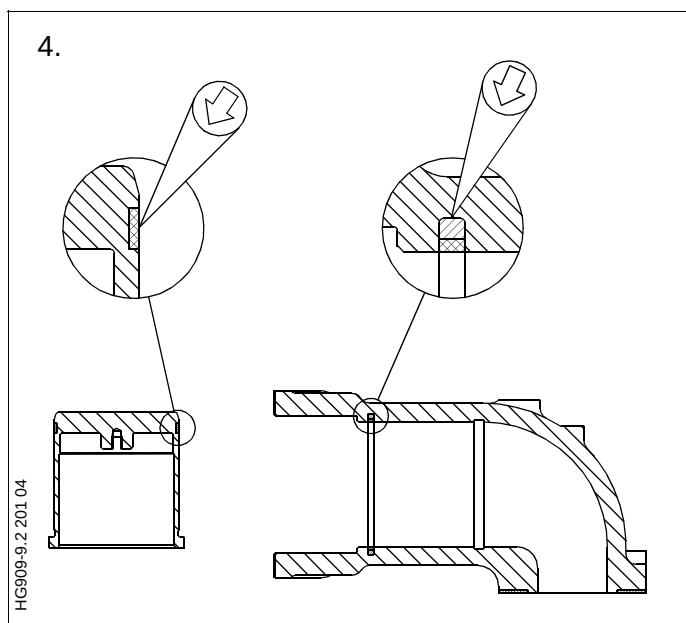
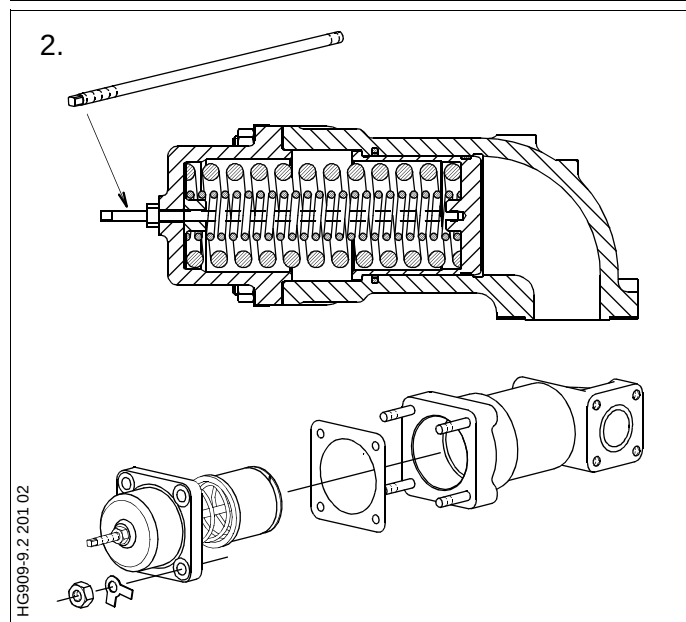
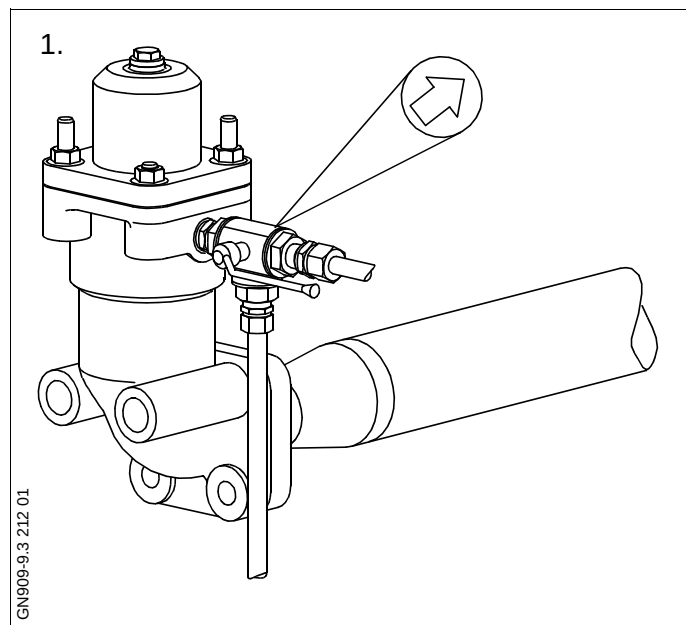
The air pulses occur when the shock absorber piston is working and the air below the piston is being compressed and forced out through the threaded hole.

3. To check the 'tightness' of the piston, measure the amount of leakage oil flowing from the drain pipe below the 3-way valve. Note down the amounts measured and compare with earlier measurements.

Increasing amounts indicate a deteriorating piston sealing. Excessive amounts indicate that the shock absorber requires overhauling.

See Procedure 909-9.3.



**Note !**

Before dismantling any part of the fuel oil system, make sure that the system has been relieved of pressure and that all oil has been drained off.

1. Remove the 3-way valve and the drain pipes. Loosen the screws holding the shock absorber and remove it from the fuel oil system.
2. Remove the plug from the end cover of the shock absorber. Fit a threaded rod between the piston and the end cover to counteract the force of the springs. Remove the locking plates and loosen the screws of the end cover. Remove the end cover and piston. Discard the gasket.
3. Loosen the nut on the threaded rod to relieve the springs and remove the threaded rod. Remove and discard the sealing ring in the shock absorber housing and the wear ring on the piston.
4. Clean all parts and polish the sliding surfaces of the piston and the shock absorber housing. Mount a new sealing ring in the shock absorber housing and a new wear ring on the piston.
5. Mount the springs and the spring guide between the end cover and the piston, using the threaded rod.

Lubricate all sliding surfaces and assemble the shock absorber. Take care not to damage the sealing ring or wear ring when mounting the piston. Check that the piston slides easily in the shock absorber housing.

6. Mount the end cover. Remove the threaded rod and insert the plug.

7. Fit a new gasket and mount the shock absorber. Mount the 3-way valve and the drain pipes on the fuel oil system.

If an excessive amount of drain oil is still observed after the overhaul, the shock absorber must be replaced and the damaged one sent to an authorized MAN B&W repair shop for reconditioning.

SAFETY PRECAUTIONS

☒	Stopped engine
☒	Block the starting mechanism
☒	Shut off starting air supply
☐	Engage turning gear
☐	Shut off cooling water
☒	Shut off fuel oil
☐	Shut off lubricating oil
☐	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit
D09-40	Fuel valve opening pressure	350 - 380	bar
D09-41	Fuel valve tightening torque	25	Nm
D09-42	Inlet seat, max. diameter	18	mm
D09-43	Fuel valve	8	kg

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P90951 refers to chapter 909.

Plate	Item No.	Description
P90151	71	Dismantling tool for fuel valve
P90951	94	Milling tool for fuel oil pipe seats
P90951	165	Grinding mandrel for valve head
P90951	177	Grinding mandrel for thrust spindle
P90951	189	Grinding mandrel for holder - outside
P90951	190	Grinding mandrel for holder - inside
P90951	248	Hook wrench
P909		Assembling tool for fuel valve
P90961		Fuel valve tester, complete

1. Fuel Valve

The fuel valves must be given the utmost attention and care, as the greater part of irregularities that may occur during the running of the engine can be attributed to defects in these valves.

If the engine gives normal performance in accordance with diagrams and exhaust temperatures, it is only necessary to inspect the fuel valves after the service period stated in the Checking and Maintenance Schedules. See *Chapter 900-1*.

In order to obtain reliable results during testing of the fuel valves, all fuel valves that are dismantled from the engine must be disassembled, cleaned, inspected and re-assembled before testing. See *Procedures 909-11.3 and 909-12.3*.

Note!

In the event that the slide-type fuel valve is pressure tested without being cleaned between the fuel nozzle and the cut-off slide, the opening pressure value measured might be considerably lower

All fuel valves must be function-tested before being mounted in the cylinder cover.

Pressure testing pump

If an air supply of 10 bar is not available, working air of 7 bar can be used. *Regarding the operation of the pressure testing pump, see the supplier's instructions.*

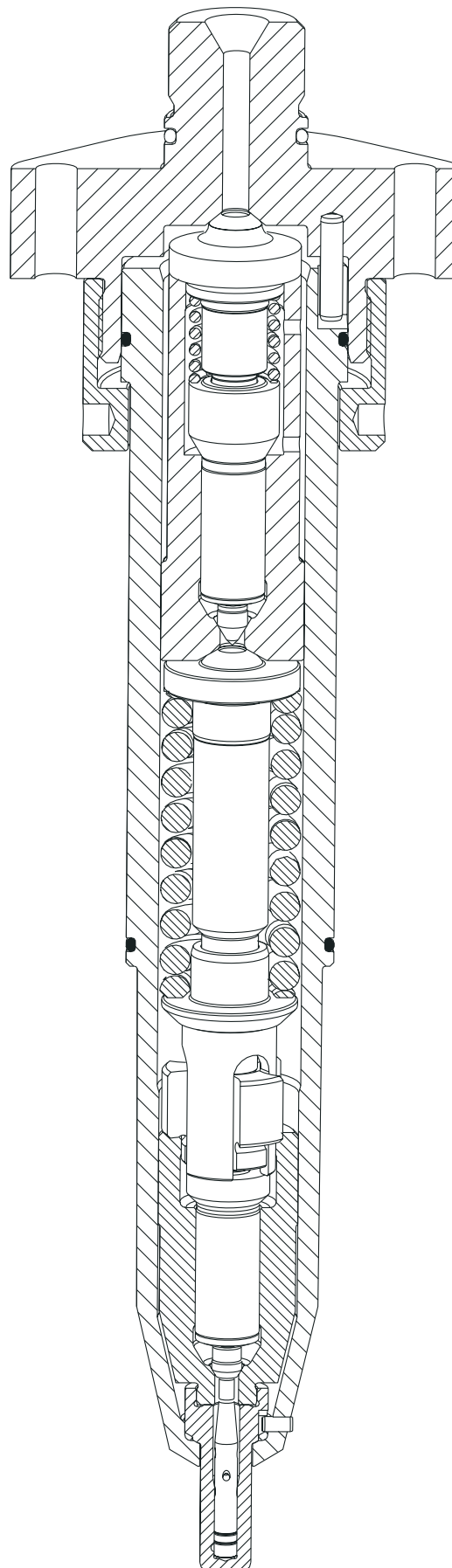
Note!

The high-pressure pump should be periodically checked in accordance with the

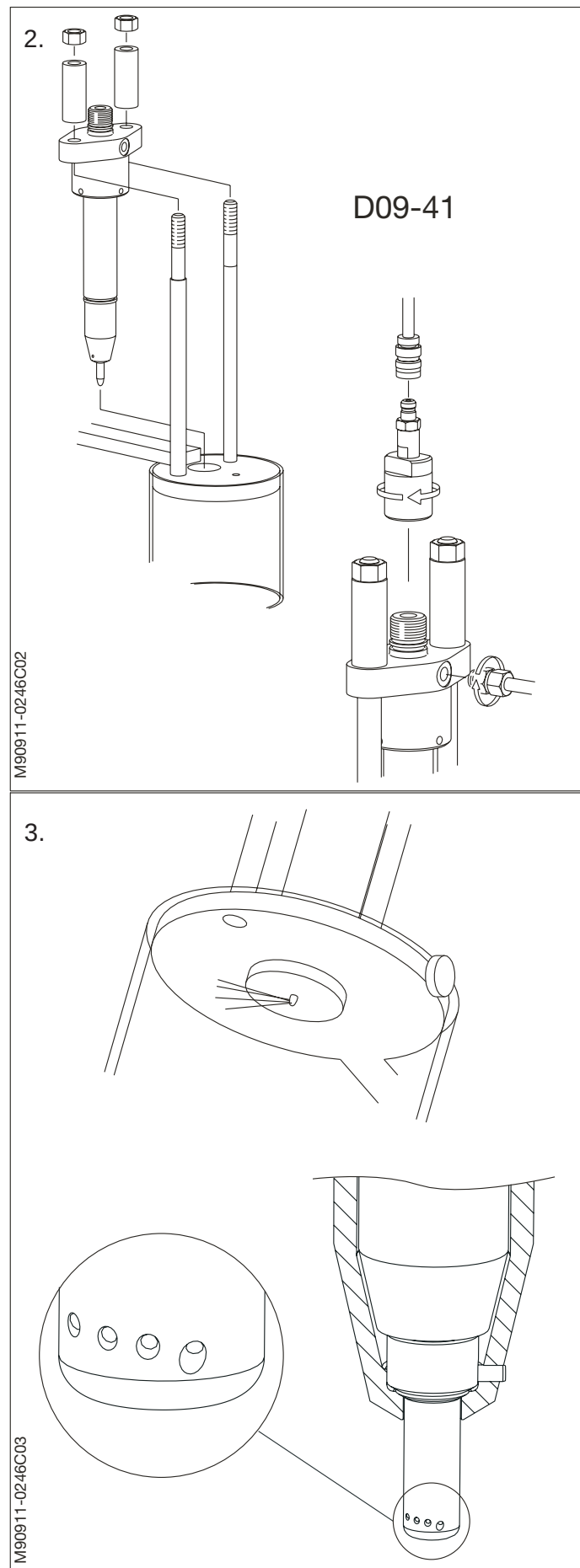
Prescribed oil: Hydraulic oil (rust-preventing) with a viscosity of between 7 and 10 cSt at 50°C.

Note: use clean oil only.

1.



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2. Setting-up the fuel valve

Place the fuel valve in the test rig and secure it with the tubular spacers and nuts.

Tighten the nuts to the torque stated in Data. Mount the oil pipe between the pressure testing pump and the fuel valve.

3. Pressure testing procedure

The following functions of the fuel valve must be checked:

- Flushing and jet control
- Opening pressure
- Sealing test and sliding function
- Pressure test, O-ring sealing

4. Flushing and jet control

Remove air in the system and check the fuel jet in the following way.

Slowly increase the oil pressure until straight jets of oil are ejected from the nozzle holes (no atomization).

Acceptance criteria:

There is to be a continuous jet of oil through at least one of the nozzle holes.

Owing to the geometry of the internal part of the nozzle - and because of the height to which the spindle is lifted during pressure testing is lower than the height it is lifted during normal engine operation - the fuel oil will not necessarily flow from all of the nozzle holes.

Cause of fault:

If the jets do not fulfil the above point, the cause may be:

Dirt in the nozzle holes

The nozzle is not mounted correctly

5. Opening pressure

To check the opening pressure, increase the oil pressure until oil is admitted through the nozzle holes.

Acceptance criteria:

Check the opening pressure on the pressure gauge and compare with Data D09-40 on the datasheet.

Note!

Do NOT attempt to carry out an atomization test on the slide type fuel valves, as this may damage the cut-off slide and nozzle.

Cause of fault:

If the opening pressure is higher than specified in D09-40, the cause may be that a wrong type of spring is used - replace the spring on the thrust spindle, if necessary, replace the complete thrust spindle.

If the opening pressure is lower than specified in D09-40, the cause may be that the spring has sagged - replace the spring, or add a special thin disc.

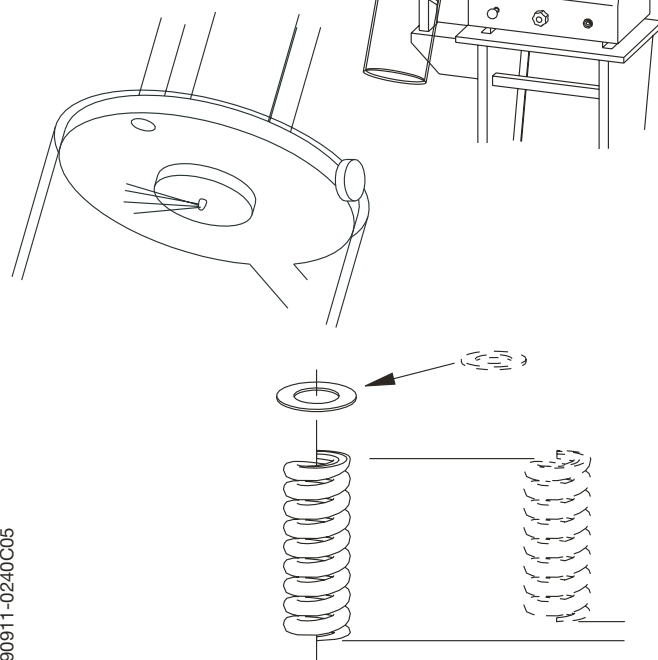
Note!

Special thin discs are available as spares.

If a spring or a disc has been changed, the pressure testing procedure of the fuel valve must be repeated.

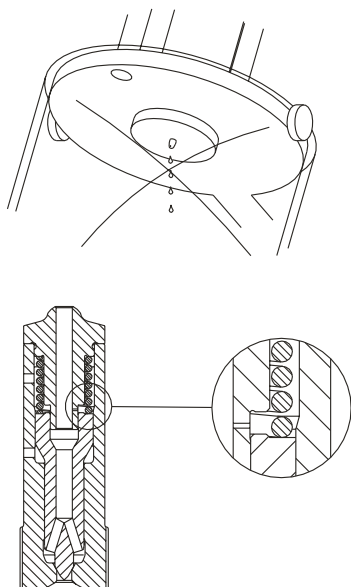
5.

D09-40



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6.



6. Sealing test and sliding function

To check the needle valve seat for tightness and the slide for correct closing.

Slowly increase the oil pressure to about 50 bar below the opening pressure. Maintain the built-up pressure by closing for the oil supply.

Acceptance criteria:

Oil must not flow from the nozzle holes.

The pressure drops relatively slowly to about 15 bar, after which it drops quickly to 0 (the slide is pressed against the conical seat and opens for circulation oil).

Note!

Oil flows out of the leak oil outlet when the fuel valve is full of oil.

Cause of fault:

If oil flows out of the nozzle holes, the cause is either:

- Defective spindle guide at needle seat, or a sticking spindle. Examine and/or replace the spindle guide.
See Procedure 909-12.3.
- Too quick pressure drop:

- the clearances of the movable parts, both of the spindle guide and of the non-return valve, are too large, or the seats between the thrust piece/spindle in the spindle guide or thrust piece/valve slide in the non-return valve are damaged.

Examine and/or replace both the spindle guide and non-return valve. *See Procedure 909-12.3. See Procedure 909-13.3.*

If a quick pressure drop from 15 to 0 bar cannot be registered:

- The valve slide is sticking; or
- The vent hole in the thrust piece is blocked.

If so, disassemble and examine the spindle guide, replace if necessary. *See Procedure 909-12.3.*

7. Pressure test, O-ring sealing

To ensure that the leak oil (circulation oil) remains in the closed system, build up a working pressure of about max. 10 bar until oil flows out of the leak oil outlet.

Close the leak oil outlet with a gasket and plug screw.

Increase the working pressure to about 100 bar.

Acceptance criteria:

The built-up pressure of about 100 bar should be maintained.

Cause of fault:

If oil leaks out at the union nut, the O-ring inside the fuel valve head is defective, and must be replaced.

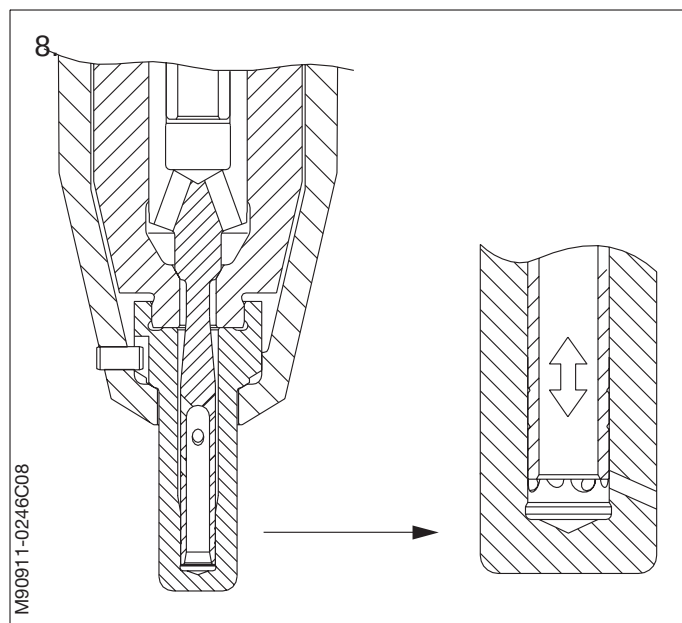
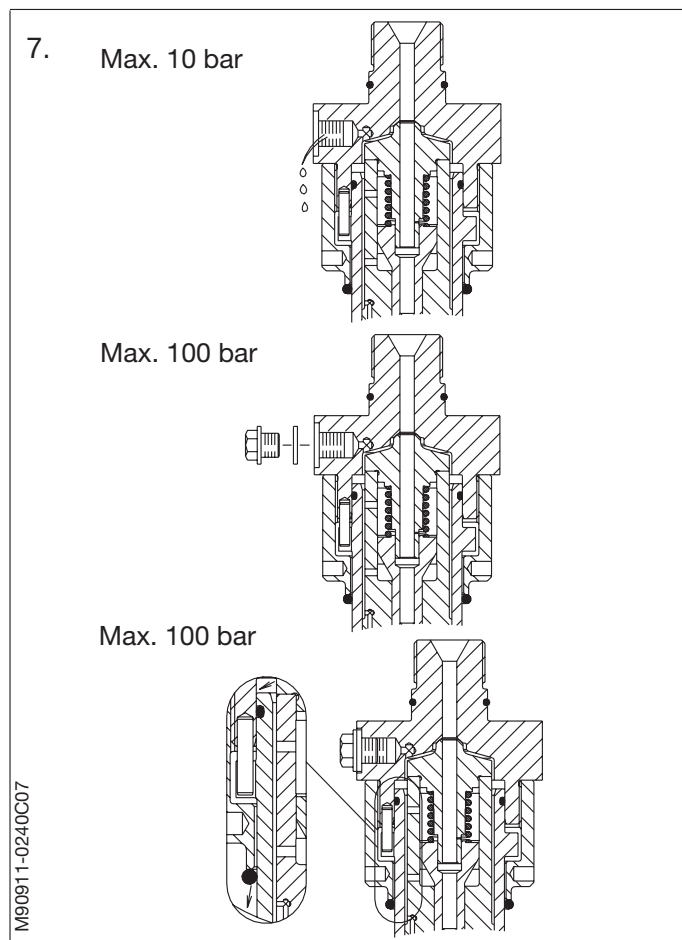
8. Atomization test

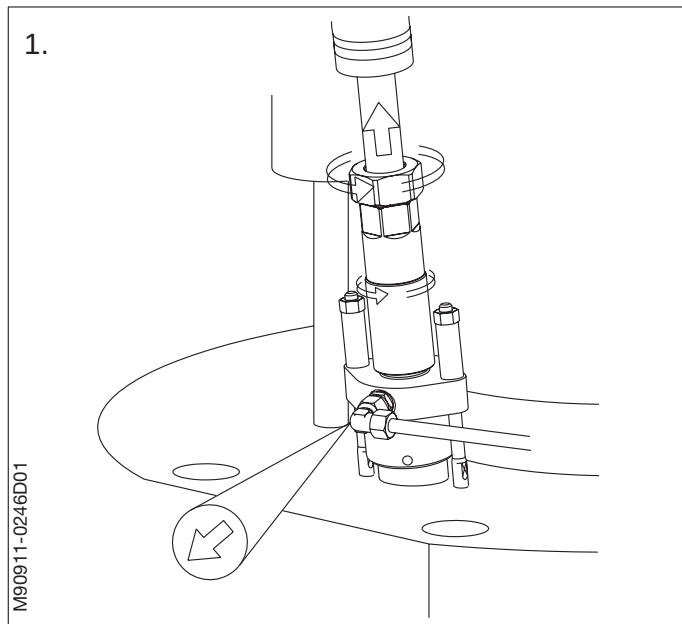
Note!

Do **NOT** attempt to carry out an atomization test on slide type fuel valves, as this may damage the cut-off slide and nozzle.

The atomization test may damage the valve because it makes the needle oscillate, with a small lift at a very high frequency. The high pressure drop across the cut-off edge and the high contact pressure between slide and fuel nozzle, in combination with the poor lubricity of the test oil, increase the risk of seizures between cut-off slide and nozzle.

All of these conditions involve the risk of seizure between the cut-off slide and the nozzle.





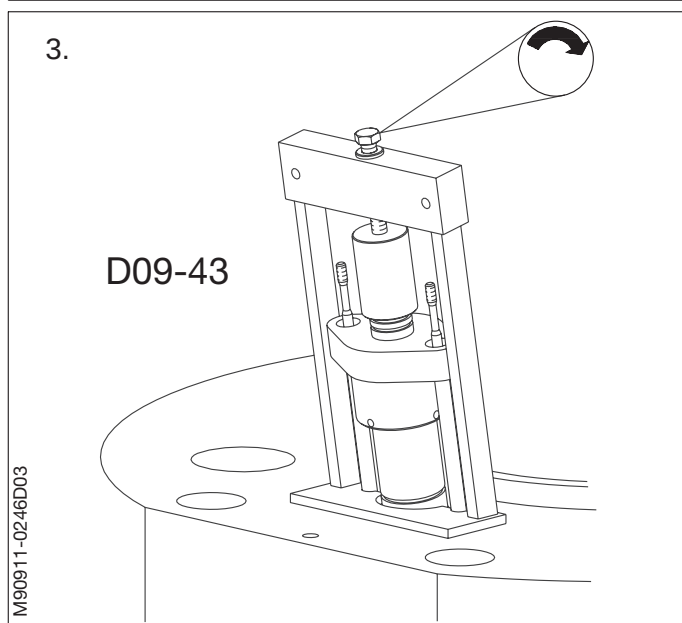
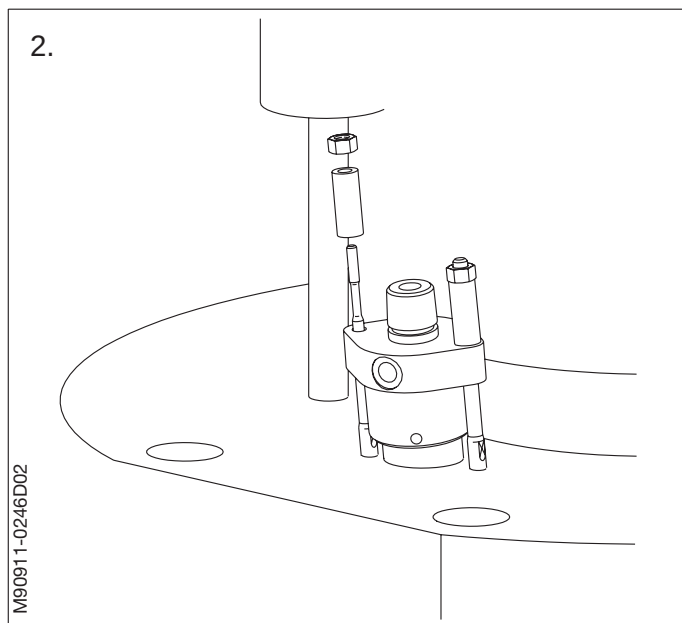
1. Close the fuel oil inlet and outlet valves.

Dismantle the high-pressure pipe. See *Procedure 909-14.2*.

Disconnect the return oil pipe from the fuel valve.

2. Remove the nuts and the tubular spacers.
3. Take out the valve. If the valve is sticking, use the fuel valve dismantling tool to pull the valve clear of the top cover.

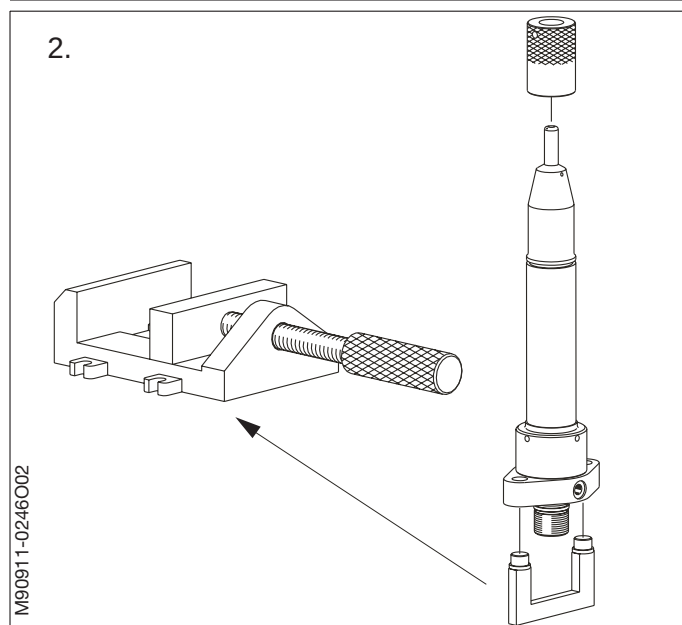
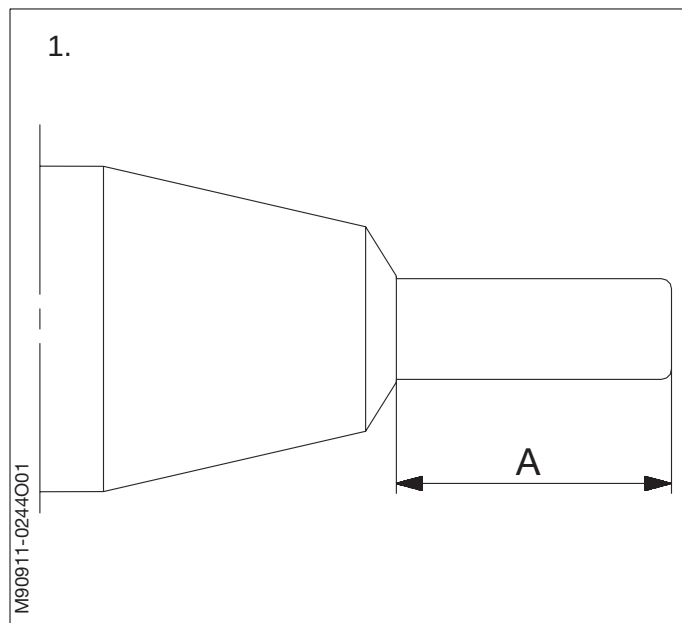
If the valve is not to be overhauled immediately, the valve should be immersed in diesel oil until overhauling.

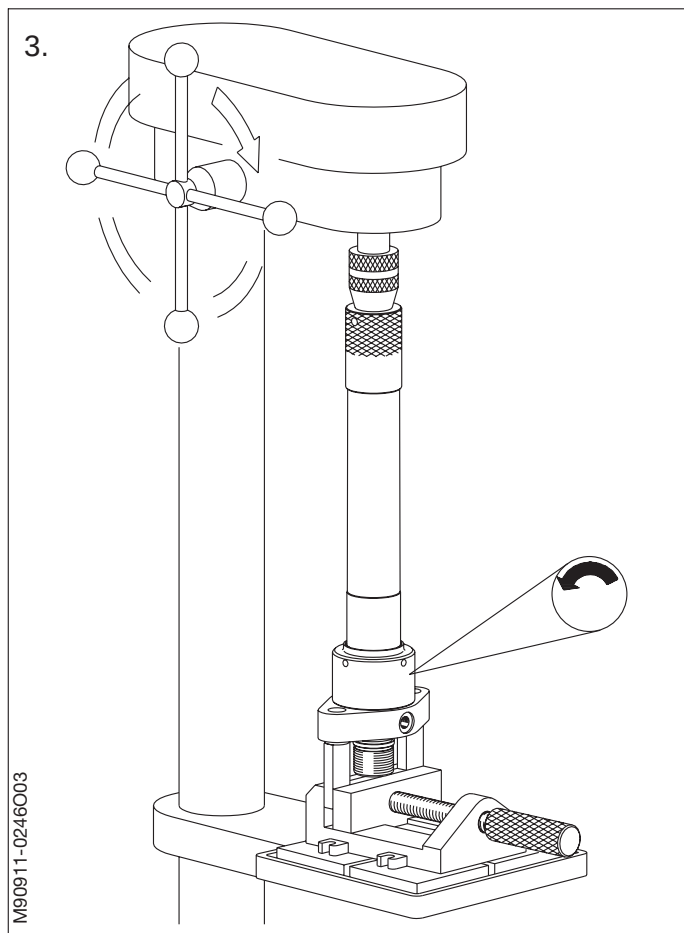


When fuel valves are overhauled, all parts should be handled carefully and be kept clean.

Use only clean, non-fluffy rags for wiping purposes. Make sure to remove all liquid or solid impurities. Whenever fuel valves are overhauled, all sealing rings should be discarded and replaced by new, faultless sealing rings before reassembly.

1. Measure the length **A** of the protruding part of the nozzle, and write down the result for correct re-assembly of the valve.
2. Place the valve holder in a machine vice, mount the fuel valve in the holder and fit the valve with the guide disc from the grinding tool.





3. Compress the fuel valve and the spring inside, by means of a drilling machine, to avoid seizures in the union thread. Hold the fuel valve compressed and unscrew the union nut with a hook spanner.

Remove the valve from the valve holder.

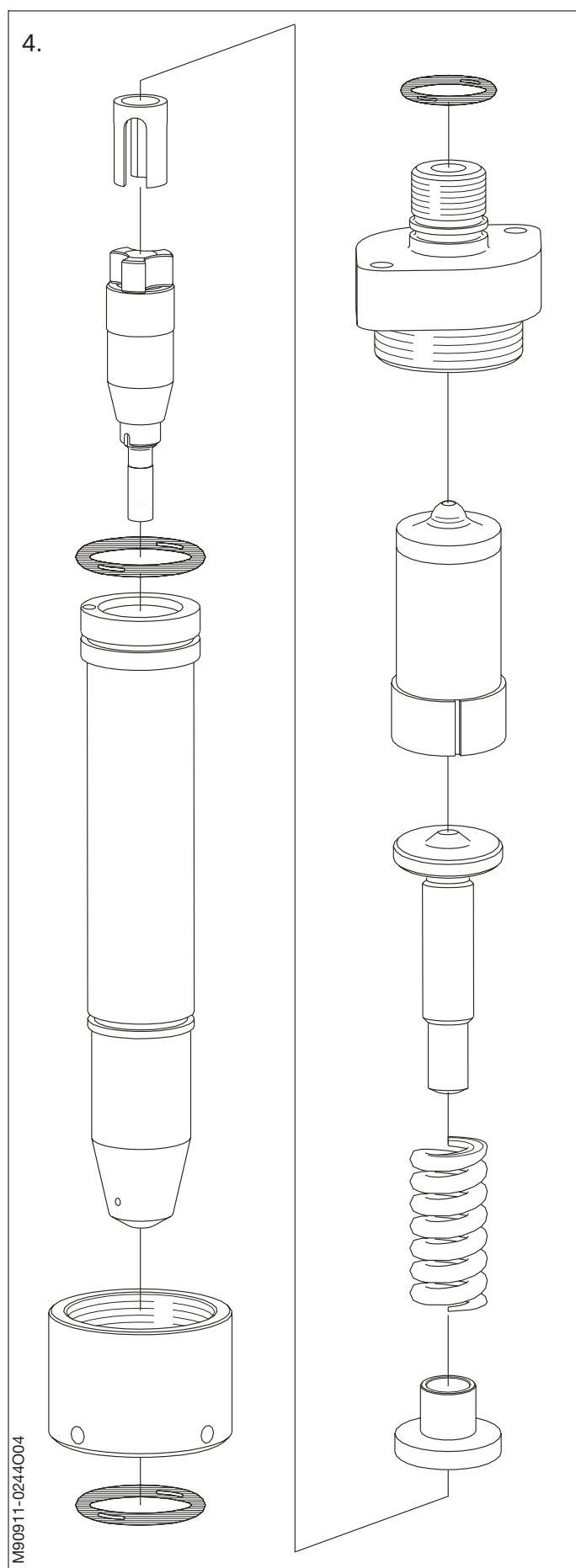
4. Pull the valve head clear of the valve housing.

Remove the:

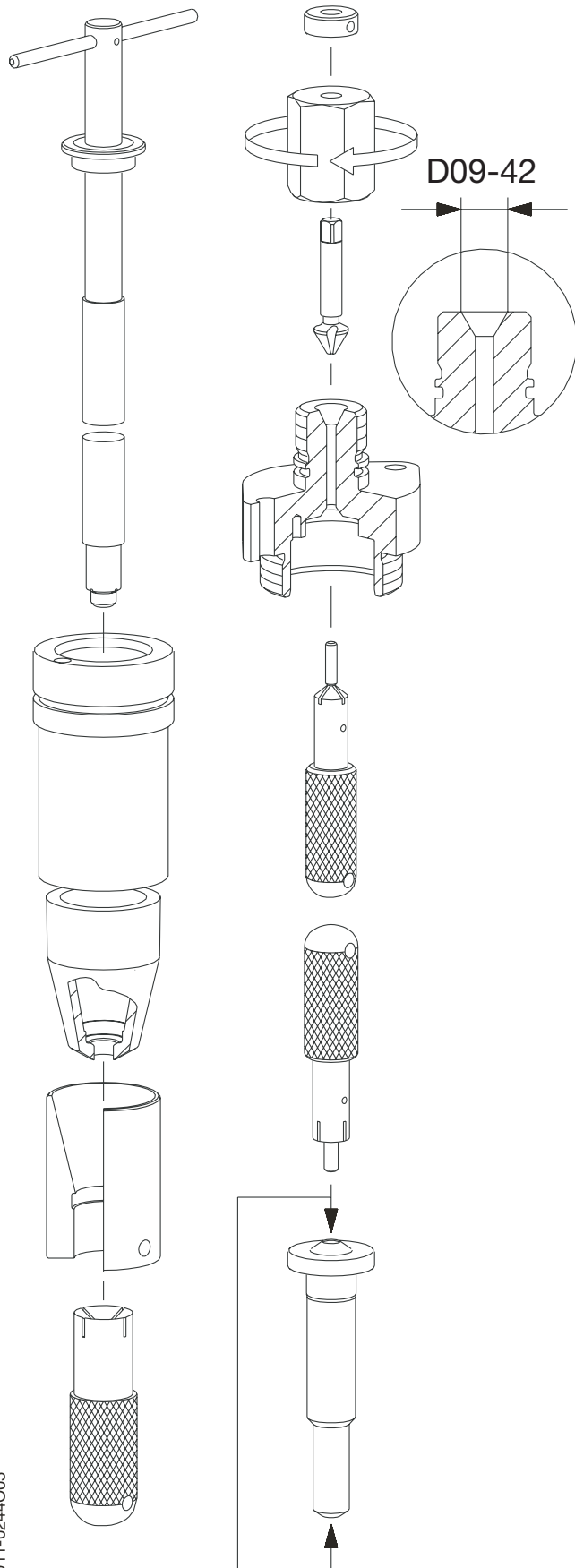
- Non return valve
- Thrust spindle parts
- Thrust foot
- Spindle guide and fuel nozzle

from the valve housing.

Remove and discard all the O-rings.



5.



5. Carefully clean and examine all surfaces of the:

- Fuel valve housing
- Fuel valve head
- Thrust spindle

If necessary, grind the seating surfaces by means of the grinding mandrels supplied and a fine-grain abrasive (such as Carborundum No. 500).

This grinding must only be carried out manually.

After the grinding, wash the parts in gas oil and blow clean by means of compressed air to remove any remains of the grinding compound.

In the event of more serious damage to the seating surface for the high-pressure pipe in the valve head, the milling tool can be used.

Normally, the milling tool is turned by hand, but it may be fitted in the chuck of a column-type drilling machine provided that the number of revolutions is kept at a minimum (not exceeding approx. 100 r/min). An ample supply of cutting emulsion must be used.

Note!

Take care not to exceed the maximum diameter of the seat, see *Data*.

6. The complete spindle guide, including the fuel nozzle, should be sent to an authorised MAN B&W repair shop for overhaul. If this is not possible, the spindle guide may be overhauled on board.
See *Procedure 909-12.3*.

Note!

Do **not** attempt to remove the fuel nozzle from the spindle guide unless the spindle guide has been dismantled. Otherwise the cut-off slide on the spindle may be damaged.

7. The non-return valve should be sent to an authorised MAN B&W repair shop for overhaul. If this is not possible, the non-return valve may be overhauled on board.
See *Procedure 909-13.3*.

8. Mount the complete spindle guide, including the fuel nozzle, in the fuel valve housing.

Carefully slide the spindle guide down into the valve holder, and turn the nozzle until it engages correctly with the guide pin.
Check that distance A corresponds to the measurement taken before the valve was disassembled.

Note!

Make sure that the fuel nozzle engages correctly with the guide pin in the fuel valve housing. This can be ascertained by attempting to turn the nozzle by hand after mounting. It must **not** be possible to turn the nozzle.

6.

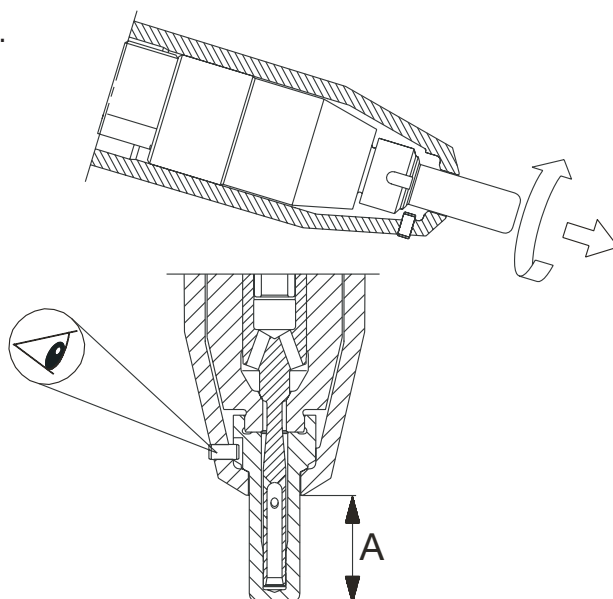
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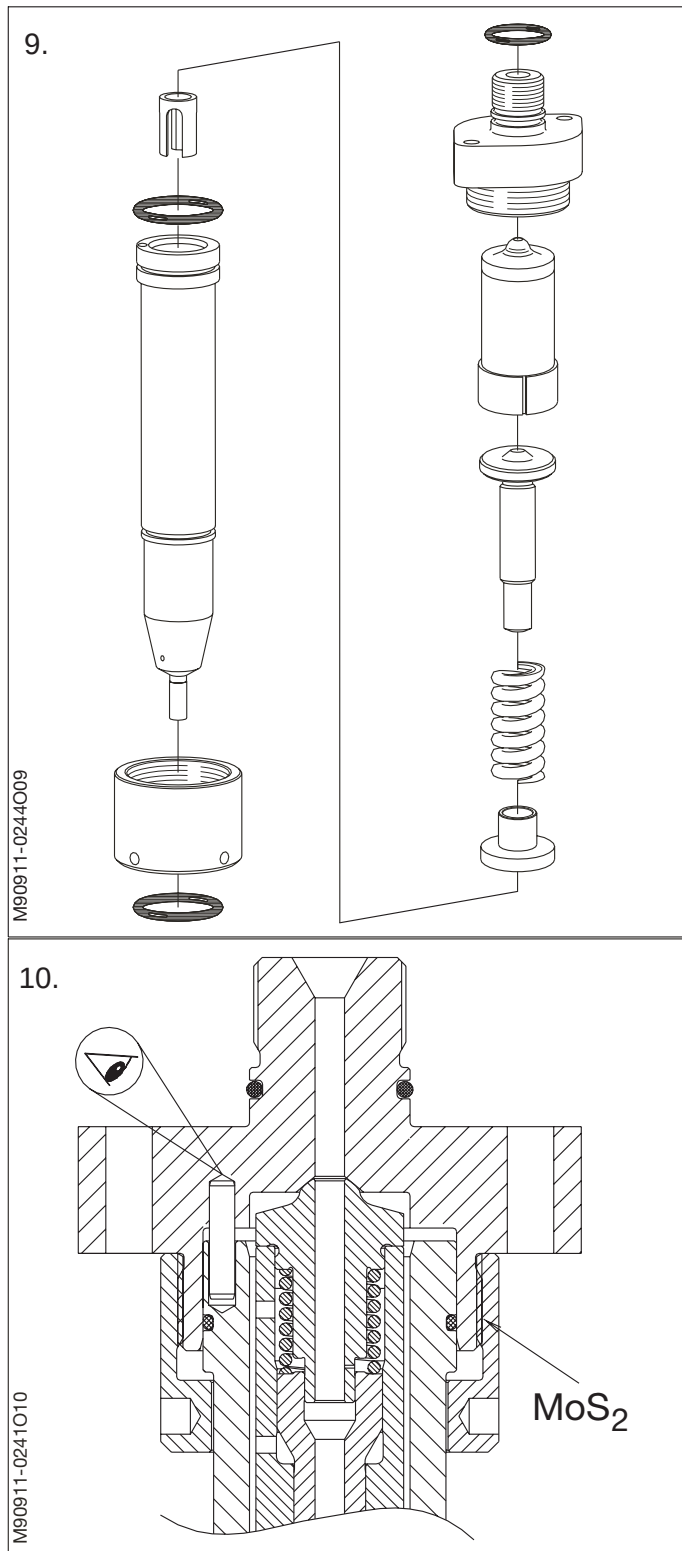
7.

M90911-0240O06

8.

M90911-0244O08





9. Mount:

- the thrust foot
- the parts of the thrust spindle
- the non-return valve

in the fuel valve housing.

Mount a new O-ring in the uppermost groove of the fuel valve housing.

Lubricate the thread of the valve head with molybdenum disulphide (MoS₂).

For the correct use of this lubricant, see Procedure 913-11.

Fit the valve head with new O-rings.

10. Make sure that the guide pin between valve housing and valve head is intact. Press the valve head down into the valve housing and check that the guide pin engages correctly so as to prevent relative turning of the parts.

11. Assemble the valve by means of the union nut.

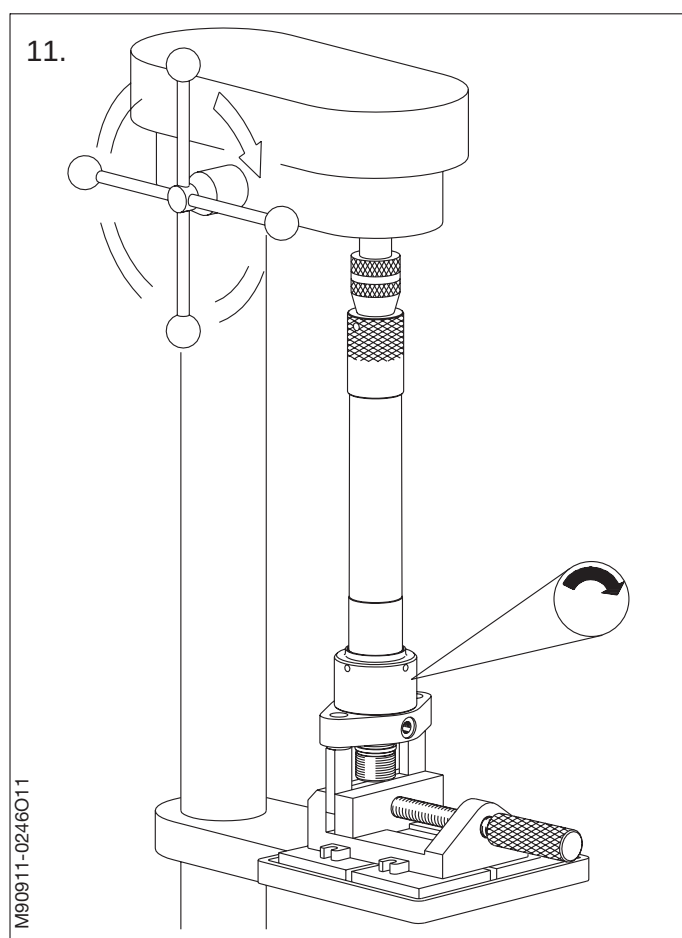
Place the valve in the valve holder in a drilling machine.

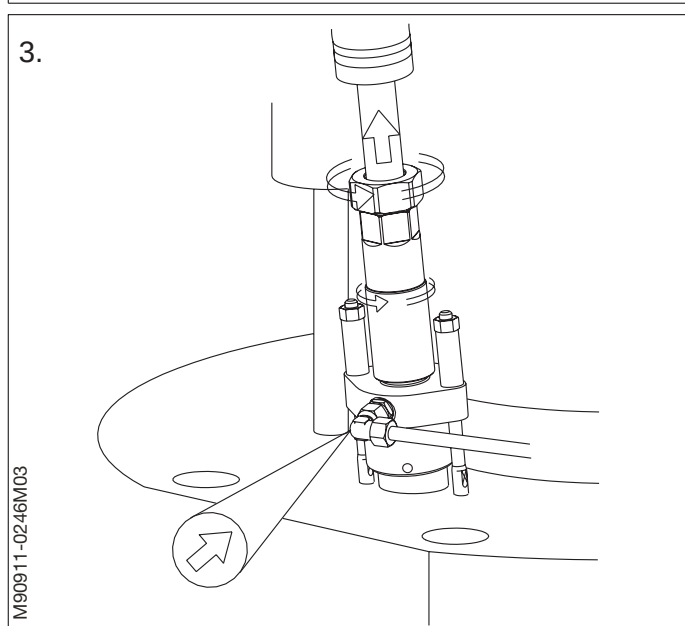
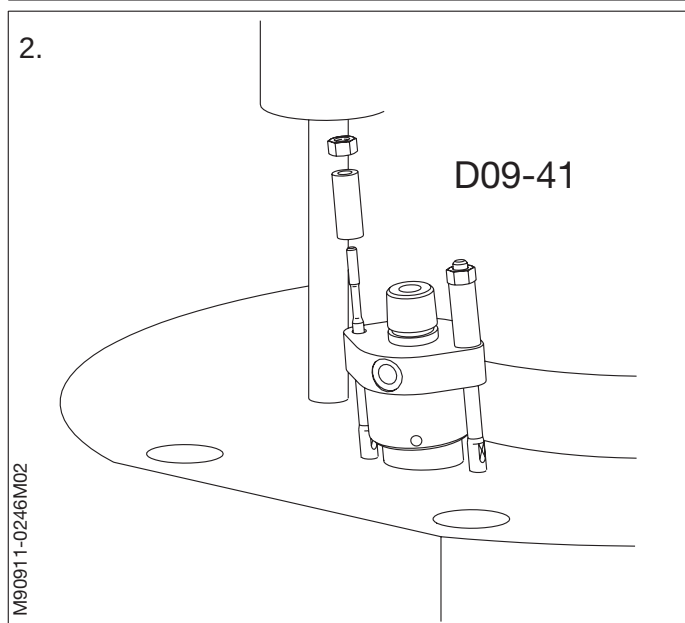
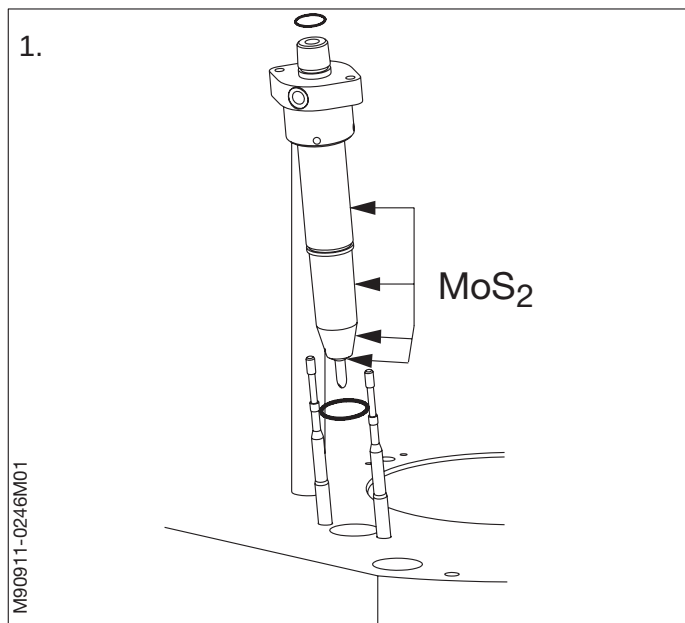
Compress the fuel valve and the spring inside. Keep the valve compressed and tighten the union nut with a hook spanner.

After overhaul, the fuel valve must be tested in the test rig.

See *Procedure 909-11.1*.

12. If the fuel valve is not to be mounted in the engine immediately after the overhaul, cover all openings of the valve with plastic to prevent dirt from entering the valve during storage.



**Note!**

All fuel valves must be function-tested before being mounted in the cylinder cover, see *Procedure 909-11.1*.

1. Before mounting the fuel valve, thoroughly clean the valve bore in the cylinder cover and check the seating in the bore for marks which, if any, must be eliminated. (For reconditioning of valve bore in cylinder cover, see *Procedure 901-1*).

If not already done, mount new O-rings on the fuel valve. Lubricate the valve with molybdenum Disulphide (MoS₂).

2. Mount the valve in position in the cylinder cover.

Mount the tubular spacers and nuts. Tighten the nuts as stated in Data.

3. Lubricate the thread on the union nipple of the fuel oil pipe with a heat resistant anti seize grease before mounting.

Note!

It is recommended to overhaul the fuel oil pipe before mounting. See *Procedure 909-14.3*.

Mount the overhauled fuel oil pipe. See *Procedure 909-14.4*.

Mount and tighten the union nuts.

Reconnect the return oil pipe to the fuel valve.

Turn on the fuel oil supply.

SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

☐	Stopped engine
☐	Shut off starting air supply – <i>At starting air receiver</i>
☐	Block the main starting valve
☐	Shut off starting air distributor/distributing system supply
☐	Shut off safety air supply – <i>Not ME engines</i>
☐	Shut off control air supply
☐	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
☐	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☐	Stop lubricating oil supply
☐	Lock the turbocharger rotors

Data

Ref.	Description						Value	Unit
D09-70	Nozzel spray hole cleaning tool table							
	Nozzel spray hole	Cleaning drill	Test pin	Nozzel spray hole	Cleaning drill	Test pin		
	0.90	0.85	0.94	1.60	1.55	1.65		
	0.95	0.90	0.99	1.65	1.60	1.70		
	1.00	0.95	1.04	1.70	1.65	1.74		
	1.05	1.00	1.09	1.75	1.70	1.80		
	1.10	1.05	1.14	1.80	1.75	1.85		
	1.15	1.10	1.20	1.85	1.80	1.89		
	1.20	1.15	1.24	1.90	1.85	1.94		
	1.25	1.20	1.30	1.95	1.90	1.99		
	1.30	1.25	1.34	2.00	1.95	2.05		
	1.35	1.30	1.39	2.05	2.00	2.09		
	1.40	1.35	1.44	2.10	2.05	2.14		
	1.45	1.40	1.49	2.15	2.10	2.19		
	1.50	1.45	1.54	2.20	2.15	2.24		
	1.55	1.50	1.59	2.25	2.20	2.29		



The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P90966		Fuel Valve Nozzle - Tools

Note!

This instruction is only valid for spindle guides of the slide valve design.

Extreme care and accuracy should be exercised when carrying out this operation.

Based on service experience it is recommended to replace the complete spindle guide after 16000 hours of operation.

1. Clean the outside of the spindle guide in pure gas oil or similar. The individual parts of the spindle guide are not interchangeable, therefore only one guide is to be disassembled at a time.

Note!

The spindle guide, thrust piece and spindle are matched parts and may not be re-placed individually.

2. Place the spindle guide in a bench vice provided with "soft" jaws, and use the brass mandrel as shown to disassemble the spindle guide.
3. Mount the pulling tool around the fuel nozzle on the spindle guide. Turn the nut to pull the fuel nozzle off the spindle guide.

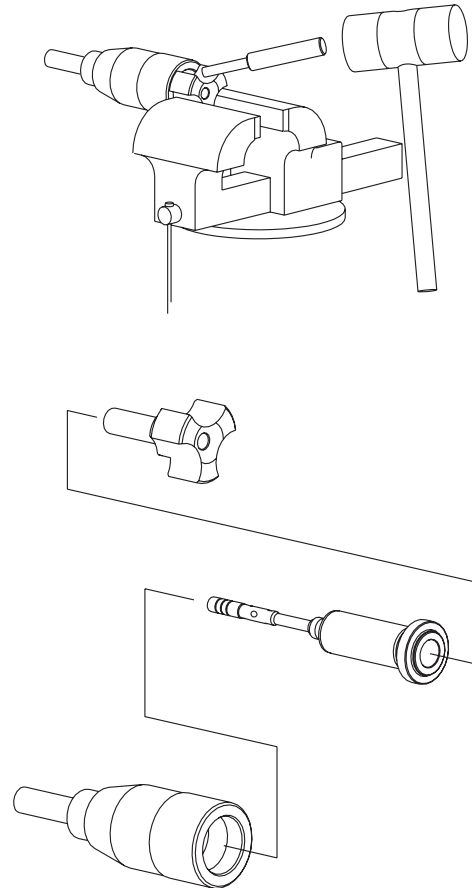
Note!

The pulling tool is not standard for all engines, but may be delivered as an optional extra.

4. If no pulling tool is available, the fuel nozzle can be dismantled from the spindle guide using two screwdrivers.

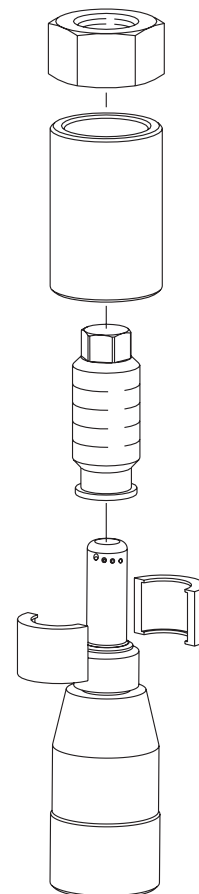
Place the screwdrivers opposite each other in the small gap between the fuel nozzle and the spindle guide and very carefully force the fuel nozzle off the spindle guide.

2.



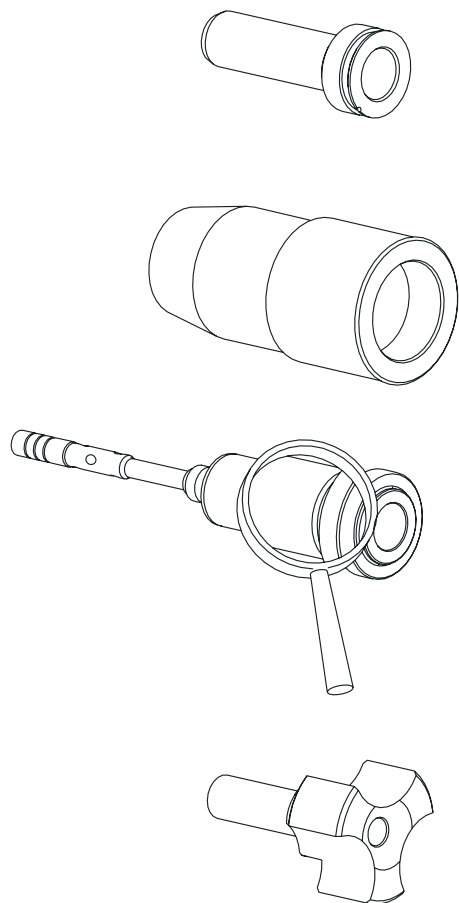
M90912-0229002

4.



M90912-0229004

5.



5. Clean all the parts of the spindle guide in gas oil and wipe dry with a clean piece of cloth.

Clean all parts again in gas oil or 'Electrocleaner' and wipe dry with a clean piece of cloth.

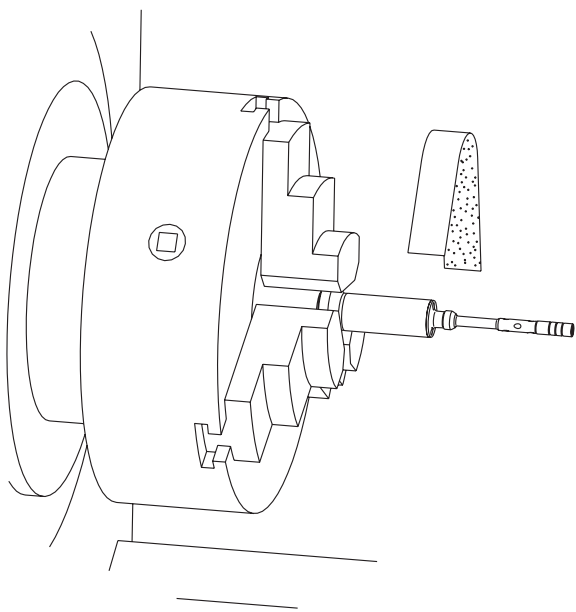
Place all the parts on a clean, soft, lint-free cloth and examine them through an 8-10 times magnification magnifying glass and an inspection lamp.

During the examination, pay special attention to the seating surfaces and sliding surfaces of the parts.

6. Remove any deposits or very fine scratches by placing the spindle, thrust piece or spindle guide respectively in a lathe, as shown, and polishing with a very fine conventional polishing linen 'grade 360'.

Use also a little oil for the polishing (a coarser polishing linen must **absolutely not** be used).

6.



Note!

The sliding surface of the cut-off slide may only be polished **VERY** carefully. The sliding surface must not be damaged.

After polishing, clean the parts again and re-check the seat on thrust piece/spindle, the seat on slide valve/spindle, and the seat on spindle/guide. Use an inspection lamp and an 8-10 times enlargement magnifying glass.

If the seats are not in order, i.e. if there are pressing-in marks or similar on the seats, the complete spindle guide must be discarded.

7. Clean any carbon deposits from the central bore of the fuel nozzle by means of the special brass brush. Clean the fuel nozzle with gas oil and wipe dry with a clean cloth.
8. Read the nominal hole size on the cylindrical part of the nozzle, as shown in the figure. The hole size is given in 1/100 mm. Depending on the engine model, there can be one or more different sizes of the holes. Find the correct drill size and matching test pin in the table, *D09-70*, on the data sheet, via the nominal hole size.

Clean the spray holes, using gas oil and the special drills supplied.

Note!

During this operation be very careful not to push the drill too far to avoid scratching the snug-fit surface on the inside of the fuel nozzle.

Then test the spray holes with the test pin. If the test pin is able to enter just one of the holes, the fuel nozzle must be discarded.

This also applies to nozzles with oval holes (can be ascertained with a magnifying glass).

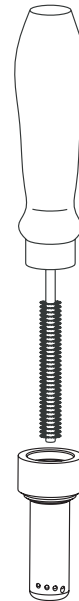
Check the fuel nozzle before mounting on the spindle guide, the cut-off slide must be able to move freely inside the nozzle.

Note!

It is recommended that the fuel nozzle is changed whenever the spindle guide is being overhauled or at maximum 8000 running hours.

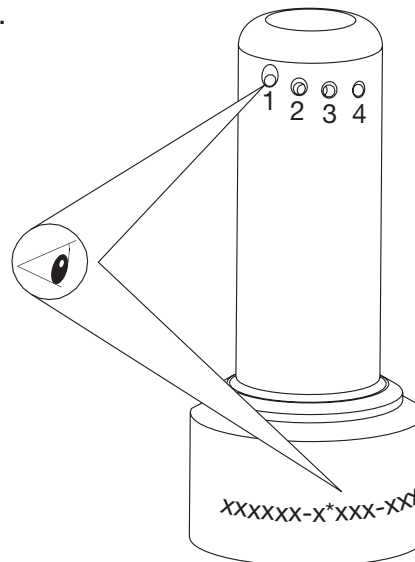
7.

M90912-0229O07



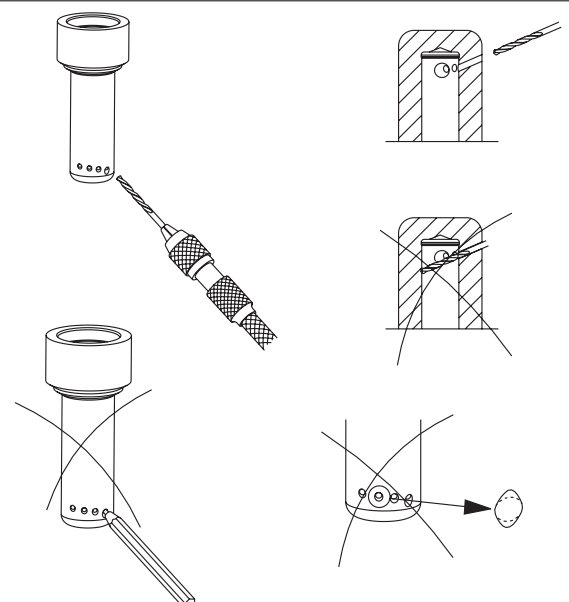
8.

M90912-0229O08

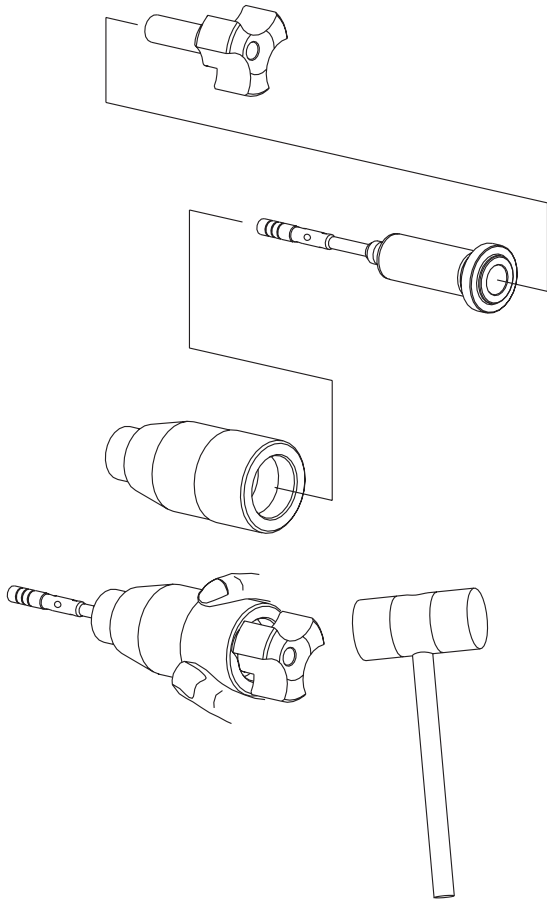


9.

M90912-0229O09



11.

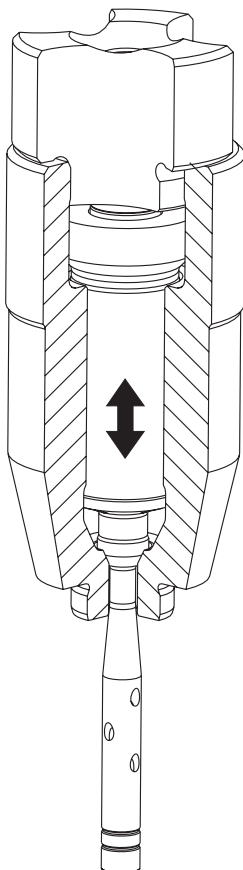


10. Lubricate the spindle and the thrust piece with the cut-off slide with a little Molybdenum Disulphide (MoS_2), see *Procedure 913-11*.

11. Assemble the thrust piece, the spindle and the spindle guide and carefully knock the parts together using a soft hammer.

12. Shake the spindle guide back and forth. The spindle with the cut-off slide must be able to slide freely back and forth inside the spindle guide, with a 'clicking' sound.

12.



M90912-0228O11

M90912-0229O12

13. Lubricate the sliding surfaces of the nozzle and the spindle with a little Molybdenum Disulphide (MoS_2). See *Procedure 913-11*.

Mount the nozzle on the spindle guide.

Make sure that the half-circle shaped groove on the nozzle and the spring pin are in line.

If nozzle and spindle guide are provided with scratch marks opposite the spring pin, they must be in line.

Place the parts on the plane of a drilling machine or hydraulic press and position the mounting tool over the parts. Make sure that all the parts are perfectly aligned.

Press the nozzle on to the spindle guide.

Note!

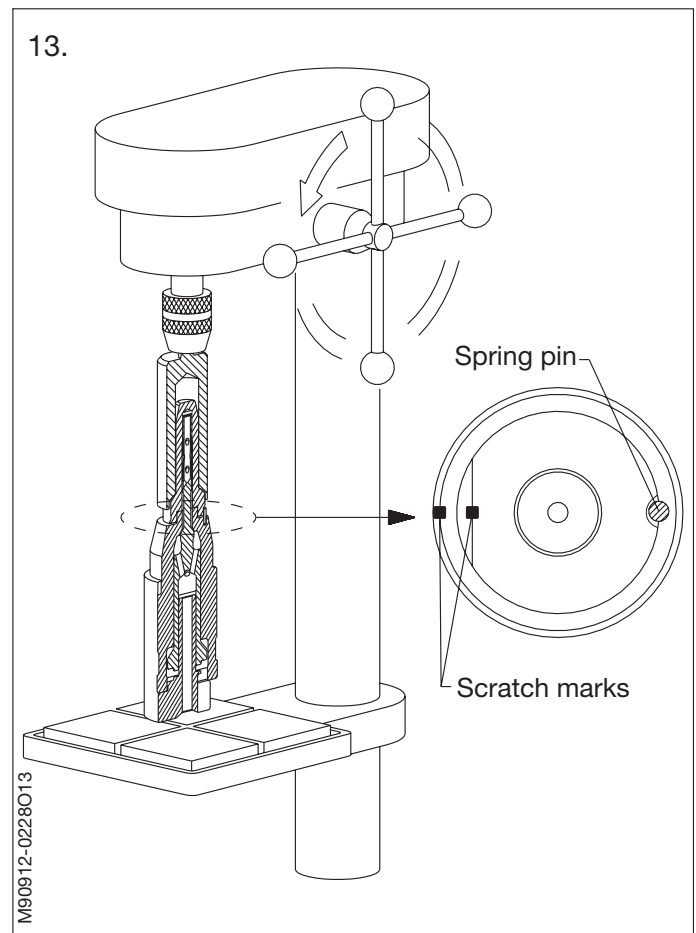
The mounting tools are not standard for all engines but may be delivered as optional extras.

If no mounting tools are available, the nozzle can be mounted on the spindle guide using a short piece of pipe.

Place the pipe around the nozzle, so that the lower end of the pipe rests on the 'foot' of the fuel nozzle. Then press the parts together the same way as when using the mounting tools.

Check that the spindle inside the spindle guide is able to move freely as in step 10.

14. If the spindle guide is not to be mounted in a fuel valve immediately after the overhaul, cover all openings of the spindle guide with plastic to prevent dirt from entering the spindle guide during storage.



SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

☐	Stopped engine
☐	Shut off starting air supply – <i>At starting air receiver</i>
☐	Block the main starting valve
☐	Shut off starting air distributor/distributing system supply
☐	Shut off safety air supply – <i>Not ME-engines</i>
☐	Shut off control air supply
☐	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
☐	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☐	Stop lubricating oil supply
☐	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P90951 P90968	153	Flange for dismantling of non return valve Fuel Valve - Inspection Tools

Note!

Extreme care and accuracy should be exercised when carrying out this operation.

1. Clean the outside of the non-return valve with pure gas oil. The individual parts are not interchangeable, therefore only one guide is to be disassembled at a time.

Note!

Except for the slide valve spring, defective parts cannot be replaced individually by new ones.

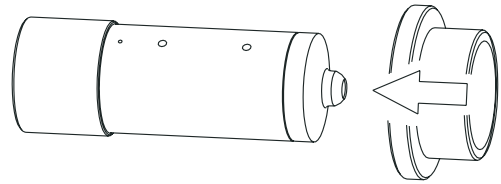
Place the non-return valve as shown in Fig. 1 in a bench vice provided with "soft" jaws, and disassemble the non-return valve, using the disassembling tool and a hammer as shown.

2. Clean all the parts for the spindle guide in gas oil and then blow clean.

Finally, clean in either gas oil, kerosene or 'Electrocleaner', and blow the parts dry with compressed air as shown in Fig. 2.

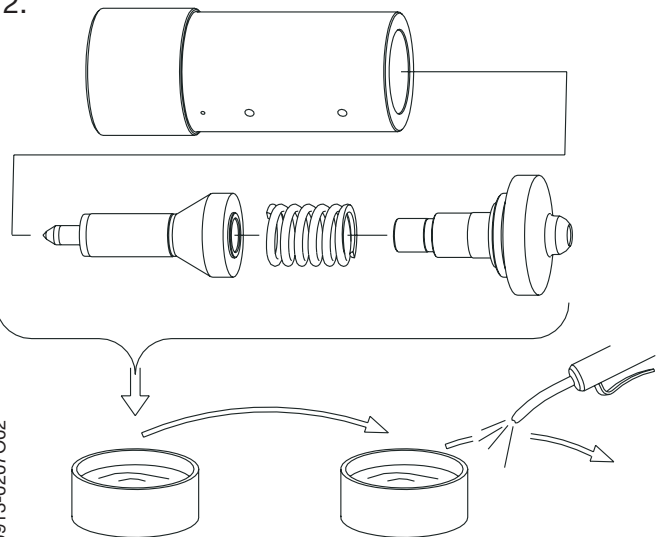
3. Now place the parts on clean, lint-free rags and examine with an 8-10 times enlargement magnifying glass, and an inspection lamp with magnifying glass as shown in Fig. 3.

1.



M90913-0207001

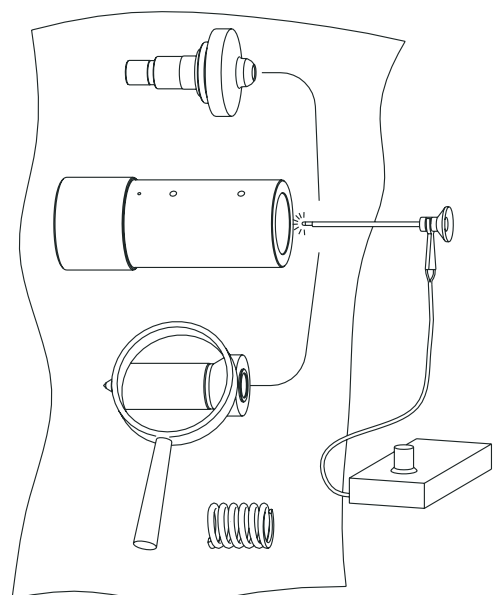
2.

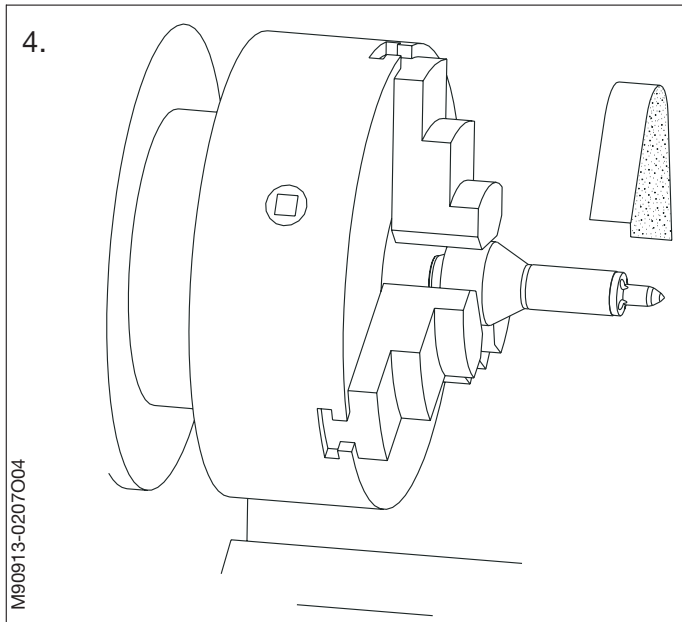


M90913-0207002

3.

M90913-0207003





4. Examine the slide faces of movable parts for coating. Vent slide **B**/housing **A** will be too tight if there is a coating. (The letters **A-D** refer to the letters/parts in Fig. 2).

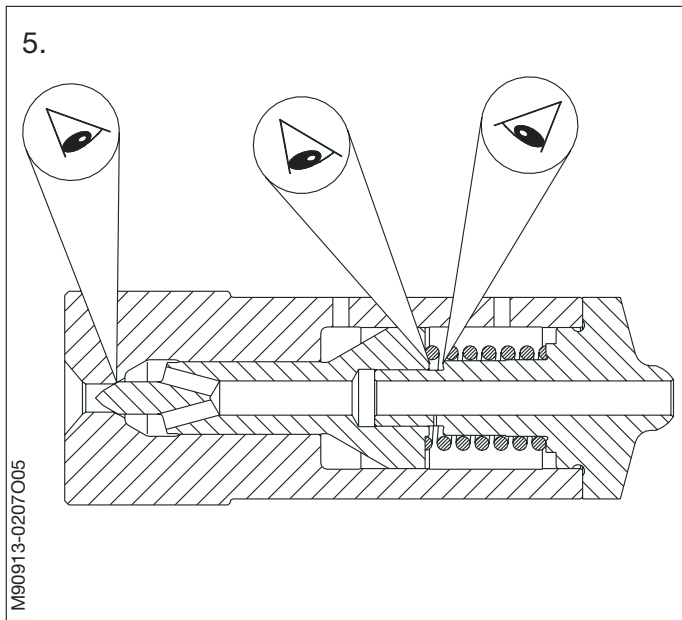
Fix vent slide **B** and, subsequently, housing **A** in a lathe as shown in Fig. 4 and remove the coating by means of very fine conventional polishing linen 'grade 360'.

Also a little oil should be used (a coarser polishing linen must **absolutely not** be used).

5. Check spring **C** for the thrust piece for outside wear marks. If defective, it should be exchanged.

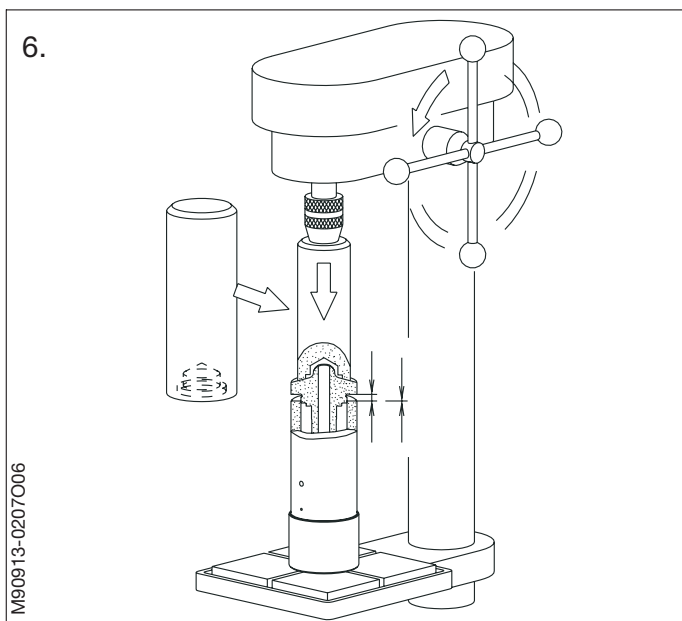
Check the seat on thrust piece **D**/vent slide, and the seat on vent slide/housing. Use an inspection lamp and an 8-10 times enlargement magnifying glass.

If the seats are not in order, i.e. if there are pressing-in marks or similar on the seats, the complete spindle guide must be discarded.



6. Mount the non-return valve as follows:

- Lubricate all movable parts with molybdenum disulphide (MoS_2).
- Place the loosely-assembled non-return valve on the plane of a drilling machine, with the tool positioned as shown in Fig. 6.
- Make sure that the thrust piece and the other parts are perfectly aligned and that the thrust piece is guided in the vent slide.
- Press the handle until the housing and thrust piece meet.



7. If the non-return valve is not to be mounted in a fuel valve immediately after the overhaul, cover all openings of the non-return valve with plastic to prevent dirt from entering the valve during storage.

SAFETY PRECAUTIONS

☒	Stopped engine
☒	Block the starting mechanism
☒	Shut off starting air supply
☐	Engage turning gear
☐	Shut off cooling water
☒	Shut off fuel oil
☐	Shut off lubricating oil
☐	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit
D09-47	Fuel oil high-pressure pipe, tightening torque	100	Nm
D09-51	Distance from pipe end to thrust bushing lower edge (fuel pump end and fuel valve end)	11	mm
D09-52	Fuel oil pipe	8	kg

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P90951 refers to chapter 909.

Plate	Item No.	Description
P90951	116	Crowfoot wrench head
P90951	165	Grinding mandrel for valve head

High-Pressure Pipe with Protective Hose

1. Close the fuel oil inlet and outlet valves.
2. Using a hook spanner, unscrew the union nut at the fuel valve end of the high-pressure pipe.

Pull the union nut and the protective hose clear of the union nipple.

Unscrew the union nipple from the fuel valve.

3. Using a hook spanner, unscrew the union nut at the fuel pump end of the high-pressure pipe.

Pull the union nut and the protective hose clear of the union nipple.

Unscrew the union nipple from the fuel pump.

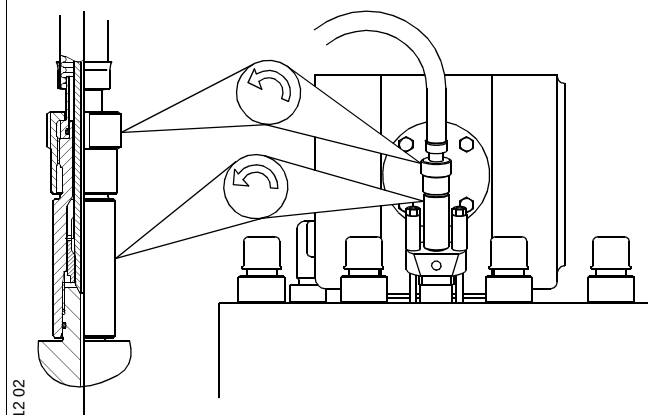
4. Lift the pipe ends clear of the fuel valve/fuel pump, and remove the high-pressure pipe from the engine.

Note!

It is recommended always to overhaul the high-pressure pipe before remounting it on the engine.
See *Procedure 909-14.3*.

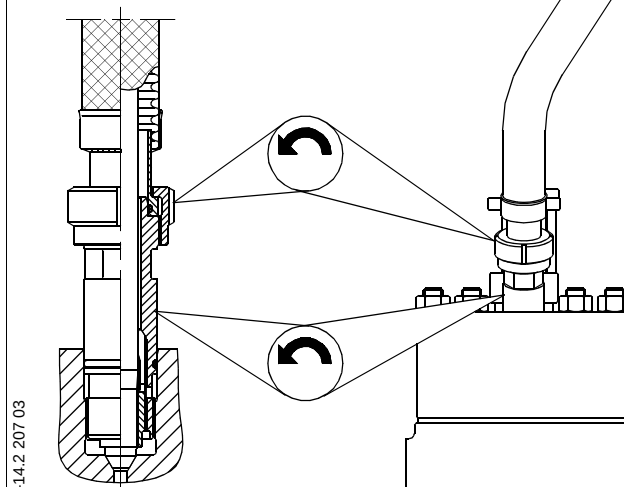
The overhaul should preferably take place immediately after the high-pressure pipe has been dismantled.

2.



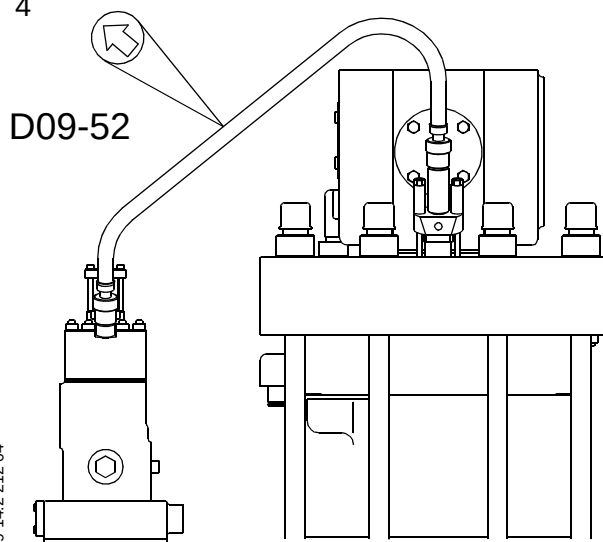
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3.

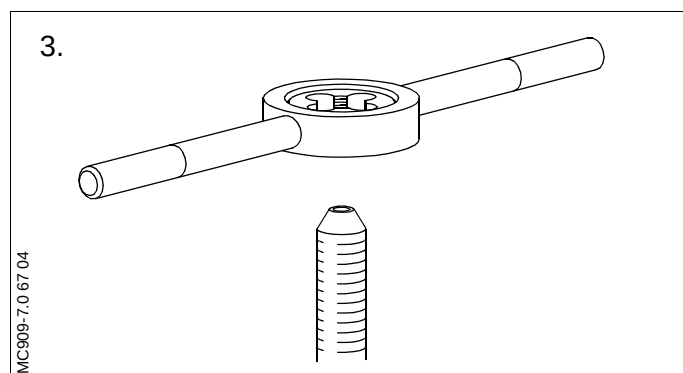
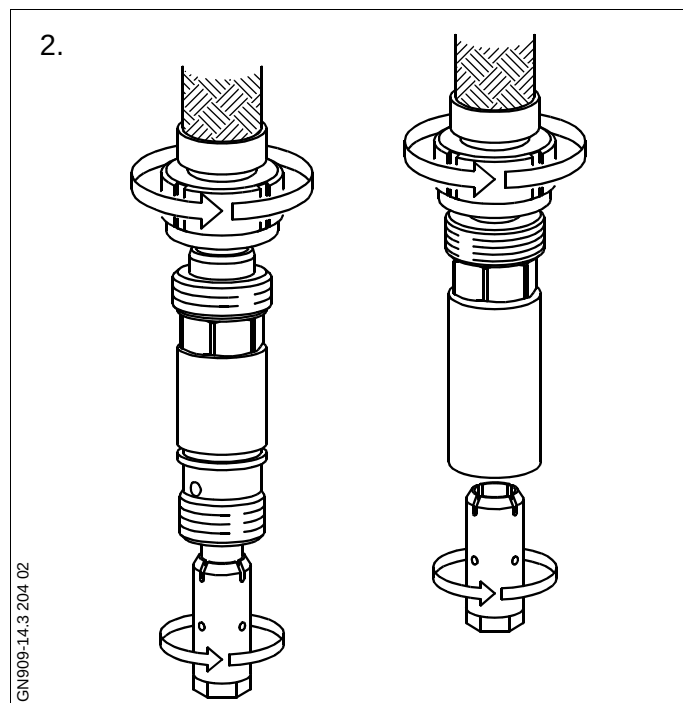
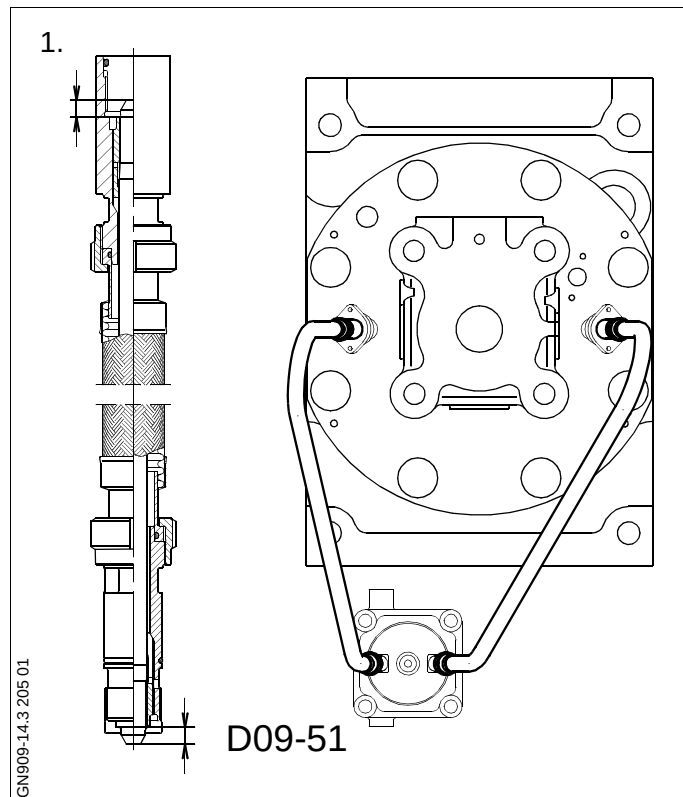


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4



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1. Whenever the fuel oil high-pressure system has been dismantled, it is necessary, before remounting the high-pressure pipes, to carefully inspect the tapered contact surfaces of the pipe ends, together with their seats in fuel valves and fuel pump top cover.

Furthermore, the position of the thrust bushing on the pipe end must be checked. If the distance is incorrect, compared with the measurement stated in Data, it should be adjusted by screwing the thrust bushing up or down the pressure pipe.

2. If any of the pipe ends requires reconditioning, dismantle the high-pressure pipe as follows:

- Lift up the union nut with flexible protective hose, together with the coupling nut, on the high-pressure pipe and screw off the thrust bushing.
- Remove the coupling nut and, when the thrust bushings at both ends of the pipe have been dismantled, pull the flexible protective hose with union nuts off the high-pressure pipe.

3. Shape-up the threads on the pipe ends with the nut die.

4. Mount the guide on the pipe end, place the miller in the guide and lightly screw on the union nut.
5. Turn the miller with, for instance, a tap wrench while lightly tightening the union nut to provide a suitable pressure between miller and pipe end. During the milling, add drilling oil emulsion liberally.

After completing the milling, carefully clean the high-pressure pipe with compressed air.

6. When assembling the high-pressure pipe, screw the thrust bushings so high up on the thread of the pressure pipe that the distance between the pipe end and the bottom edge of the thrust bushing is as stated in Data.

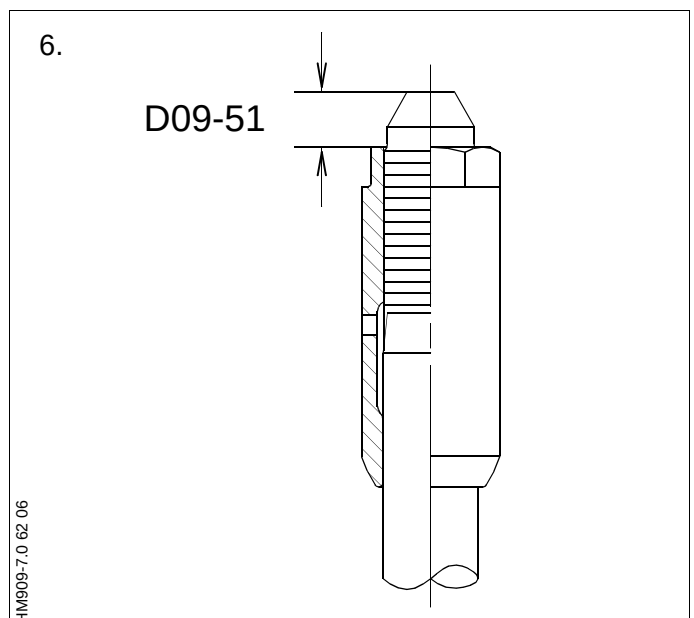
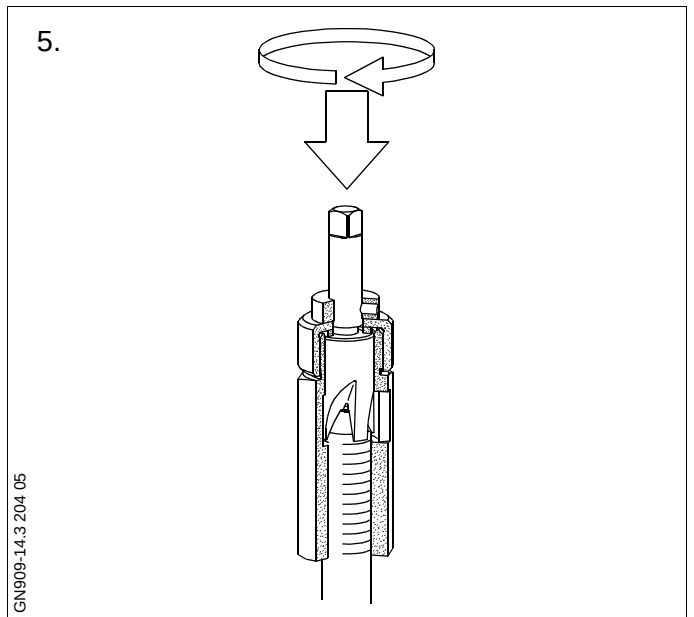
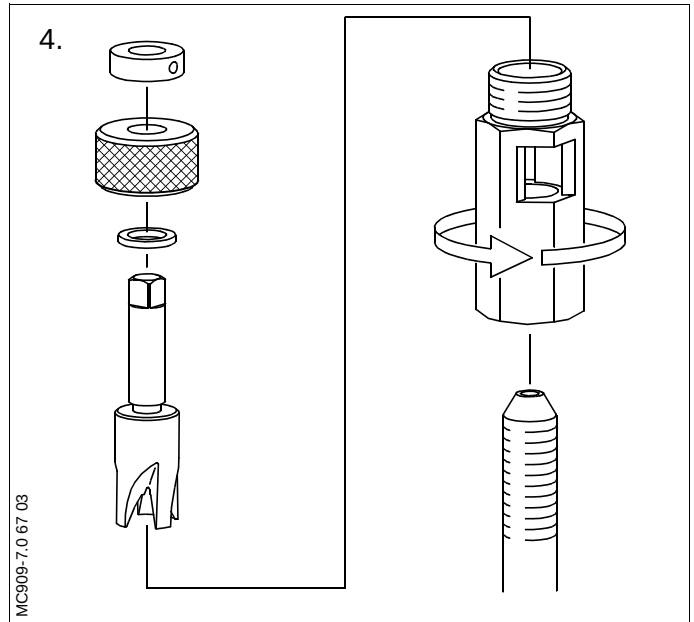
Replace the O-rings.

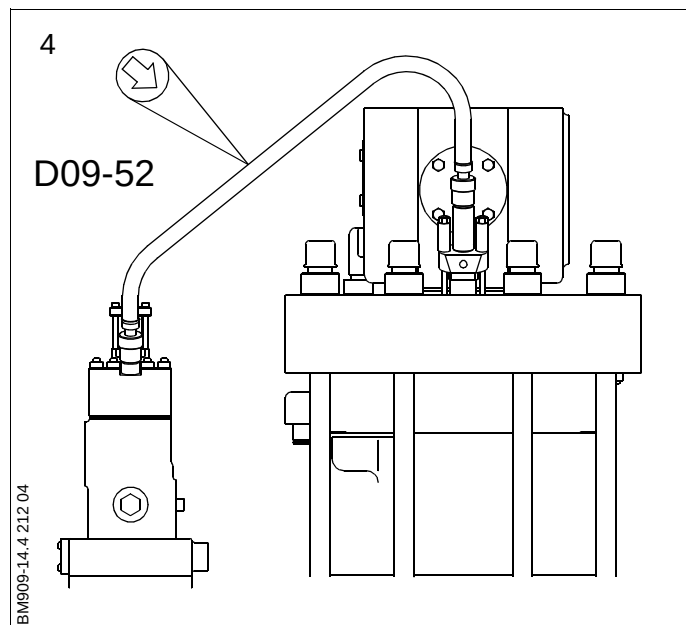
Before mounting a high-pressure pipe, check the centre distances between pipe ends and seats, and lubricate the threads of the union nuts with molybdenum disulphide (MoS_2).

It is of great importance for the mounting, and a condition for obtaining tight joints, that the pipes fit the seats accurately.

For reconditioning of seat in fuel valve, see Procedure 909-11.3.

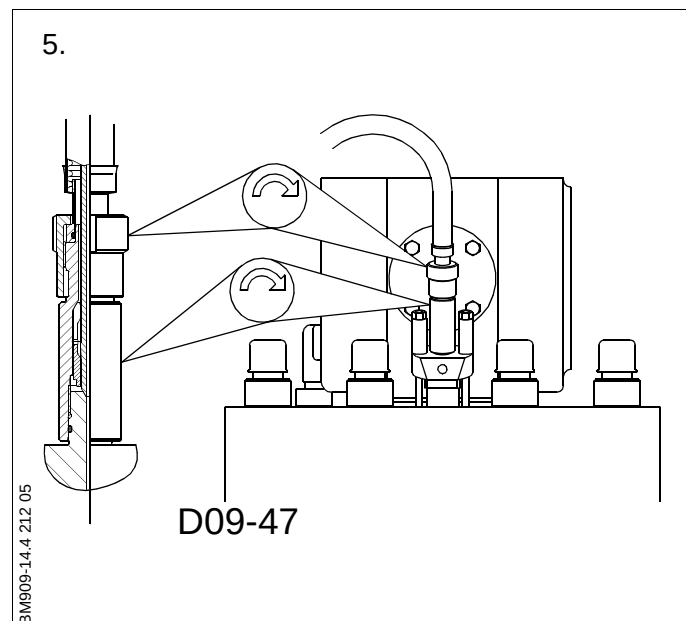
For reconditioning of seat in fuel pump top cover, see Procedure 909-5.3.





1. Check that the distance between the thrust bushing and the pipe end is correct at both ends of the pipe. See *Procedure 909-14.3*.
2. Ensure that the fuel valve is fitted with a new O-ring. Lubricate the thread of the fuel valve with a heat resistant grease.
3. Ensure that the union nipple at the fuel pump end of the high-pressure pipe is fitted with a new O-ring. Lubricate the thread on the union nipple with a heat resistant grease.
4. Mount the high-pressure pipe between the fuel valve and the fuel pump.

Screw the union nipples onto the fuel valve and into the fuel pump top cover, by hand.

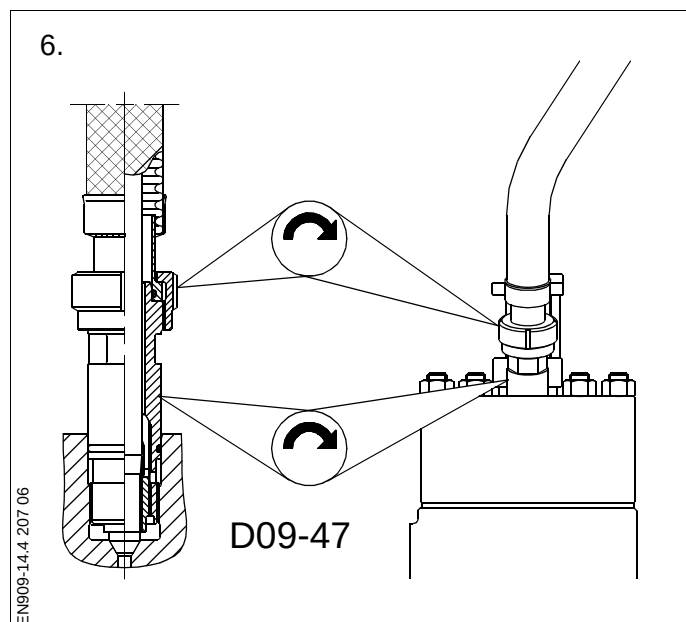


Note!

The nipples must be screwed by **hand**, as the threads of the union nipples, fuel valve or fuel pump top cover might otherwise be damaged. Tightening with a crowfoot wrench should only be done after both of the union nipples have been screwed into position.

5. Use a crowfoot wrench and a torque spanner to tighten the union nipple at the fuel valve to the torque stated in Data.

Screw the union nut onto the union nipple by hand, and tighten lightly with a hook spanner.



6. Use a crowfoot wrench and a torque spanner to tighten the union nipple at the fuel pump to the torque stated in Data.

Screw the union nut onto the union nipple by hand, and tighten lightly with a hook spanner.

7. If dismantled, re-mount the return oil pipe on the fuel valve. Open the fuel oil inlet and outlet valves.

X	Stopped engine
X	Shut off starting air supply – <i>At starting air receiver</i>
X	Block the main starting valve
X	Shut off starting air distributor/distributing system supply
X	Shut off safety air supply – <i>Not ME engines</i>
X	Shut off control air supply
	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
X	Engage turning gear
	Shut off cooling water
X	Shut off fuel oil
X	Stop lubricating oil supply
	Lock the turbocharger rotors

Ref.	Description	Value	Unit
D09-57	Roller/bushing/shaft pin, max. clearance	0.5	mm
D09-60	Housing for sealing arrangement	70	kg
D09-61	Fuel pump roller guide	40	kg
D09-62	Pump base	32	kg

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P90851	173	Lifting tool for exhaust valve roller guide

SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

- ☐ Stopped engine
- ☐ Shut off starting air supply – *At starting air receiver*
- ☐ Block the main starting valve
- ☐ Shut off starting air distributor/distributing system supply
- ☐ Shut off safety air supply – *Not ME engines*
- ☐ Shut off control air supply
- ☐ Shut off air supply to exhaust valve – *Only with stopped lubricating oil pumps*
- ☐ Engage turning gear
- ☐ Shut off cooling water
- ☐ Shut off fuel oil
- ☐ Stop lubricating oil supply
- ☐ Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit



The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P90966		Fuel Valve Nozzle - Tools
P90968		Fuel Valve - Inspection Tools
P90974		Fuel Valve Nozzle - Cleaning Tools

Note!

This instruction is only valid for spindle guides of the slide valve design.

Extreme care and accuracy should be exercised when carrying out this operation.

Based on service experience it is recommended to replace the complete spindle guide after 8000 hours of operation.

1. Clean the outside of the spindle guide in pure gas oil or kerosene. The individual parts of the spindle guide are not interchangeable, therefore only one guide is to be disassembled at a time.

Note!

The spindle guide, thrust piece and spindle are matched parts and may not be replaced individually.

2. Place the spindle guide in a bench vice provided with "soft" jaws, and use the brass mandrel as shown to disassemble the spindle guide.
3. Mount the pulling tool around the fuel nozzle on the spindle guide. Turn the nut to pull the fuel nozzle off the spindle guide.

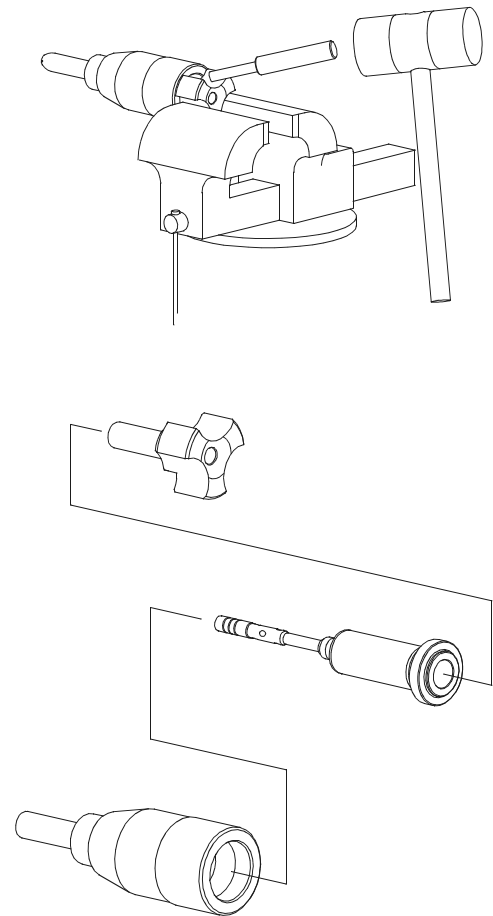
Note!

The pulling tool is not standard for all engines, but may be delivered as an optional extra.

4. If no pulling tool is available, the fuel nozzle can be dismantled from the spindle guide using two screwdrivers.

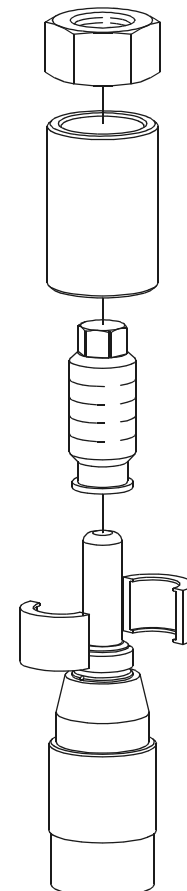
Place the screwdrivers opposite each other in the small gap between the fuel nozzle and the spindle guide and very carefully force the fuel nozzle off the spindle guide.

2.



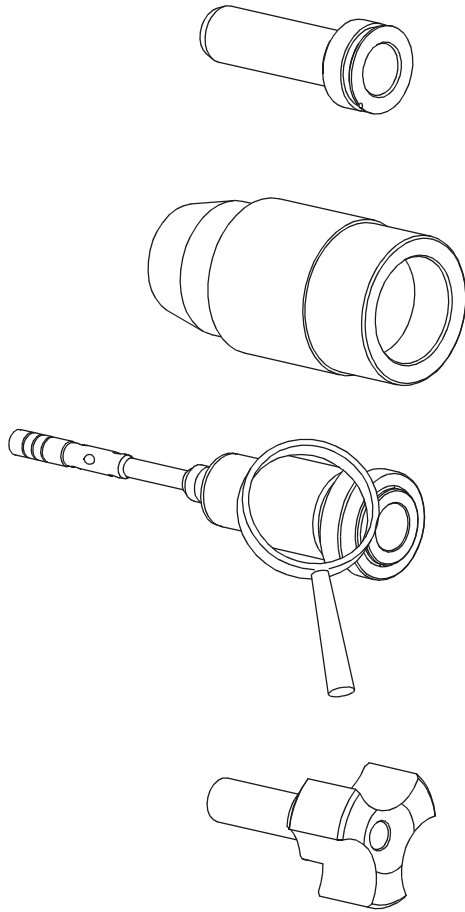
GW909-12.3 222 02

4.



GW909-12.3 222 04

5.



5. Clean all the parts of the spindle guide in gas oil and wipe dry with a clean piece of cloth.

Clean all parts again in kerosene or 'Electrocleaner' and wipe dry with a clean piece of cloth.

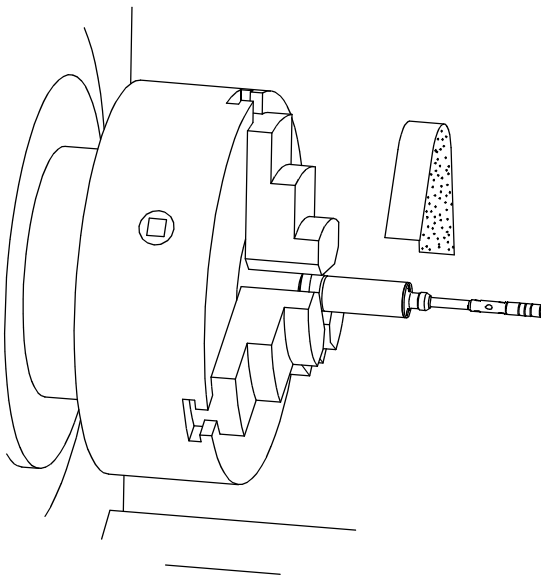
Place all the parts on a clean, lint-free cloth and examine them through an 8-10 times magnification magnifying glass and an inspection lamp.

During the examination, pay special attention to the seating surfaces and sliding surfaces of the parts.

6. Remove any deposits or very fine scratches by placing the spindle, thrust piece or spindle guide respectively in a lathe, as shown, and polishing with a very fine conventional polishing linen 'grade 360'.

Use also a little oil for the polishing (a coarser polishing linen must **absolutely not** be used).

6.



Note!

The sliding surface of the cut-off slide may only be polished **VERY** carefully. The sliding surface must not be damaged.

After polishing, clean the parts again and re-check the seat on thrust piece/spindle, the seat on slide valve/spindle, and the seat on spindle/guide. Use an inspection lamp and an 8-10 times enlargement magnifying glass.

If the seats are not in order, i.e. if there are pressing-in marks or similar on the seats, the complete spindle guide must be discarded.

7. Clean any carbon deposits from the central bore of the fuel nozzle by means of the special brass brush. Clean the spray holes, using gas oil and the special drills supplied.

Note!

During this operation be very careful not to push the drill too far to avoid scratching the snug-fit surface on the inside of the fuel nozzle.

Clean the fuel nozzle with kerosene and wipe dry with a clean cloth.

Then test the spray holes with the test pin. If the test pin is able to enter just one of the holes, the fuel nozzle must be discarded.

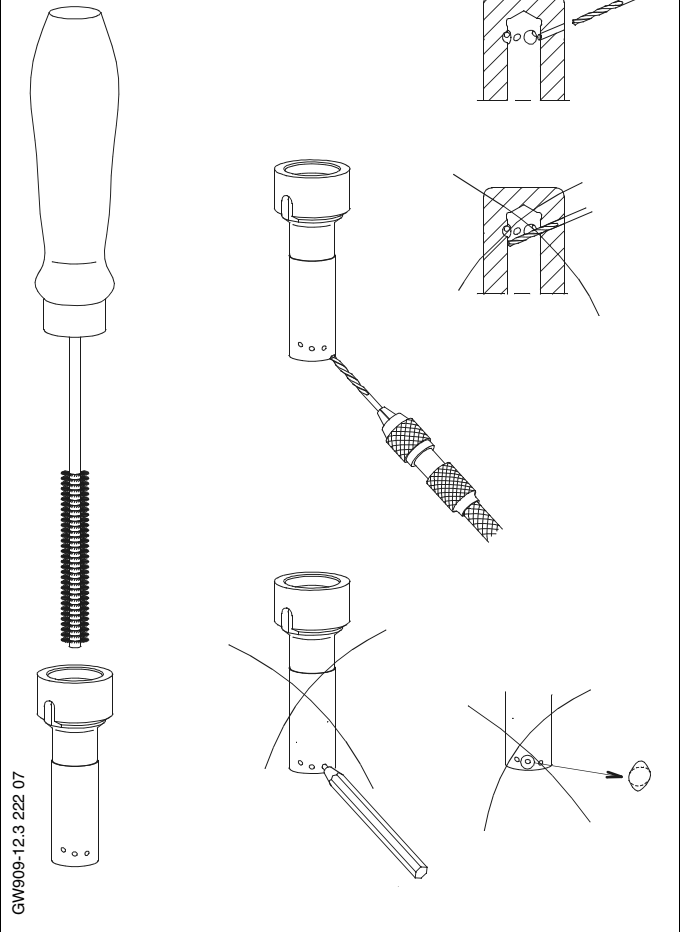
This also applies to nozzles with oval holes (can be ascertained with a magnifying glass).

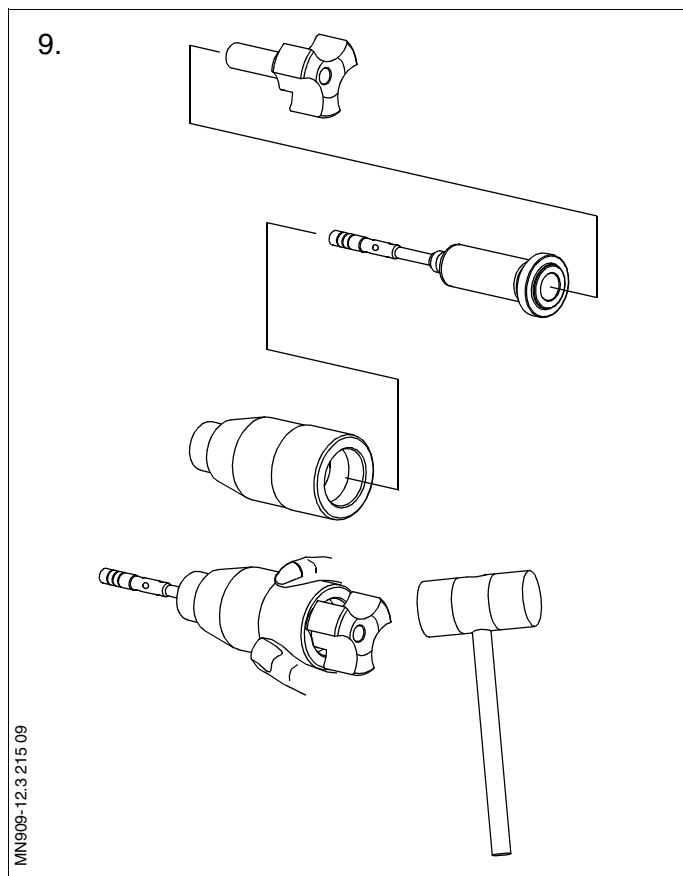
Check the fuel nozzle before mounting on the spindle guide, the cut-off slide must be able to move freely inside the nozzle.

Note!

It is recommended that the fuel nozzle is changed whenever the spindle guide is being overhauled.

7.

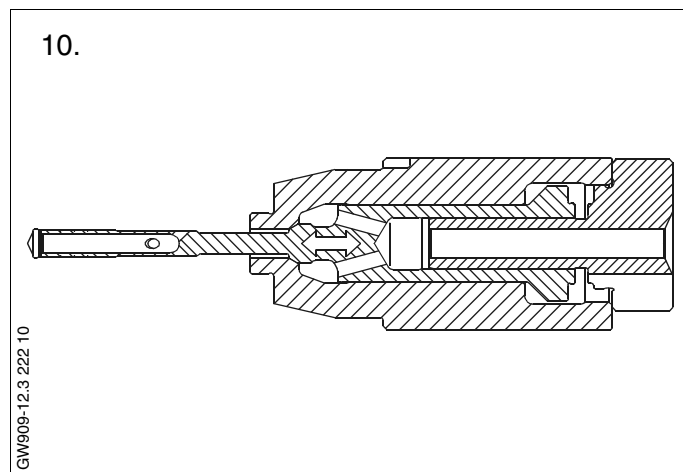




8. Lubricate the spindle and the thrust piece with the cut-off slide with a little Molybdenum Disulphide (MoS_2), see *Procedure 913-11*.

9. Assemble the thrust piece, the spindle and the spindle guide and carefully knock the parts together using a soft hammer.

10. Shake the spindle guide back and forth. The spindle with the cut-off slide must be able to slide freely back and forth inside the spindle guide, with a 'clicking' sound.



11. Lubricate the sliding surfaces of the nozzle and the spindle with a little Molybdenum Disulphide (MoS_2). See Procedure 913-11.

Mount the nozzle on the spindle guide.

Place the parts on the plane of a drilling machine or hydraulic press and position the mounting tool over the parts. Make sure that all the parts are perfectly aligned.

Press the nozzle on to the spindle guide.

Note!

The mounting tools are not standard for all engines but may be delivered as optional extras.

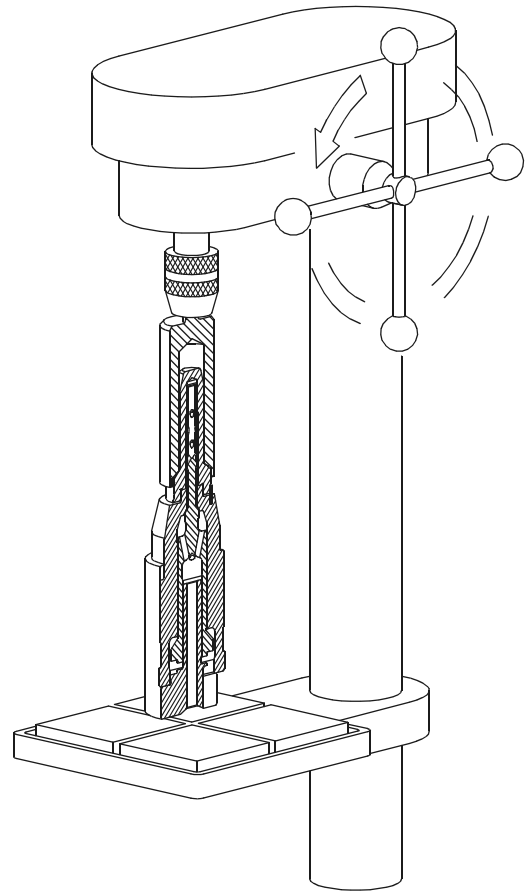
If no mounting tools are available, the nozzle can be mounted on the spindle guide using a short piece of pipe.

Place the pipe around the nozzle, so that the lower end of the pipe rests on the 'foot' of the fuel nozzle. Then press the parts together the same way as when using the mounting tools.

Check that the spindle inside the spindle guide is able to move freely, by 'shaking' the spindle guide.

12. If the spindle guide is not to be mounted in a fuel valve immediately after the overhaul, cover all openings of the spindle guide with plastic to prevent dirt from entering the spindle guide during storage.

11.



GW909-12.3 222 11

SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

- ☐ Stopped engine
- ☐ Shut off starting air supply – *At starting air receiver*
- ☐ Block the main starting valve
- ☐ Shut off starting air distributor/distributing system supply
- ☐ Shut off safety air supply – *Not ME-engines*
- ☐ Shut off control air supply
- ☐ Shut off air supply to exhaust valve – *Only with stopped lubricating oil pumps*
- ☐ Engage turning gear
- ☐ Shut off cooling water
- ☐ Shut off fuel oil
- ☐ Stop lubricating oil supply
- ☐ Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P90951 P90968	153	Flange for dismantling of non return valve Fuel Valve - Inspection Tools

Note!

Extreme care and accuracy should be exercised when carrying out this operation.

1. Clean the outside of the non-return valve with pure gas oil. The individual parts are not interchangeable, therefore only one guide is to be disassembled at a time.

Note!

Except for the slide valve spring, defective parts cannot be replaced individually by new ones.

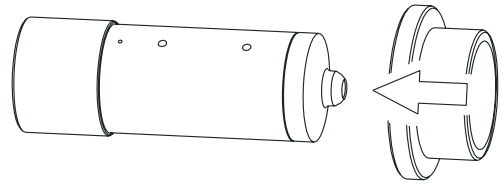
Place the non-return valve as shown in Fig. 1 in a bench vice provided with "soft" jaws, and disassemble the non-return valve, using the disassembling tool and a hammer as shown.

2. Clean all the parts for the spindle guide in gas oil and then blow clean.

Finally, clean in either gas oil, kerosene or 'Electrocleaner', and blow the parts dry with compressed air as shown in Fig. 2.

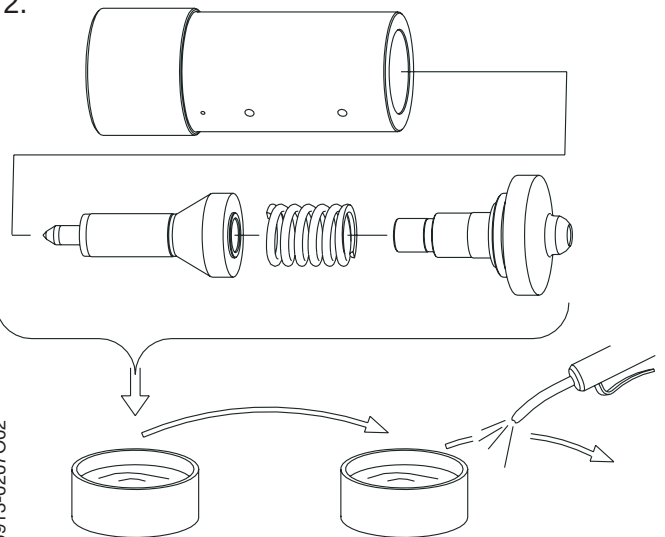
3. Now place the parts on clean, lint-free rags and examine with an 8-10 times enlargement magnifying glass, and an inspection lamp with magnifying glass as shown in Fig. 3.

1.



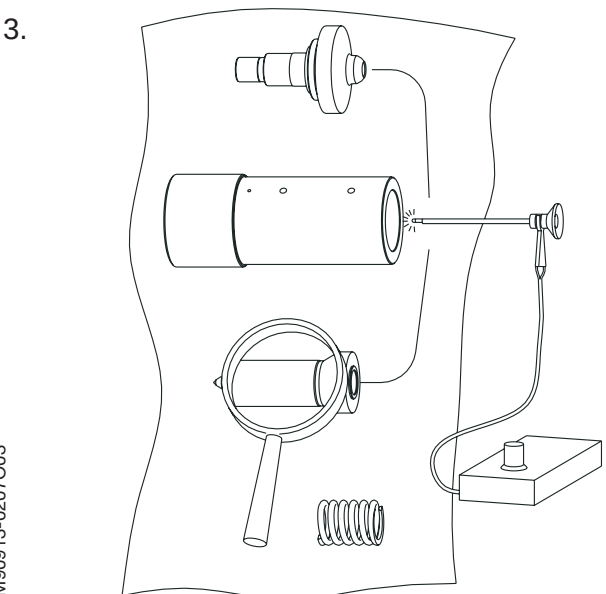
M90913-0207001

2.

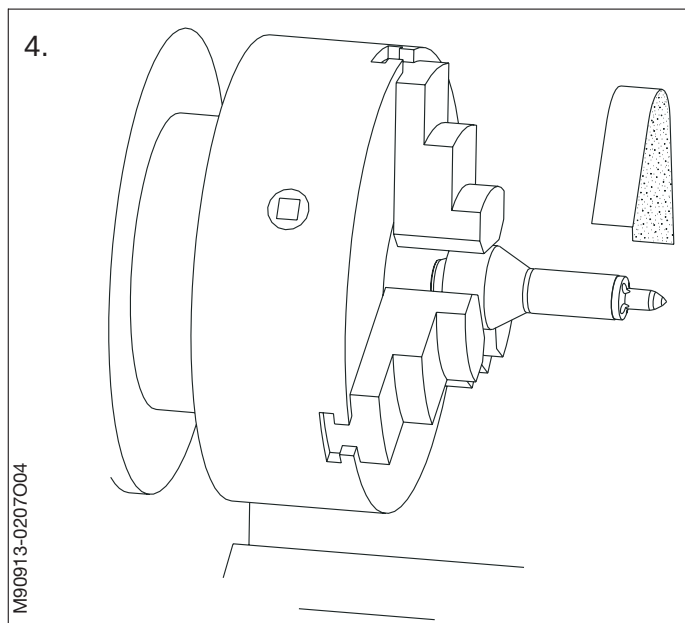


M90913-0207002

3.



M90913-0207003



4. Examine the slide faces of movable parts for coating. Vent slide **B**/housing **A** will be too tight if there is a coating. (The letters **A-D** refer to the letters/parts in Fig. 2).

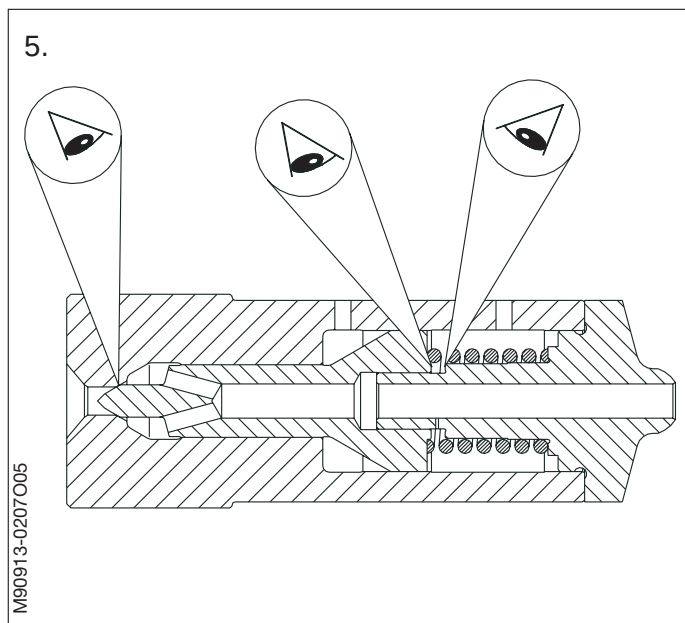
Fix vent slide **B** and, subsequently, housing **A** in a lathe as shown in Fig. 4 and remove the coating by means of very fine conventional polishing linen 'grade 360'.

Also a little oil should be used (a coarser polishing linen must **absolutely not** be used).

5. Check spring **C** for the thrust piece for outside wear marks. If defective, it should be exchanged.

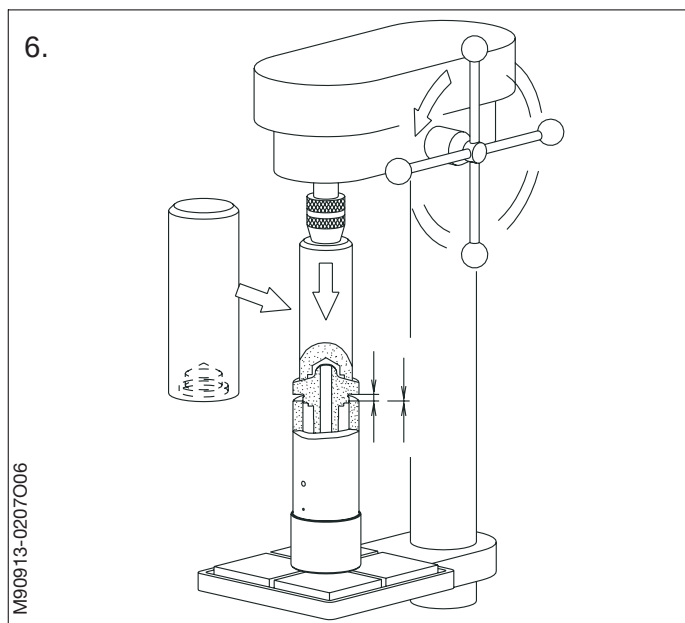
Check the seat on thrust piece **D**/vent slide, and the seat on vent slide/housing. Use an inspection lamp and an 8-10 times enlargement magnifying glass.

If the seats are not in order, i.e. if there are pressing-in marks or similar on the seats, the complete spindle guide must be discarded.



6. Mount the non-return valve as follows:

- Lubricate all movable parts with molybdenum disulphide (MoS_2).
- Place the loosely-assembled non-return valve on the plane of a drilling machine, with the tool positioned as shown in Fig. 6.
- Make sure that the thrust piece and the other parts are perfectly aligned and that the thrust piece is guided in the vent slide.
- Press the handle until the housing and thrust piece meet.



7. If the non-return valve is not to be mounted in a fuel valve immediately after the overhaul, cover all openings of the non-return valve with plastic to prevent dirt from entering the valve during storage.

X	Stopped engine
X	Block the starting mechanism
X	Shut off starting air supply
	Engage turning gear
	Shut off cooling water
X	Shut off fuel oil
	Shut off lubricating oil
	Lock turbocharger rotors

[illegible]

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P90951 refers to chapter 909.

Plate	Item No.	Description
P90951	104	Milling tool for fuel oil pipe
P90951	128	Crowfoot wrench head

High-Pressure Pipe with Double Pipe

1. Close the fuel oil inlet and outlet valves.
2. Unscrew the union nipple from the fuel valve.
3. Unscrew the union nipple from the fuel pump.
4. Lift the pipe ends clear of the fuel valve/fuel pump, and remove the high-pressure pipe from the engine.

Note!

It is recommended always to overhaul the high-pressure pipe before remounting it on the engine.
See Procedure 909-14.3.

The overhaul should preferably take place immediately after the high-pressure pipe has been dismantled.

2.

BM909-14.2 211 02

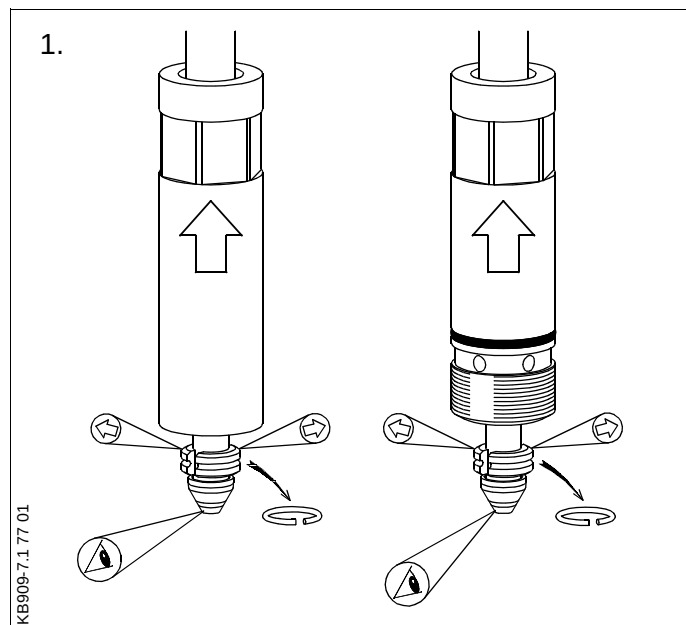
3.

GN909-14.2 210 03

4.

D09-52

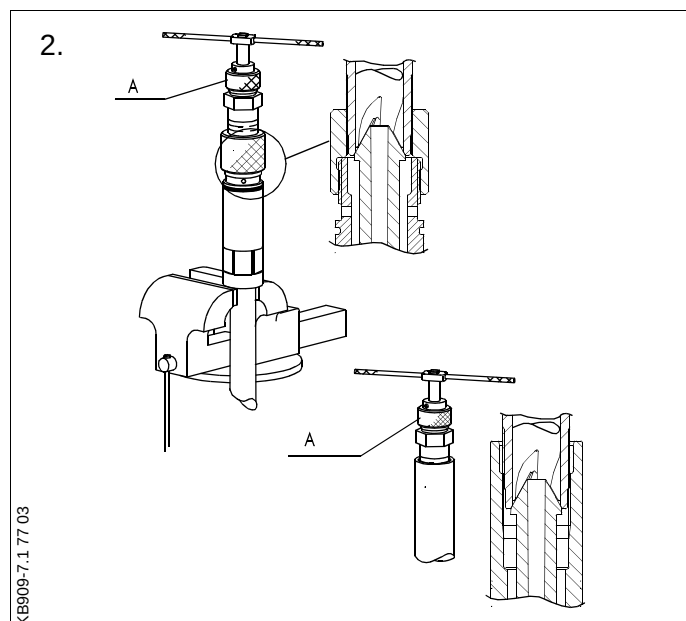
BM909-14.2 211 04



Whenever the fuel oil high-pressure system has been dismantled, it is necessary, before re-mounting the high-pressure pipes, to carefully inspect the tapered contact surfaces of the pipe ends, together with their seats in fuel valves and fuel pump top cover.

If the pipe ends require reconditioning, proceed as follows:

1. Lift up the union nut. Remove the spring ring from the two-part sleeve. Remove the sleeve.
2. Fasten the pipe in a vice with soft jaws. Mount the miller on the pipe end.



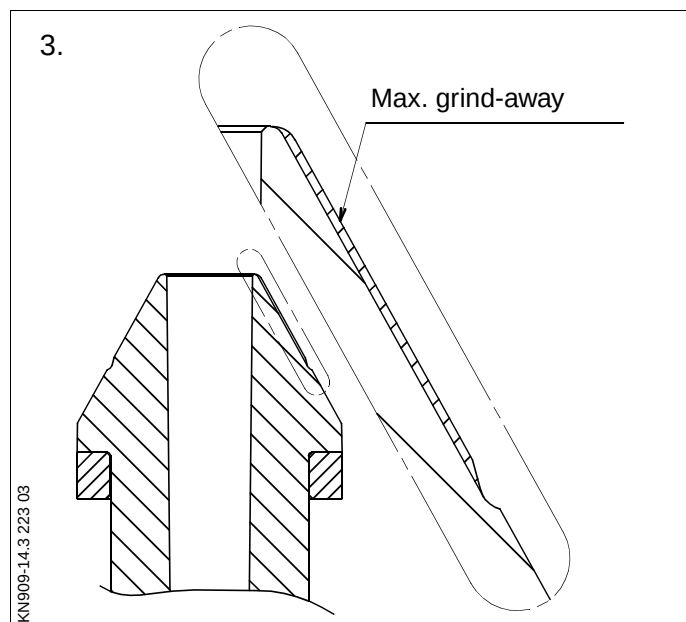
Note!

The two pipe ends are of different designs – see *sketch*.

Turn the upper tightening ring (A) until the miller presses against the pipe end.

Turn the miller with, for instance, a tap wrench while lightly tightening the upper tightening ring (A) to provide a suitable pressure between the miller and the pipe end.

During the milling process, add drilling oil emulsion liberally.



3. On both pipe ends a small groove has been ground to indicate the maximum allowable grinding of the pipe ends.

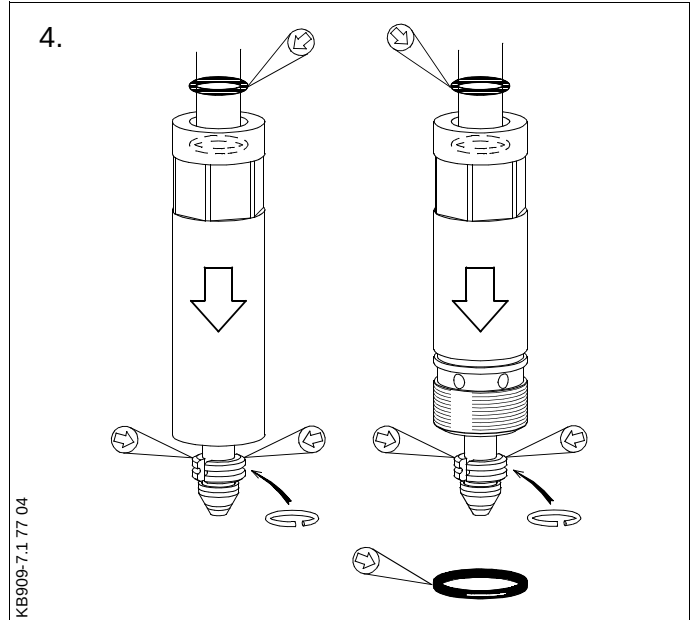
The pipe ends may only be ground until the ground surface is flush with the bottom of the groove.

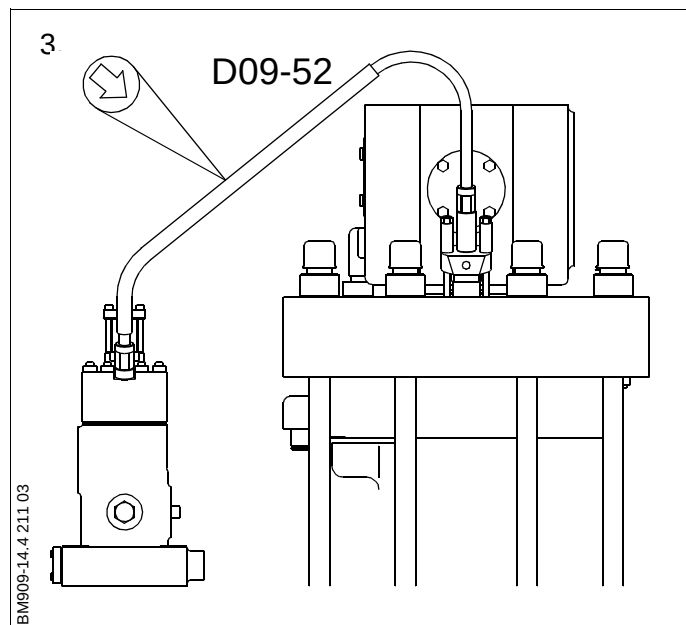
If the pipe ends are still not in order, the pipe must be discarded.

After completing the milling, carefully clean the high-pressure pipe, and blow through the bore with compressed air.

4. Remove and discard both the internal and external O-rings. Mount new O-rings.

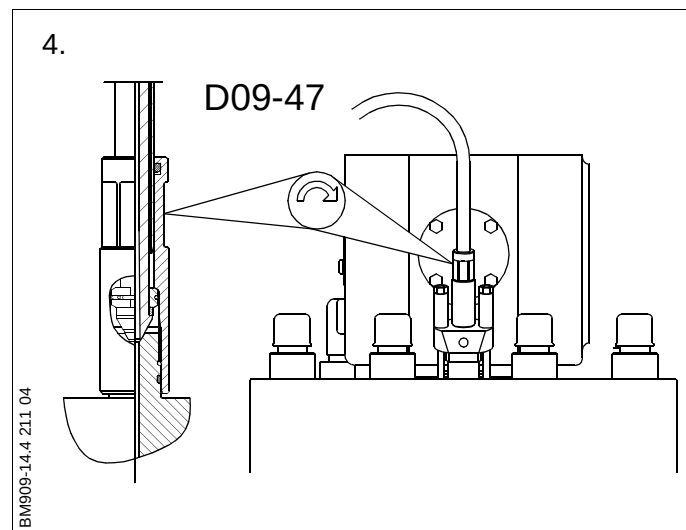
Mount the two-part sleeve and lock the sleeve with the spring ring. Ensure that all parts are completely clean before mounting the high-pressure pipe.





1. Ensure that the fuel valve is fitted with a new O-ring. Lubricate the thread of the fuel valve with a heat resistant grease.
2. Ensure that the union nipple at the fuel pump end of the high-pressure pipe is fitted with a new O-ring. Lubricate the thread on the union nipple with a heat resistant grease.
3. Mount the high-pressure pipe between the fuel valve and the fuel pump.

Screw the union nipples onto the fuel valve and into the fuel pump top cover, by hand.

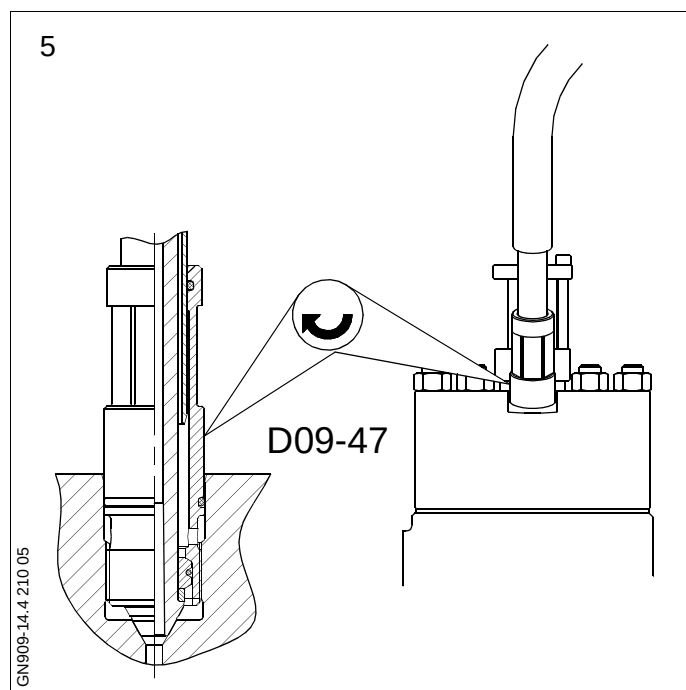


Note!

The nipples must be screwed by **hand**, as the threads of the union nipples, fuel valve or fuel pump top cover might otherwise be damaged.

Tightening with a crowfoot wrench should only be done after both of the union nipples have been screwed into position.

4. Use a crowfoot wrench and a torque spanner to tighten the union nipple at the fuel valve to the torque stated in Data.
5. Use a crowfoot wrench and a torque spanner to tighten the union nipple at the fuel pump to the torque stated in Data.
6. If dismantled, re-mount the return oil pipe on the fuel valve. Open the fuel oil inlet and outlet valves.



SAFETY PRECAUTIONS

☐	Stopped engine
☐	Block the starting mechanism
☐	Shut off starting air supply
☐	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☐	Shut off lubricating oil
☐	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit
D09-53	Nut, reversing shaft, tightening torque	80	Nm
D09-54	Clearance A	0.7 - 3.2	mm
D09-55	Clearance B	2.8 - 5.3	mm
D09-56	Clearance C, max.	0.45	mm
D09-57	Roller/bushing/shaft pin, max. clearance	0.5	mm
D09-60	Housing for sealing arrangement	70	kg
D09-61	Fuel pump roller guide	60	kg
D09-62	Pump base	31	kg

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P90951 refers to chapter 909.

Plate	Item No.	Description
P90851	173	Lifting tool for exhaust valve roller guide

Roller guide mounted in engine

1. The fuel pump roller guide can be partially checked while mounted on the engine, using the following procedure:
2. Lift the fuel pump roller guide.
See Procedure 909-16.
3. Remove the cover on the camshaft housing to inspect the fuel pump roller guide.

Turn the engine to provide the maximum space between the roller guide and the cam disc.

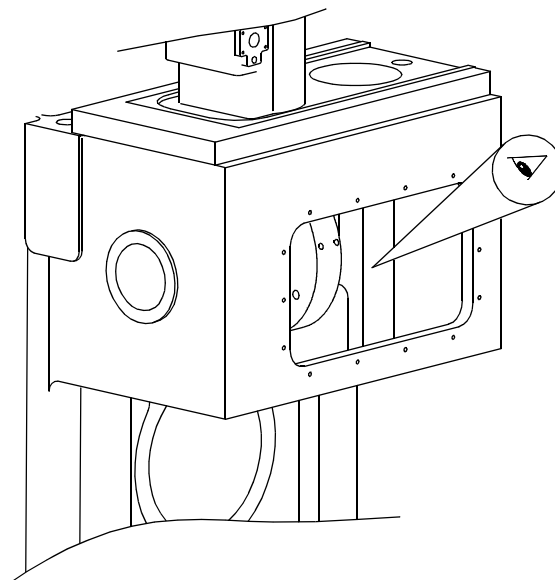
Turn the roller by hand to check that it can move freely and without unnecessary resistance or “hard” points.

Also inspect the surface of the roller for possible damage marks, seizures or scratches. Make sure that there are no traces of bearing metal left in the roller guide housing.

Check the clearance in the slide bearing by lifting the roller relative to the roller guide and measuring the difference in distance between the cam disc and the roller in the upper and lower positions of the roller.

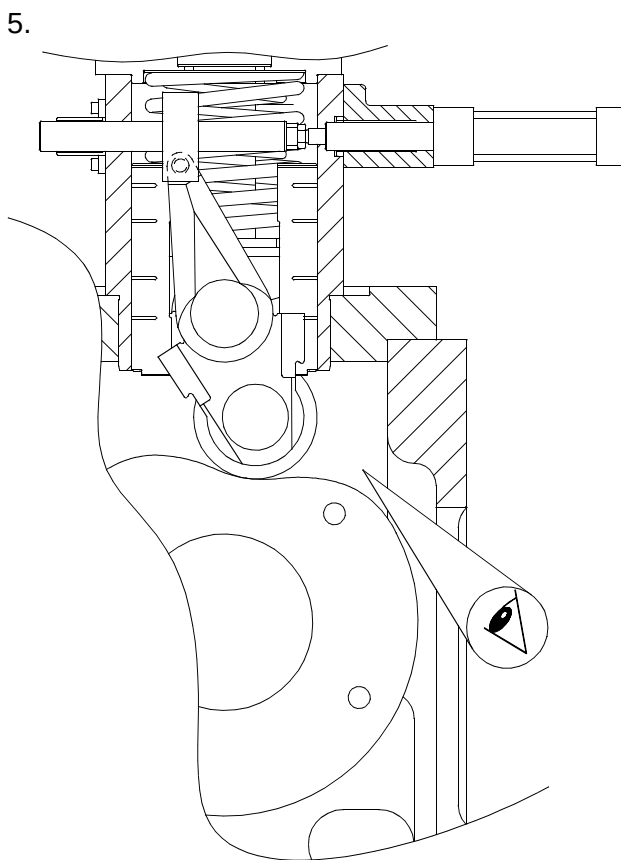
4. Lower the roller guide onto the fuel cam again.
See Procedure 909-16.

3.



GN909-16 201 03

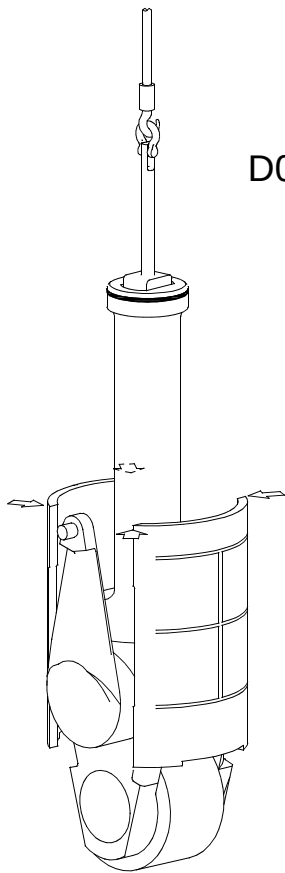
5.



GN909-15.1 203 05

7.

D09-61

**Roller guide removed from engine**

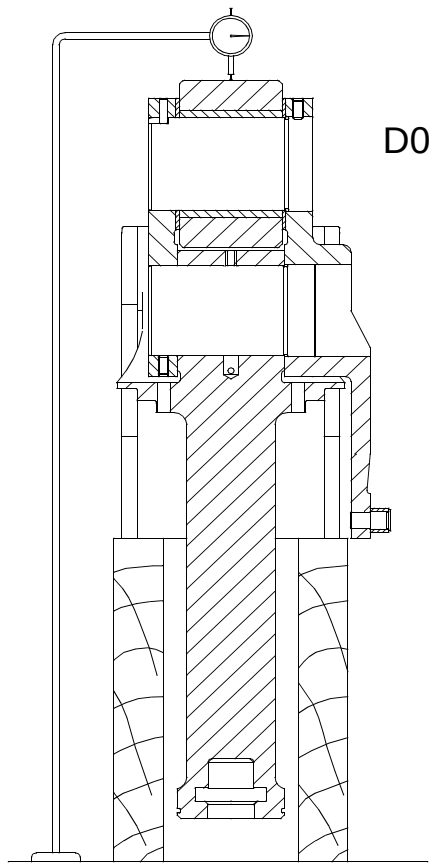
A more suitable and reliable method is to check the roller guide and the bearing clearance after the roller guide has been dismantled:

5. Dismount the roller guide from the engine.
See Procedure 909-15.2.
6. Check the surface of the roller and the sliding surfaces of the roller guide, and measure any ovalness.
7. Turn the roller guide upside down and land it in a vertical position on a couple of wooden planks, with the roller hanging freely.

Measure the clearance in the slide bearing by placing a dial gauge against the roller and lifting the roller as much as the clearance permits.

8.

D09-57



Note!

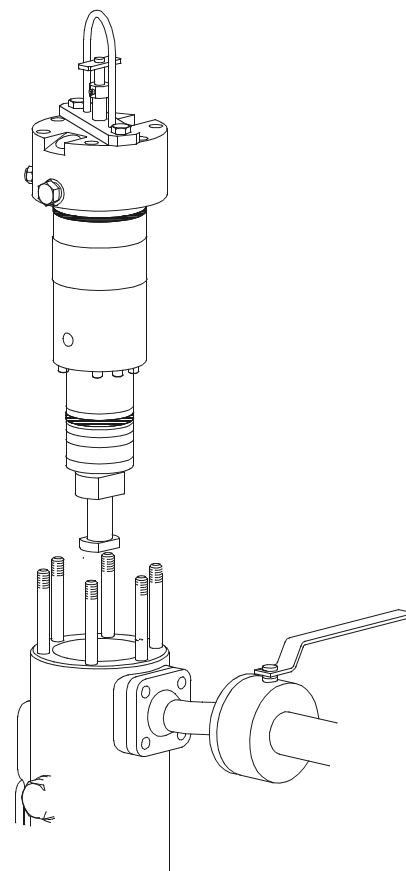
Before dismantling any part of the fuel pump, make sure that the pump has been relieved of pressure and that all oil has been drained off.

1. Dismantle the top cover with the barrel/plunger assembly.
See Procedure 909-5.2.
2. Unscrew the nuts, mount the lifting tool and remove the fuel pump housing.
See Procedure 909-4.2.

Note!

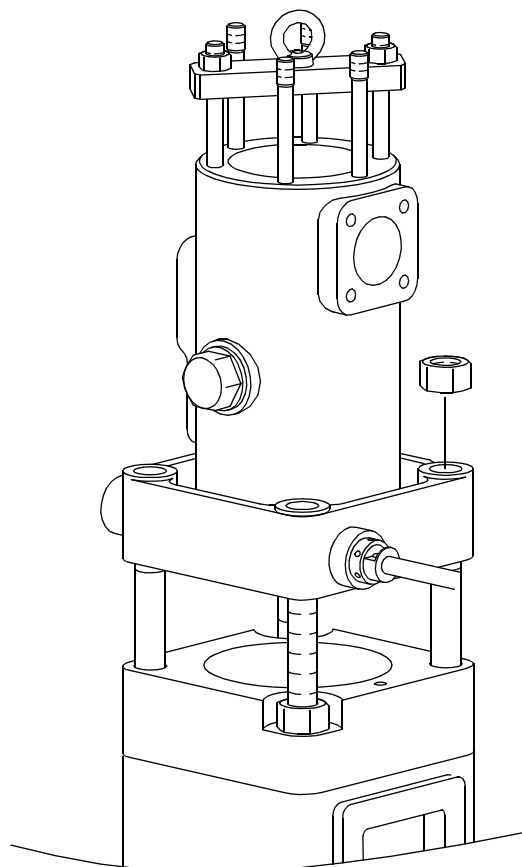
When lifting the fuel pump housing, take care not to damage the threads on the studs.

1.



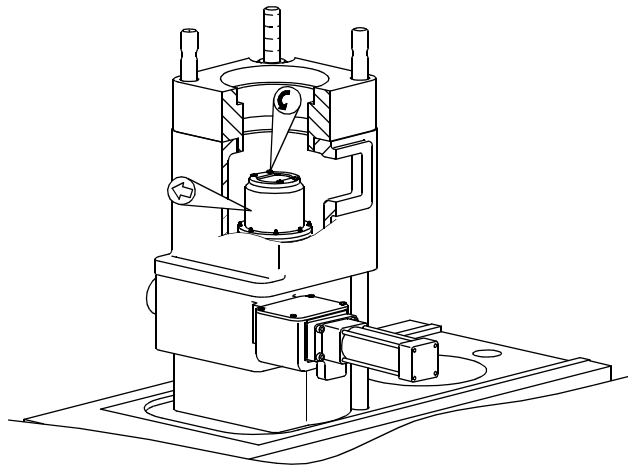
GN9095.2 205 05

2.



EN909-15.2 205 02

3.



3. Loosen and remove the screws from the sealing cap on top of the roller guide inside the pump base.

Remove the sealing cap from the roller guide/pump base.

Remove and discard the O-ring from the groove in the top of the roller guide.

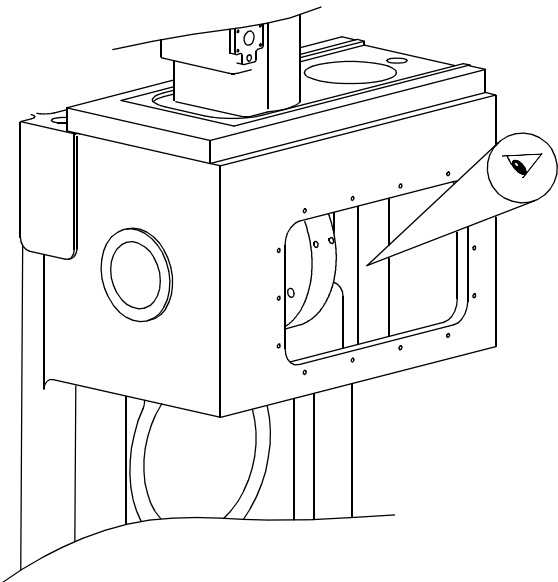
4. Remove the cover on the camshaft housing to inspect the position of the roller guide. Turn the engine until the roller guide is at BDC.

5. Loosen the nuts on the two threaded studs successively until the roller guide springs are relieved.

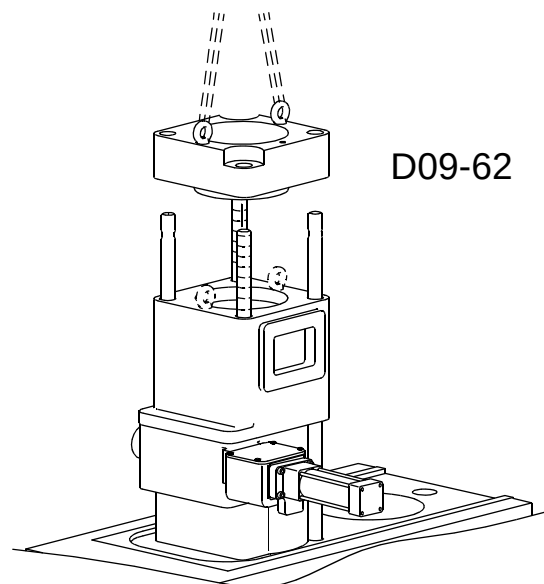
Remove the nuts, mount two eye bolts in the upper pump base and lift it away.

When lifting the pump base, take care not to damage the threads of the studs.

4.



5.



6. Mount two eye bolts in the sealing housing and lift it away.

When lifting, take care that the bushing of the sealing housing slides against the upper part of the roller guide without scratching.

Remove the springs from the roller guide.

7. Mount the roller guide lifting tool as follows:

- Lift the lock plate of the tool up on the shaft of the tool.
- Position the tool so that its foot rests on the thrust piece in the bayonet joint of the roller guide.
- Turn the tool 90° so as to allow the foot to engage properly in the bayonet joint.
- Lower the lock plate, thereby securing the foot in the locked position.

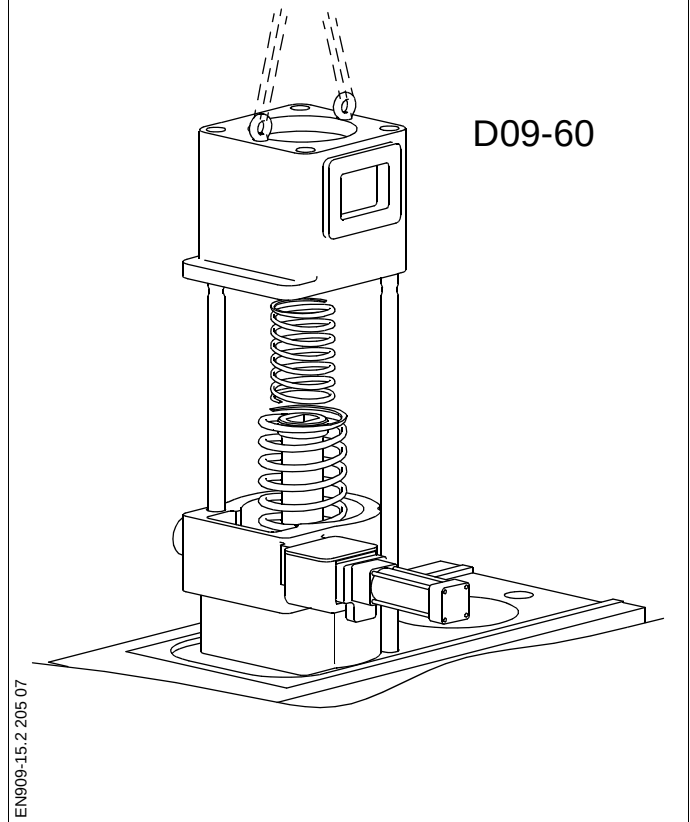
Lift the roller guide out of the roller guide housing, using the crane.

When lifting, take care that the slide surfaces of the roller guide do not scrape against the roller guide bushing.

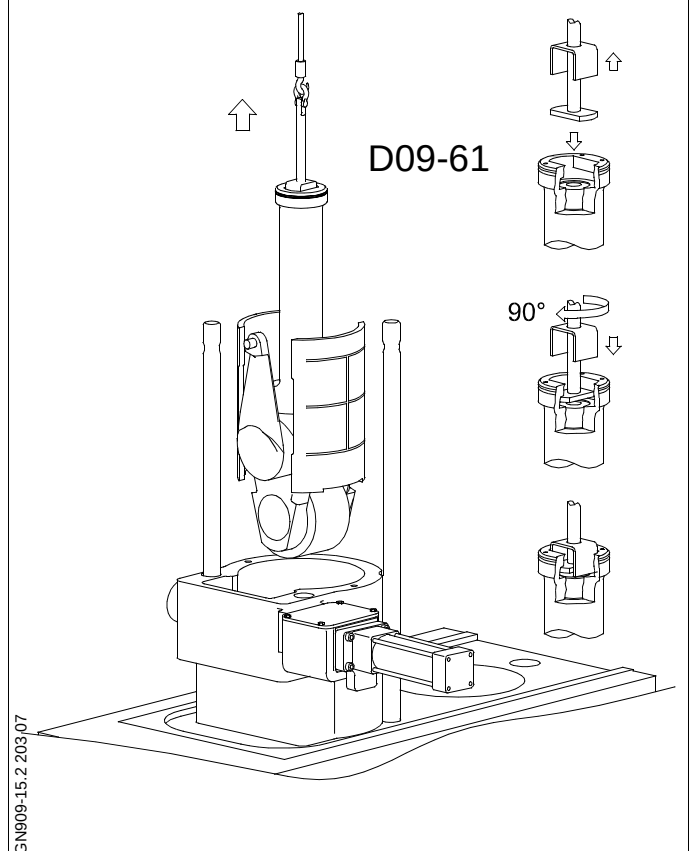
Land the roller guide on wooden planks.

8. Check the roller guide.
See *Procedure 909-15.1*.

6.

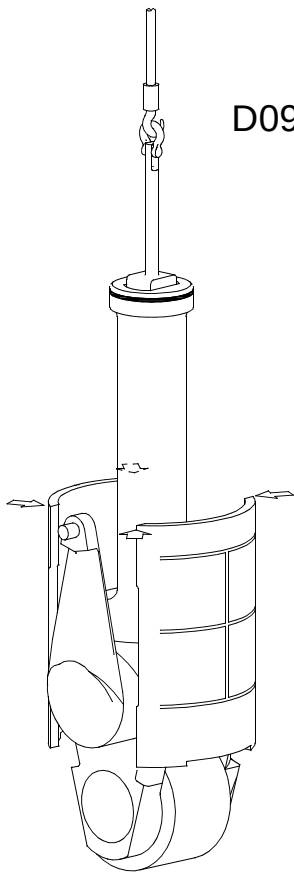


7.



1.

D09-61



1. Inspect and check the fuel pump roller guide.
See *Procedure 909-15.1*.

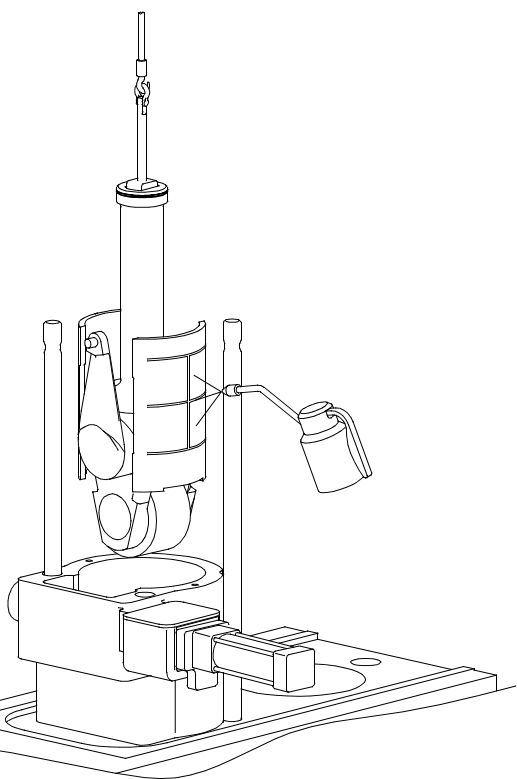
2. It is recommended that the roller guide should **ONLY** be dismantled if

- irregularities when turning the roller,
- damage to the roller,
- larger clearance than stated on the Data sheet,
- seizure marks on the slide surfaces,

have been observed when checking the roller guide.

Note!
For disassembling of the roller guide, contact an MAN B&W authorized repair shop.

3.



3. If no irregularities are found, just clean and lubricate the roller guide before remounting it in the roller guide bushing.

1. Lubricate the roller guide with plenty of camshaft lubricating oil, and mount it in the roller guide bushing, using the lifting tool.

During mounting, take care not to scratch the sliding surfaces of the roller guide.

Make sure that the pin of the reversing link fits in the guide of the reversing shaft.

2. After mounting the roller guide with the reversing link, check the clearance between the roller guide and the guide plate mounted in the roller guide bushing.

The clearance **C** between the roller guide and the guide plate must be the same at both ends. See *Data*.

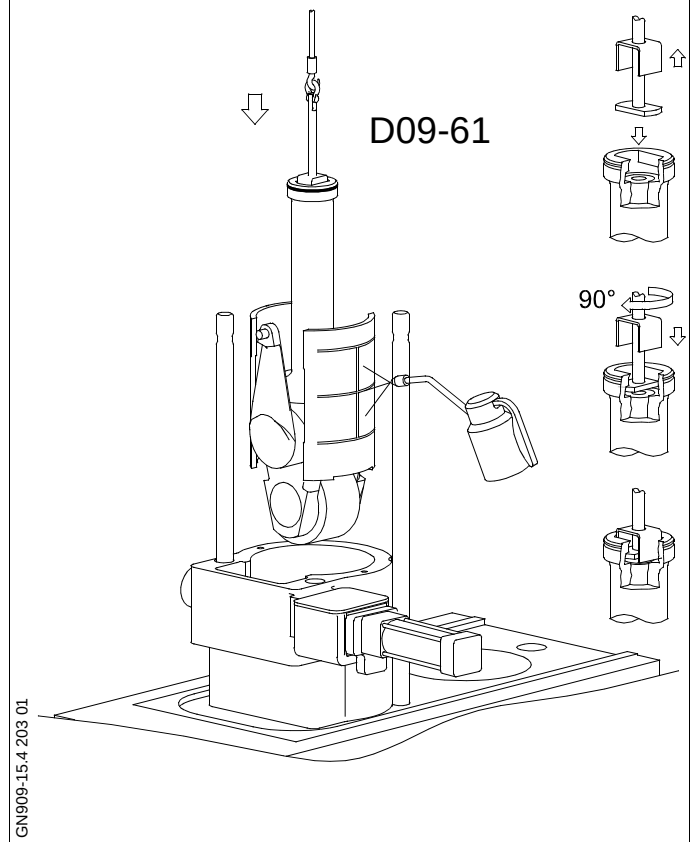
The clearances **A** and **B** between the guideway and the pin of the reversing link (arm) must be checked in the AHEAD and ASTERN positions.

Hint!

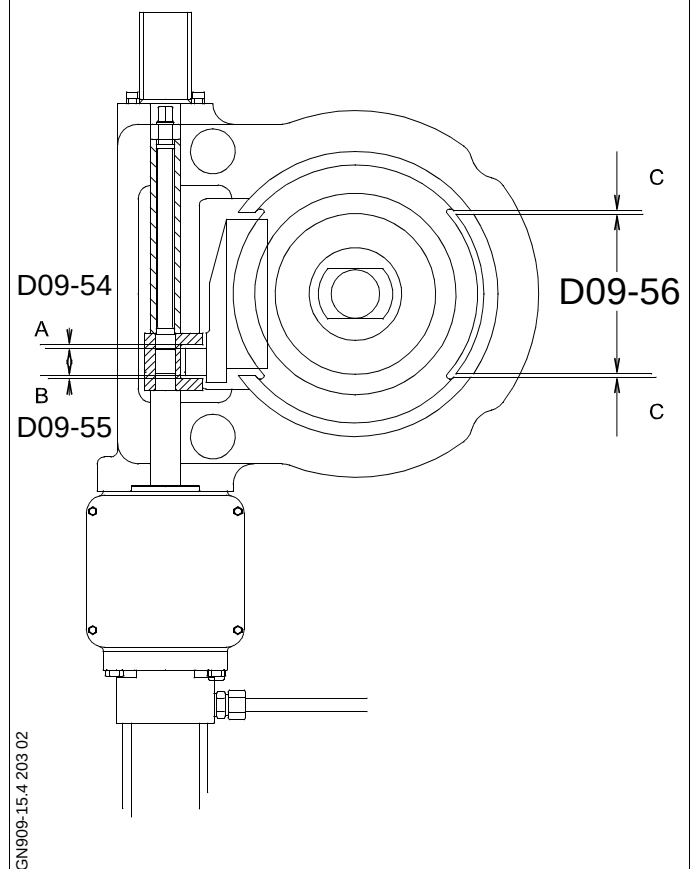
Remove the pipe connections to the air cylinder, and connect working air to the air cylinder, and check that the reversing mechanism is working smoothly.

The clearances **A** and **B** must be approx. the same in both positions. See *Data*.

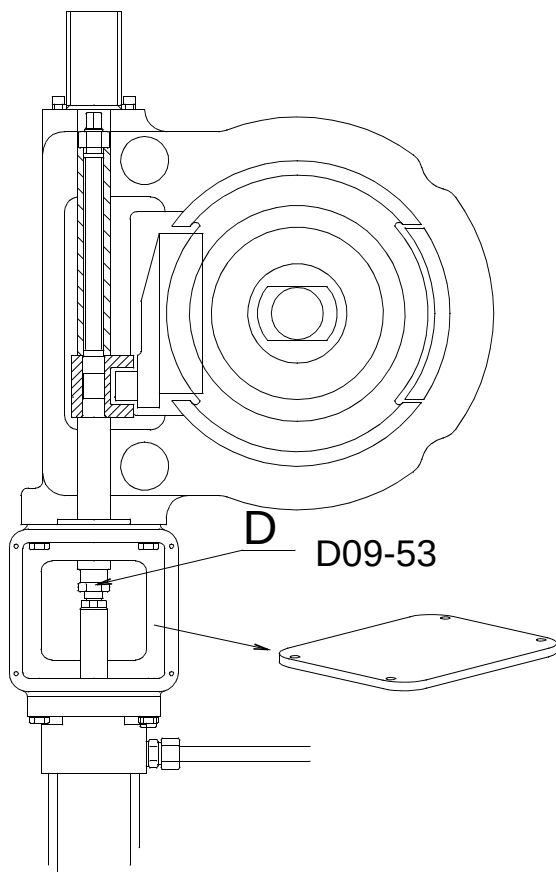
1.



2.



3.



3. If adjustment is necessary, it must take place in the ASTERN position.

Remove the cover over the shaft connection.

Loosen nut **D** and carry out the adjustment.

After adjustment, shift to AHEAD position and check the clearances **A** and **B** once more.

Finally, shift again to ASTERN.

Tighten nut **D** (see data), and lock the nut with Loctite type: Screw Lock No. 243.

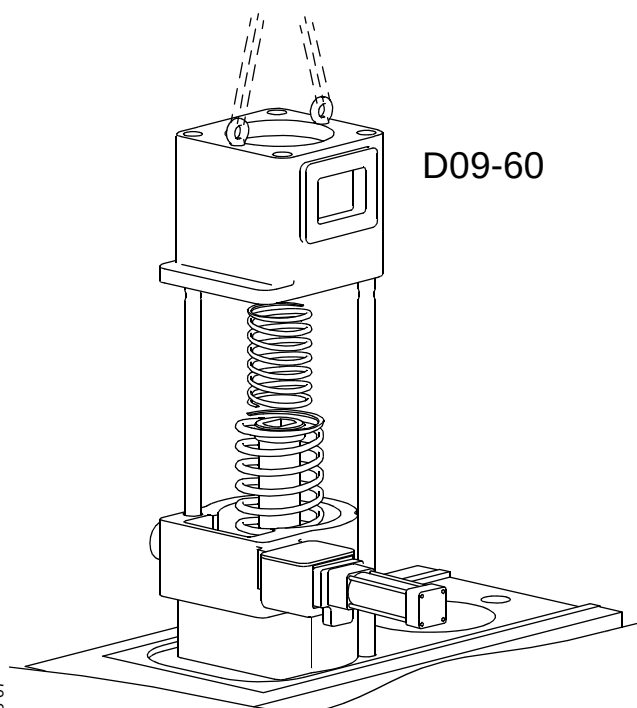
Mount the cover over the shaft connection.

4. Turn the engine until the roller guide is at BDC.
5. Mount the springs and the spring seat discs over the roller guide.

Mount the sealing housing over the roller guide and the springs.

Remove the eye bolts.

5.



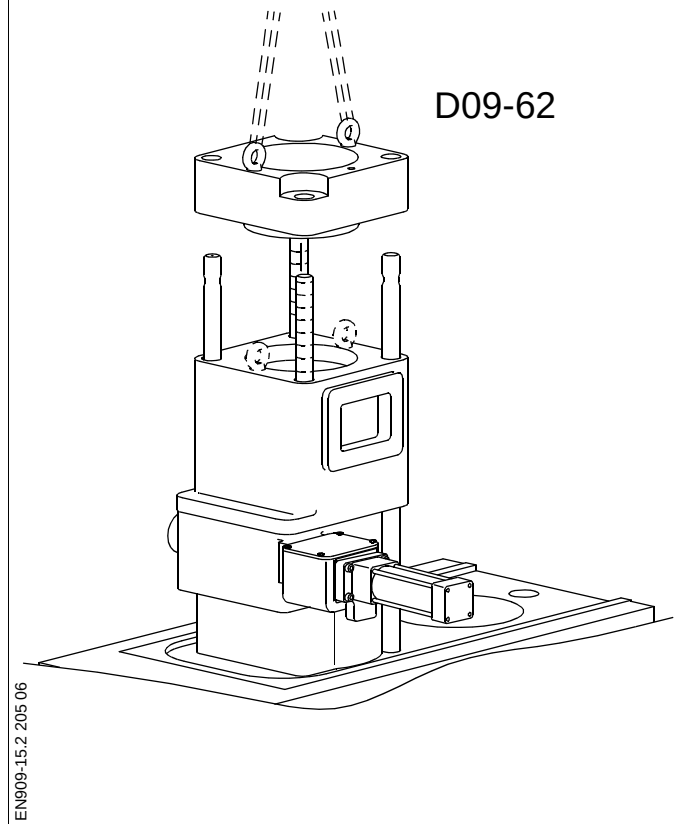
6. Mount the pump base.

Mount the two nuts on the threaded studs.

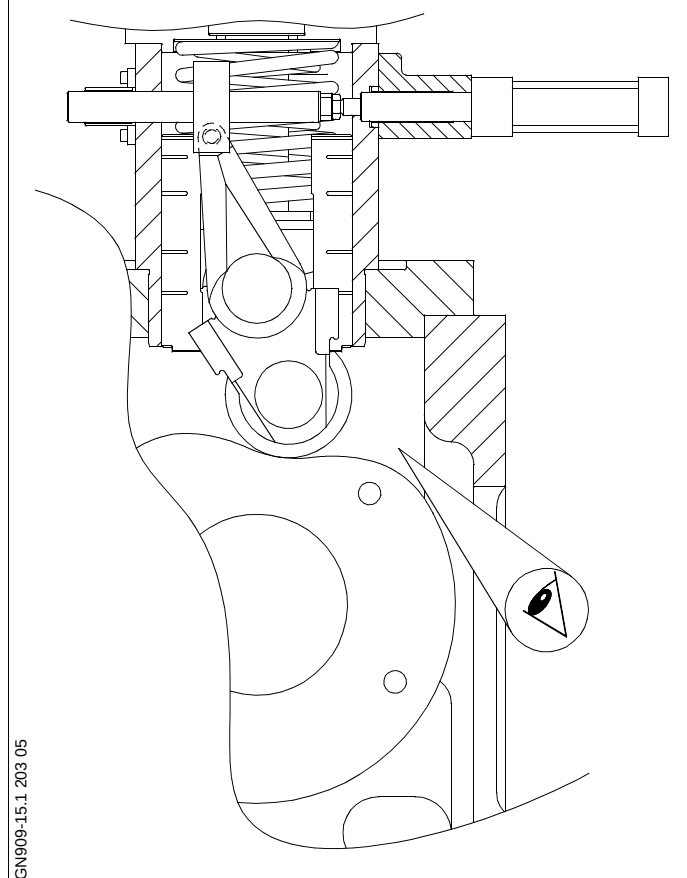
Tighten the nuts until the lower pump base is fastened tightly to the roller guide bushing.

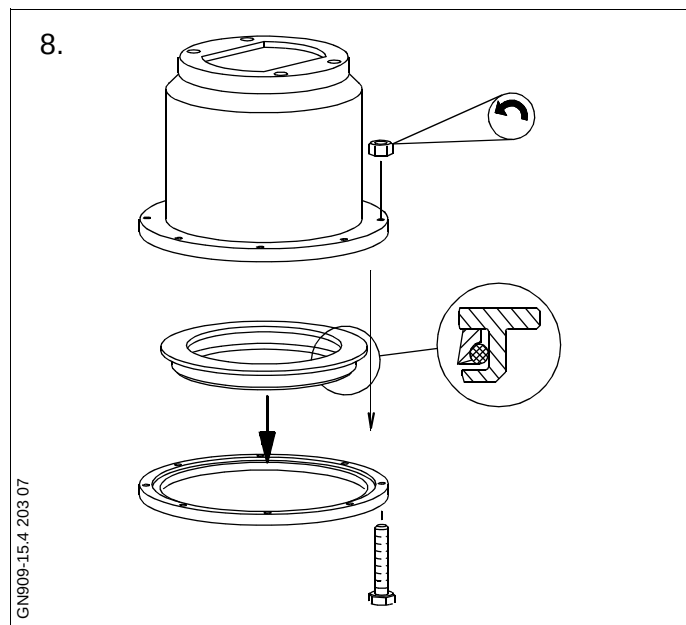
7. Check the movement of the reversing link. See *Procedure 909-15.1*.

6.



7.





8. Replace the scraper rings in the sealing cap as follows:

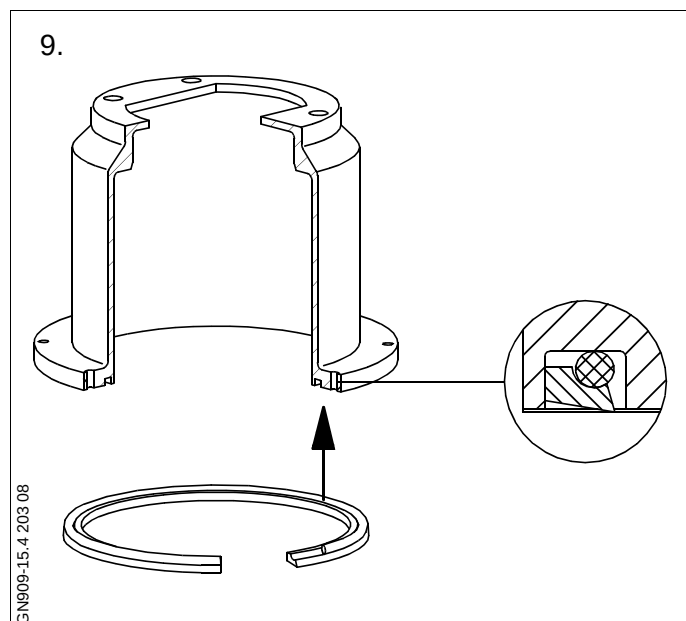
Loosen and remove the screws and nuts from the bottom of the cap, remove the retaining flange and ring holder with scraper ring.

9. Replace the axial scraper ring in the bottom of the cap. Check that the O-ring and scraper ring are mounted correctly. See the sketch.

10. Remove the scraper ring and O-ring from the ring holder.

Place the new O-ring in the ring holder.

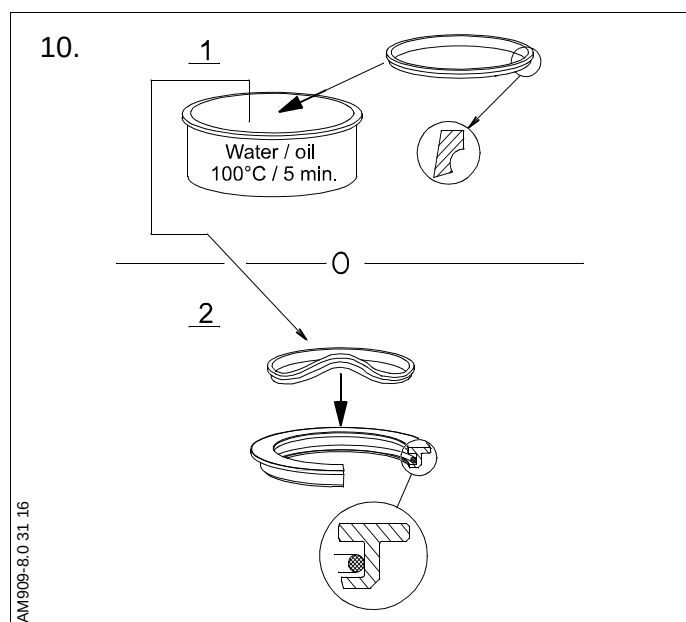
Before mounting, the new scraper ring must be heated in 100°C hot water or oil for at least five minutes.



Note!

When mounting the scraper ring in the groove, fold the ring without making any sharp edges, and be careful not to cut the scraper ring on the edge of the ring groove when pressing it in.

Check that the O-ring and scraper ring are mounted correctly. See the sketch.



11. Assemble the sealing cap, the ring holder and retaining flange.

Tighten the nuts on the screws.

After tightening, check that the holder can be moved.

12. Mount a new O-ring in the groove on top of the roller guide.

Mount the sealing cap on top of the roller guide, over the sealing bush inside the pump base.

Tighten the screws and lock with locking wire.

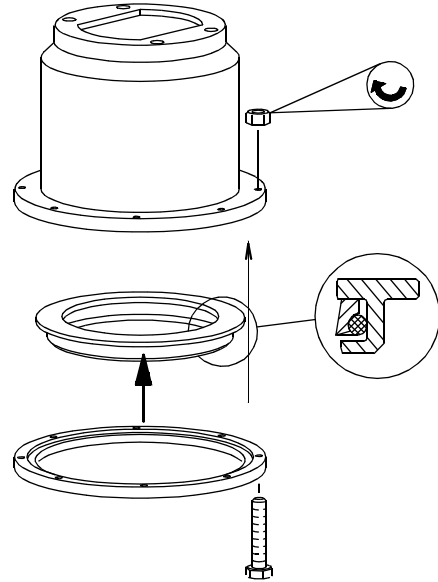
13. Mount the fuel pump housing.

See *Procedure 909-4.4*.

Mount the top cover with the barrel/plunger assembly.

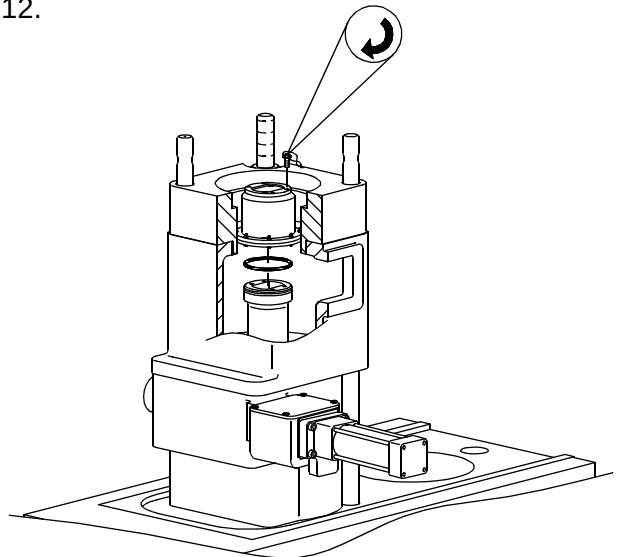
See *Procedure 909-5.4*.

11.



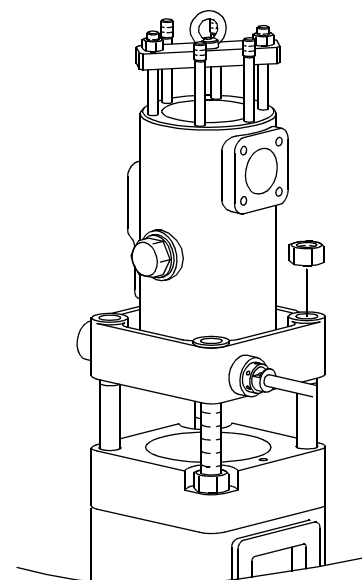
GN909-15.4 203 10

12.



EN909-15.4 205 12

13.



EN909-15.2 205 02

SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

☒	Stopped engine
☒	Shut off starting air supply – <i>At starting air receiver</i>
☒	Block the main starting valve
☒	Shut off starting air distributor/distributing system supply
☒	Shut off safety air supply – <i>Not ME engines</i>
☒	Shut off control air supply
☐	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
☒	Engage turning gear
☐	Shut off cooling water
☒	Shut off fuel oil
☐	Stop lubricating oil supply
☐	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P90951	082	Lifting tool for fuel pump roller guide
P90951	128	Crowfoot spanner
P90951	212	Extraction tool for suctions valve

Disengagement of fuel pump

Warning!

Disengagement or re-engagement of the fuel pump roller guide or the exhaust valve roller guide must **only** take place at engine standstill.

1. Engine trouble might require a cylinder to be taken out of action. The engine can continue operation with the fuel and exhaust systems of a single cylinder disconnected. See *Volume I, Section 704*.

To disconnect the fuel pump of a single cylinder, proceed as follows:

2. Remove the fuel oil high pressure pipes and the puncture valve from the fuel pump top cover.

Note!

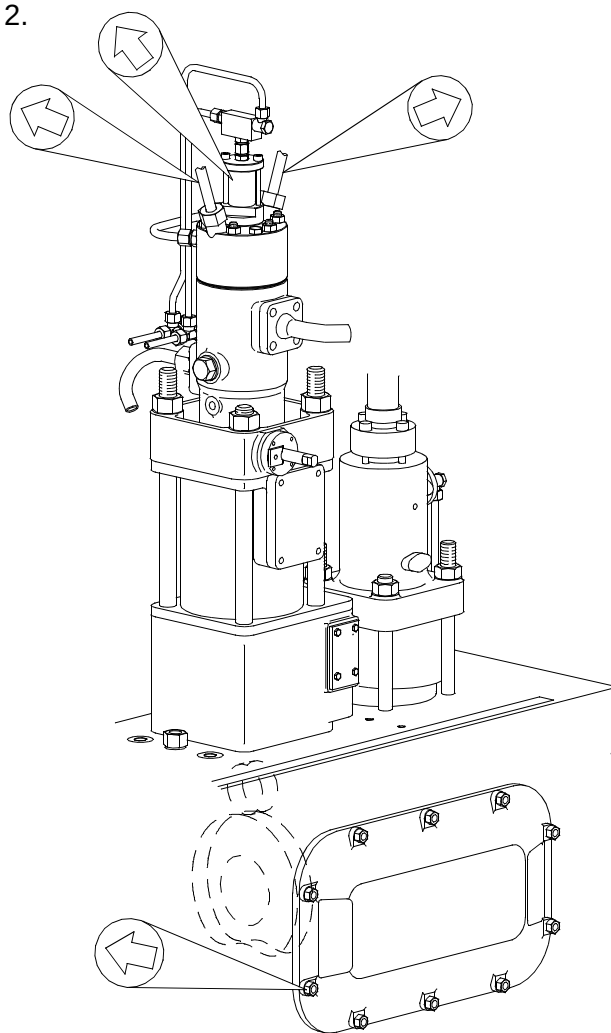
Before dismantling any part of the fuel pump, make sure that the pump has been relieved of pressure and that all oil has been drained off.

Screw a plug-screw into the control air pipe of the puncture valve.

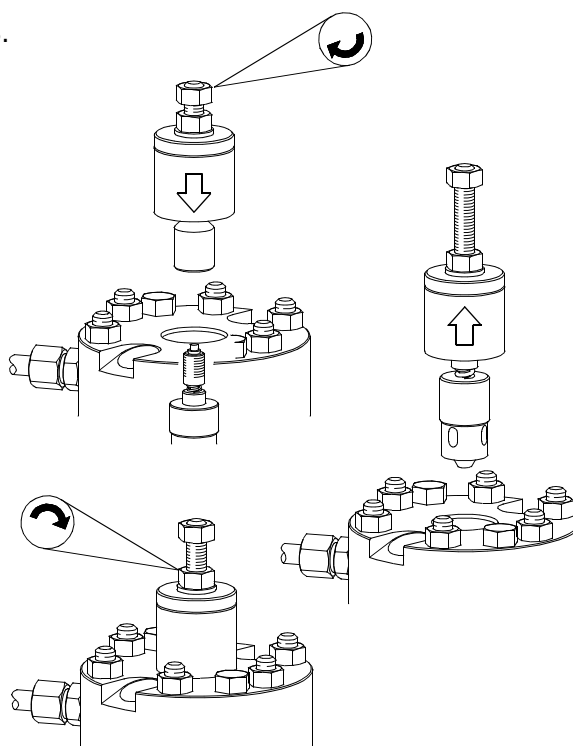
Remove the inspection cover on the camshaft housing. Turn the engine until the fuel pump roller is at on the circular part of the cam.

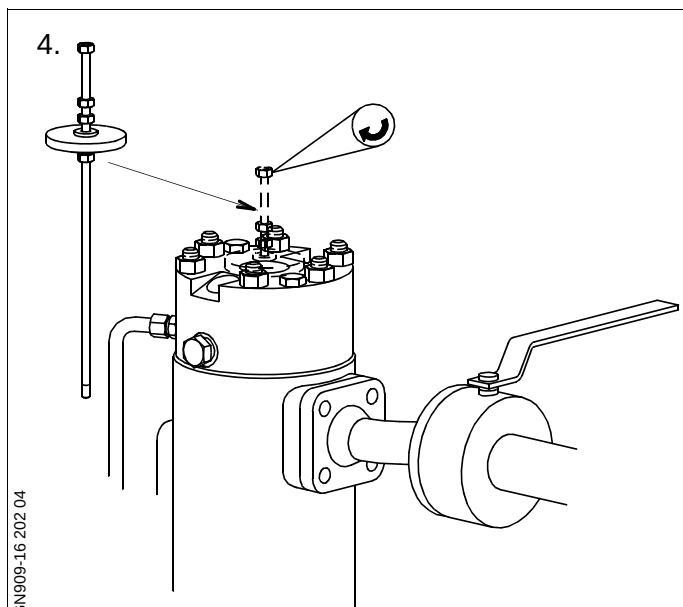
3. Dismount the suction valve by means of the extractor tool.

2.



3.





4. Mount the lifting tool on the fuel pump top cover.

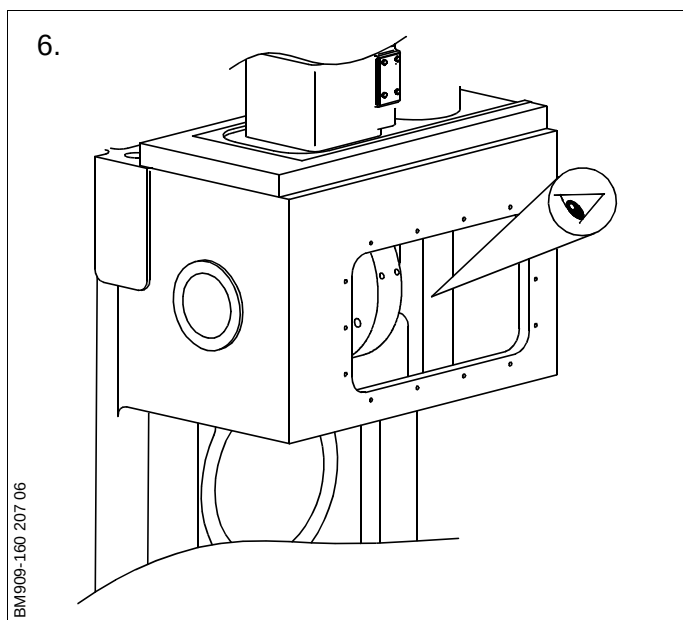
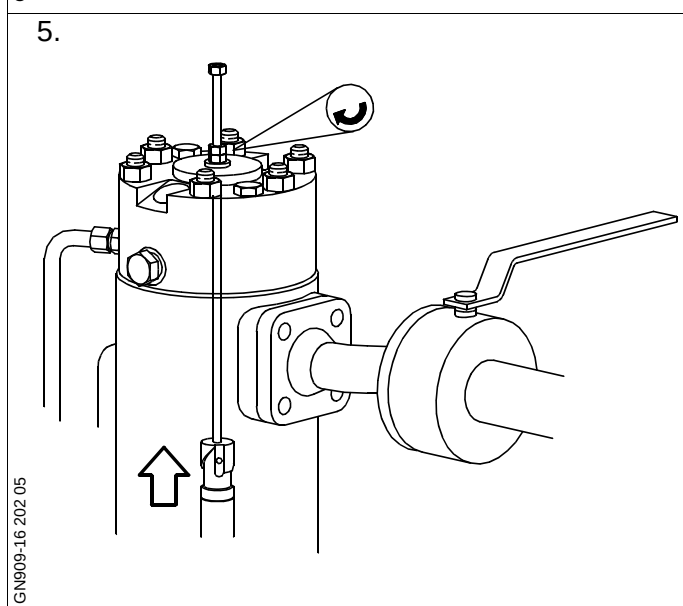
Turn the uppermost nut to screw the spindle of the lifting tool into the fuel pump plunger.

5. Tighten the lowermost nut on the tool to pull the roller guide upwards as far as possible.

Lock the tool in this position by means of the lock nut.

6. Turn the engine one full revolution and inspect through the opening in the camshaft housing to make sure that the roller guide is clear of the fuel cam.

7. The engine can now operate on the remaining cylinders. See *Volume I, Section 704*.



Re-engagement of fuel pump

Warning!

Re-engagement of the fuel pump must **only** take place at engine standstill.

8. Remove the cover on the camshaft housing. Turn the engine until the circular section of the fuel cam is positioned upwards.
9. Lower the roller guide on to the fuel cam in the reverse order to lifting and remove the lifting tool.
10. Re-mount the suction valve and the puncture valve. (See Procedure 909-8.4).

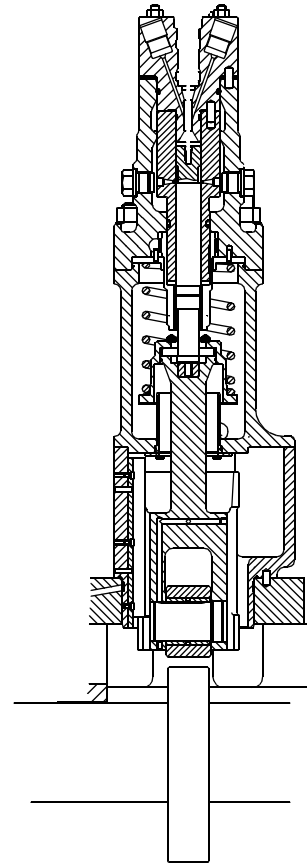
Remove the plug screw from the control air pipe.

Note!

It is recommended always to mount new or overhauled suction and puncture valves.

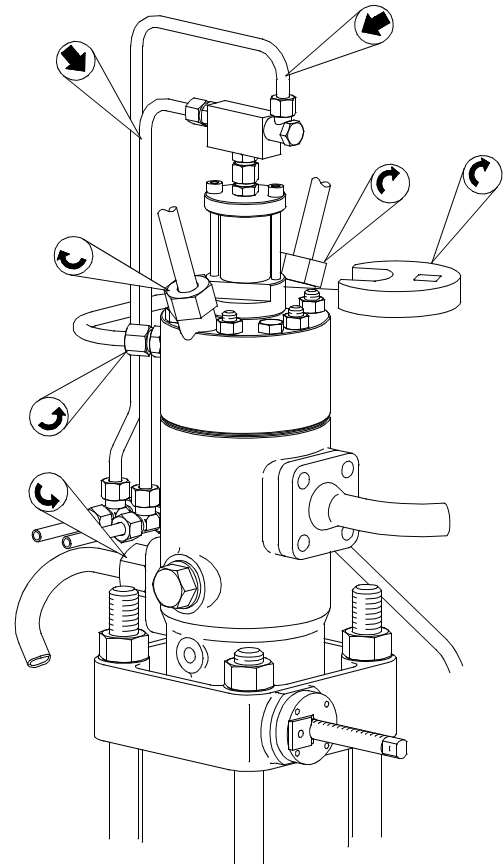
Mount the fuel oil high pressure pipes. (See Procedure 909-14.4).

8.

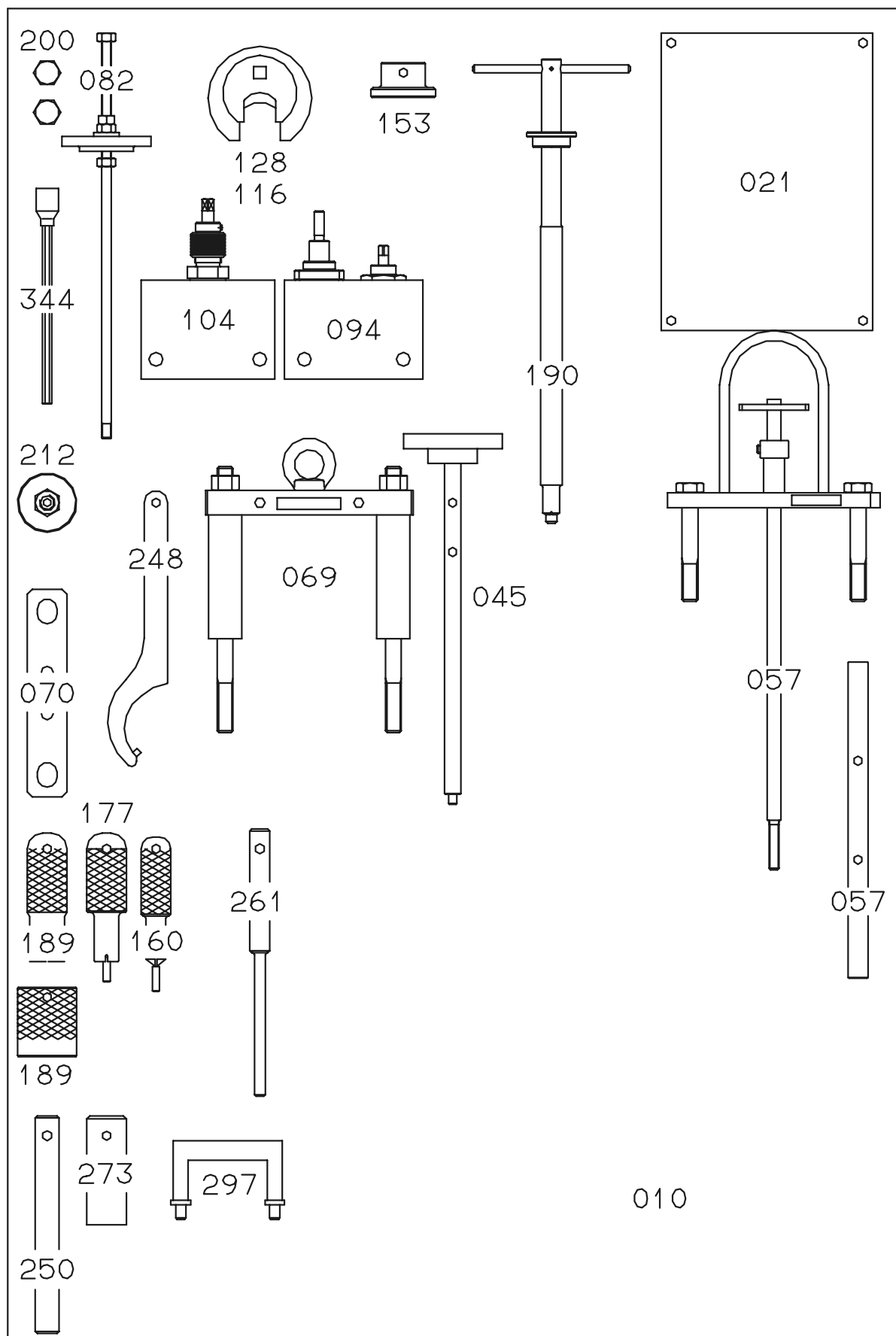


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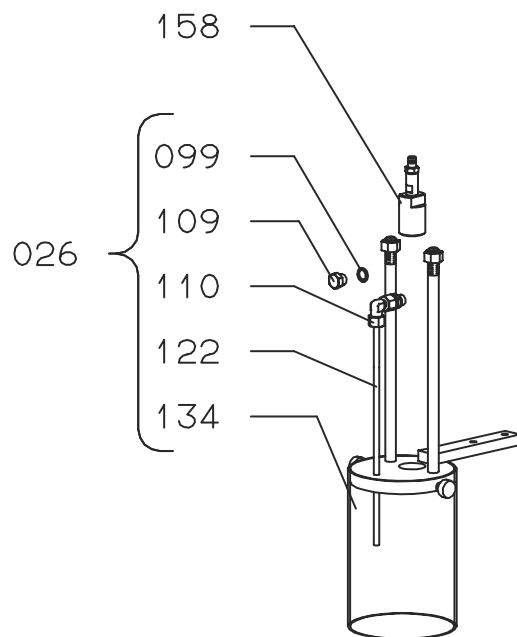
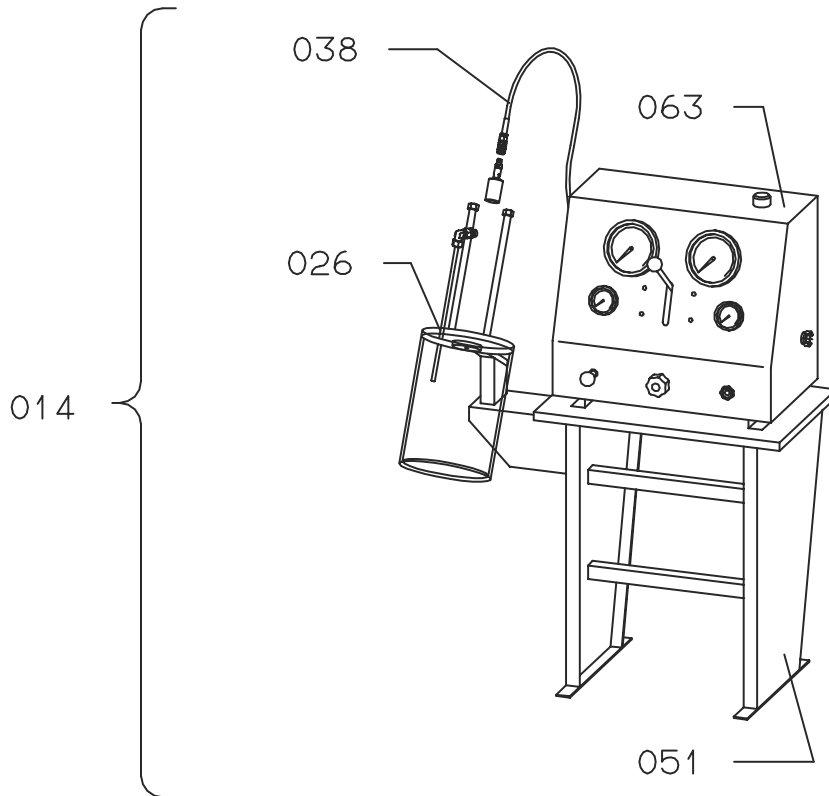
10.



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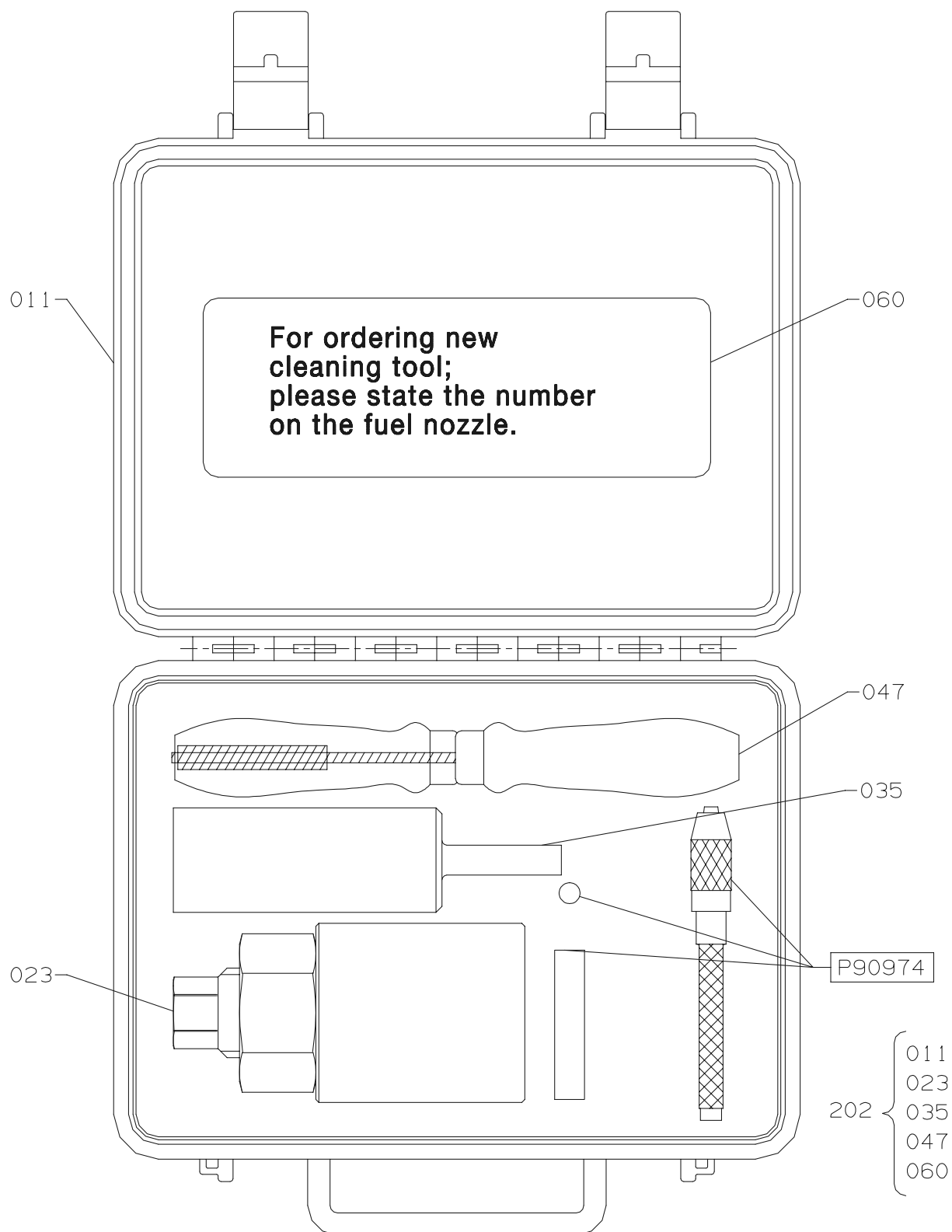


Item No.	Item Description	Item No.	Item Description
010	Panel for tools		
021	Name plate		
045	Measuring tool for fuel pump		
057	Lifting tool for fuel pump		
069	Lifting tool for fuel pump		
070	Lifting tool for fuel pump housing		
082	Lifting tool for roller guide		
094	Milling tool for fuel oil pipe seats		
104	Milling tool for fuel oil pipe		
116	Crowfoot spanner		
128	Crowfoot spanner		
153	Flange for dismantling of non-return valve		
165	Grinding mandrel for valve head		
177	Grinding mandrel for thrust spindle		
189	Grinding mandrel for holder, outside		
190	Grinding mandrel for holder, inside		
200	Screw		
212	Extractor tool for suction valve		
248	Hook spanner		
250	Drift for spindle guide		
261	Drift for puncture valve		
273	Drift for Non-return valve		
297	Assembling tool		
344	Spanner		



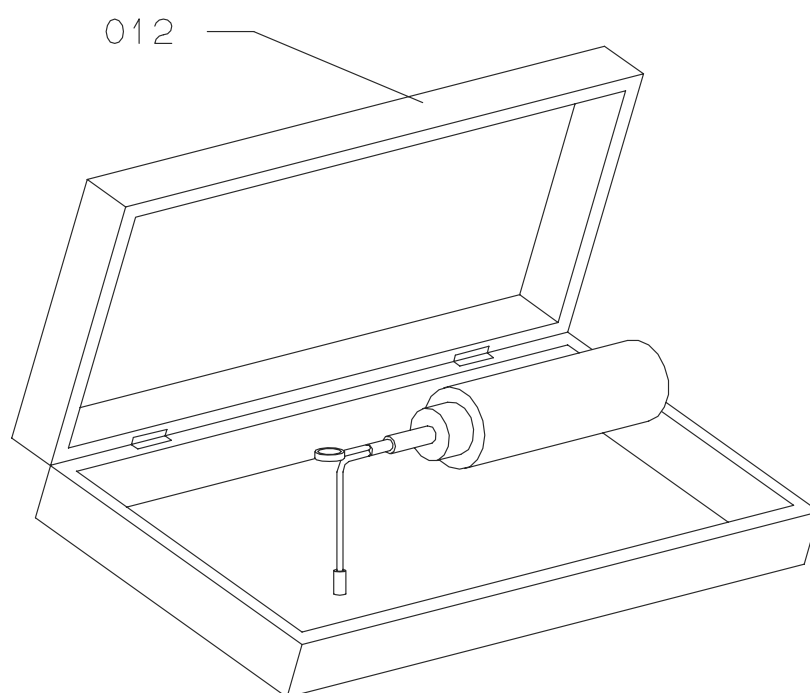


Item No.	Item Description	Item No.	Item Description
014	Test rig, complete		
026	Rig fuel oil valve		
038	High-pressure hose		
051	Table		
063	High-pressure pump		
099	Gasket		
109	Plug screw		
110	Angle union		
122	Hose		
134	Cover, test rig		
158	Connecting piece		





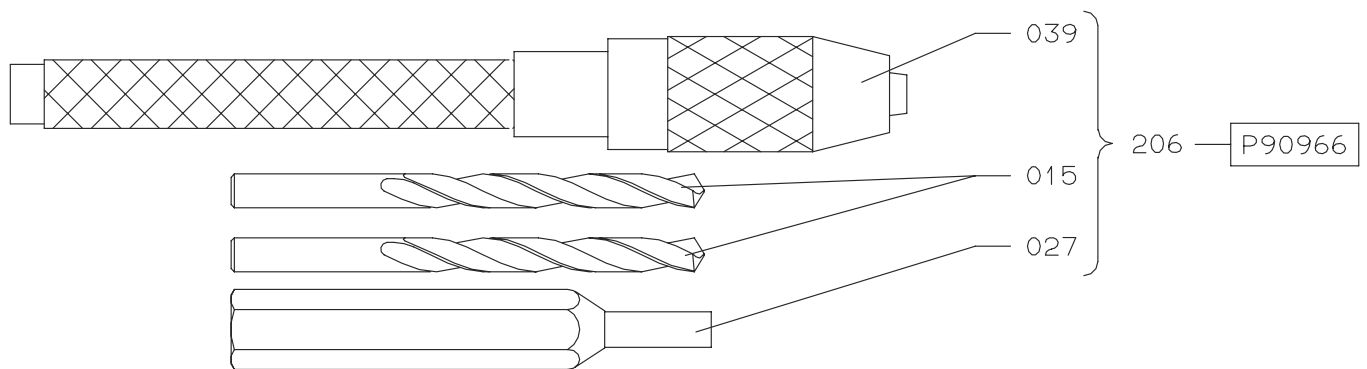
Item No.	Item Description	Item No.	Item Description
011	Tool box		
023	Extractor for atomizer		
035	Drift for atomizer		
047	Cleaning brush		
060	Instruction plate		
202	Toolset, complete		
	Note: When odering new cleaning tool, please state P/N on fuel nozzle.		





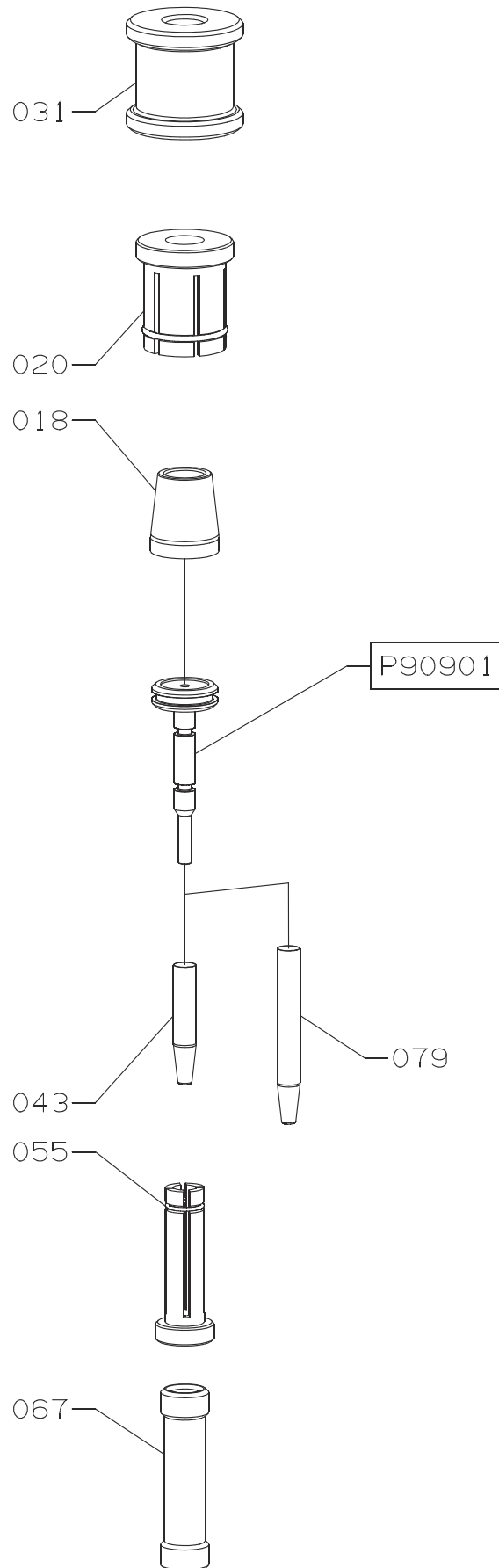
Item No.	Item Description
012	Probe light with magnifier, complete

Item No.	Item Description





Item No.	Item Description	Item No.	Item Description
015	Drill*		
027	Test mandrel*		
039	Pin vice*		
206	Cleaning tool set, complete		
	Note: * When ordering new cleaning tool, please state P/N on fuel nozzle.		



Item No.	Item Description
018	Cone
020	Pushing tool
031	Compression tool
043	Cone
055	Pushing tool
067	Compression tool
079	Cone

Item No.	Item Description